



# GlowCore

## SE Project Documentation

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|                    |                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------|
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# Contents

|           |                                         |          |
|-----------|-----------------------------------------|----------|
| <b>I</b>  | <b>Management Summary</b>               | <b>1</b> |
| <b>II</b> | <b>Product Documentation</b>            | <b>4</b> |
| 1         | Glossary                                | 5        |
| 2         | Requirements                            | 6        |
| 2.1       | Functional Requirements                 | 6        |
| 2.1.1     | Actors / Personas                       | 6        |
| 2.1.1.1   | Casual Player                           | 6        |
| 2.1.1.2   | Dedicated Player                        | 6        |
| 2.1.2     | Use Case Diagram                        | 7        |
| 2.1.3     | Use Cases                               | 8        |
| 2.1.4     | UC Overview                             | 8        |
| 2.2       | Non-Functional Requirements             | 12       |
| 2.2.1     | NFR Overview                            | 13       |
| 2.2.2     | Priority Matrix                         | 14       |
| 2.2.3     | Performance Efficiency                  | 14       |
| 2.2.4     | Usability                               | 16       |
| 2.2.5     | Reliability                             | 18       |
| 2.2.6     | Maintainability                         | 19       |
| 2.2.7     | Portability                             | 20       |
| 2.2.8     | NFR-Architecture Traceability           | 22       |
| 3         | Domain Analysis                         | 23       |
| 3.1       | Enemy State Machine                     | 24       |
| 4         | Architecture                            | 25       |
| 4.1       | Frontend Architecture                   | 25       |
| 4.1.1     | Main Menu                               | 25       |
| 4.1.2     | New Game                                | 26       |
| 4.1.3     | Pause Menu                              | 26       |
| 4.1.4     | Audio Settings                          | 27       |
| 4.1.5     | Display Settings                        | 27       |
| 4.1.6     | Unapplied Changes                       | 28       |
| 4.1.7     | Inventory                               | 28       |
| 4.1.8     | Crafting Table                          | 28       |
| 4.1.9     | More Crafting Stations                  | 29       |
| 4.1.10    | GlowCore Upgrade                        | 29       |
| 4.1.11    | Chest                                   | 29       |
| 4.2       | C4 Model                                | 30       |
| 4.2.1     | System Context Diagram                  | 30       |
| 4.2.2     | Container Diagram                       | 31       |
| 4.2.3     | Component Diagram                       | 32       |
| 4.2.3.1   | Gameplay Logic                          | 32       |
| 4.2.3.2   | Presentation Layer                      | 34       |
| 4.3       | Interfaces                              | 35       |
| 4.3.1     | Overview Diagram                        | 35       |
| 4.3.2     | Interface Contracts                     | 36       |
| 4.3.3     | Extensibility through Stable Interfaces | 36       |
| 4.4       | Technologies                            | 37       |
| 4.4.1     | Unity                                   | 37       |
| 4.4.2     | Universal Render Pipeline (URP)         | 37       |
| 4.4.3     | .NET & C#                               | 37       |
| 4.4.4     | GitHub                                  | 37       |
| 4.4.5     | Blender                                 | 37       |
| 4.4.6     | Audacity                                | 37       |

|         |                                                              |    |
|---------|--------------------------------------------------------------|----|
| 4.5     | Used Design Patterns .....                                   | 39 |
| 4.5.1   | Component-Based Architecture .....                           | 39 |
| 4.5.2   | Singleton .....                                              | 39 |
| 4.5.3   | Dependency Injection .....                                   | 39 |
| 4.5.4   | Scriptable Objects .....                                     | 39 |
| 4.5.5   | Prefabs .....                                                | 39 |
| 4.6     | Extensibility .....                                          | 40 |
| 4.7     | External Dependencies .....                                  | 40 |
| 4.8     | Game Design .....                                            | 40 |
| 4.8.1   | Core Gameplay Loop .....                                     | 40 |
| 4.8.2   | Gameplay Progression .....                                   | 40 |
| 4.8.2.1 | Early Game .....                                             | 40 |
| 4.8.2.2 | Automation Phase 1 .....                                     | 41 |
| 4.8.2.3 | Automation Phase 2 .....                                     | 41 |
| 4.8.2.4 | Automation Phase 3 .....                                     | 41 |
| 4.8.2.5 | Full Automation .....                                        | 41 |
| 4.8.3   | Automation Constraints .....                                 | 41 |
| 4.8.4   | Depth and Complexity .....                                   | 41 |
| 4.9     | Key Architectural Decisions .....                            | 42 |
| 4.9.1   | ScriptableObjects as the Data Layer .....                    | 42 |
| 4.9.2   | IInteractable for World Interaction .....                    | 42 |
| 4.9.3   | IInventoryService and Event-Driven UI .....                  | 42 |
| 4.9.4   | Use Actions and Interact Actions .....                       | 42 |
| 4.9.5   | IPlayerInventoryAware via Explicit Injection .....           | 43 |
| 4.10    | Deployment .....                                             | 43 |
| 4.10.1  | Continuous Deployment - WebGL (GitHub Pages) .....           | 43 |
| 4.10.2  | Release Deployment - Windows & Linux (GitHub Releases) ..... | 43 |
| 4.10.3  | Steam .....                                                  | 43 |
| 4.10.4  | Deployment Diagram .....                                     | 44 |
| 5       | Quality Measures .....                                       | 45 |
| 5.1     | Working Environment .....                                    | 45 |
| 5.1.1   | CI/CD Pipeline .....                                         | 45 |
| 5.1.2   | Definition of Ready and Definition of Done .....             | 46 |
| 5.1.2.1 | DoR Example .....                                            | 47 |
| 5.1.2.2 | DoD Example .....                                            | 47 |
| 5.2     | Test Concept .....                                           | 48 |
| 5.2.1   | Automated Testing (Unit & Integration) .....                 | 48 |
| 5.2.2   | System Testing .....                                         | 48 |
| 5.2.3   | Usability Testing .....                                      | 48 |
| 5.2.4   | Non-Functional Requirements (NFR) Verification .....         | 49 |
| 5.3     | Unity Testing .....                                          | 49 |
| 5.3.1   | Edit Mode Tests .....                                        | 49 |
| 5.3.2   | Play Mode Tests .....                                        | 50 |
| 5.3.3   | Coverage Summary .....                                       | 50 |
| 5.4     | Code metrics .....                                           | 50 |
| 5.4.1   | Lines of Code .....                                          | 50 |
| 5.4.2   | Test Coverage .....                                          | 53 |
| 5.5     | User Testing .....                                           | 53 |
| 5.5.1   | User Test 1 (UT01) .....                                     | 54 |
| 5.5.1.1 | Additional Feedback .....                                    | 54 |
| 5.5.1.2 | Conclusion .....                                             | 54 |
| 5.5.2   | User Test 2 (UT02) .....                                     | 55 |
| 5.5.2.1 | Additional Feedback .....                                    | 55 |
| 5.5.2.2 | Conclusion .....                                             | 55 |
| 5.6     | Coding Guidelines .....                                      | 56 |
| 5.6.1   | Naming Conventions .....                                     | 56 |
| 5.6.2   | Class Member Ordering .....                                  | 56 |

|         |                                |    |
|---------|--------------------------------|----|
| 5.6.3   | General Guidelines .....       | 56 |
| 5.6.4   | Formatting Rules .....         | 57 |
| 5.7     | Version Control Workflow ..... | 57 |
| 5.7.0.1 | Branch Naming .....            | 57 |
| 5.7.0.2 | Rebasing Workflow .....        | 57 |
| 5.7.0.3 | Pull Requests .....            | 57 |
| 6       | Savefile .....                 | 58 |

### **III Project Documentation ..... 59**

|           |                                                      |    |
|-----------|------------------------------------------------------|----|
| 7         | Initial Project Proposal .....                       | 60 |
| 7.1       | Team Members .....                                   | 60 |
| 7.2       | Availabilities .....                                 | 60 |
| 7.3       | Vision .....                                         | 60 |
| 7.4       | Proposed Realisation .....                           | 61 |
| 8         | Project Plan .....                                   | 62 |
| 8.1       | Resources .....                                      | 62 |
| 8.1.1     | People .....                                         | 62 |
| 8.1.1.1   | Project Management Roles .....                       | 62 |
| 8.1.1.2   | Developer Roles .....                                | 63 |
| 8.1.2     | Time .....                                           | 63 |
| 8.1.3     | Cost .....                                           | 63 |
| 8.2       | Organization .....                                   | 64 |
| 8.2.1     | Meetings .....                                       | 64 |
| 8.2.2     | Planning Tools .....                                 | 65 |
| 8.3       | Schedule .....                                       | 65 |
| 8.3.1     | Work Items .....                                     | 65 |
| 8.3.2     | Long Term Plan .....                                 | 66 |
| 8.3.2.1   | Milestones .....                                     | 66 |
| 8.3.2.1.1 | Initial Milestones .....                             | 66 |
| 8.3.2.1.2 | Updated Milestones as of 07.04.2026 .....            | 67 |
| 8.3.2.2   | Epics .....                                          | 68 |
| 8.3.2.3   | Initial Backlog & MVP .....                          | 70 |
| 8.3.3     | Short Term Plan .....                                | 71 |
| 8.3.3.1   | Sprint 1 (Sprint Zero) 20.02.2026 - 06.03.2026 ..... | 71 |
| 8.3.3.2   | Sprint 2 06.03.2026 - 20.03.2026 .....               | 72 |
| 8.3.3.3   | Sprint 3 20.03.2026 - 03.04.2026 .....               | 73 |
| 8.3.3.4   | Sprint 4 03.04.2026 - 17.04.2026 .....               | 74 |
| 8.3.3.5   | Sprint 5 17.04.2026 - 01.05.2026 .....               | 75 |
| 8.3.3.6   | Sprint 6 01.05.2026 - 15.05.2026 .....               | 76 |
| 8.3.3.7   | Sprint 7 15.05.2026 - 29.05.2026 .....               | 77 |
| 8.4       | Risk Management .....                                | 78 |
| 8.4.1     | Risk Matrix .....                                    | 79 |
| 8.4.2     | Risk Matrix (Mitigated) .....                        | 79 |
| 8.4.3     | Assessed Risk Matrix (14.04.2026) .....              | 79 |
| 8.4.4     | Adjustment of Current Risk Assessment .....          | 80 |
| 9         | Time Tracking Report .....                           | 81 |
| 9.1       | Time Tracking .....                                  | 81 |
| 9.2       | Time Statistics .....                                | 81 |
| 10        | Personal Reports .....                               | 82 |
| 10.1      | Nathanael Fässler .....                              | 82 |
| 10.2      | Dominik Wyss .....                                   | 82 |
| 10.3      | Cedric Cathomas .....                                | 83 |
| 10.4      | Yoris Kucera .....                                   | 84 |
| 11        | Meeting Minutes .....                                | 85 |
| 11.1      | Weekly Scrum / 28.02.2026 .....                      | 85 |
| 11.1.1    | Agenda .....                                         | 85 |



|          |                                                          |     |
|----------|----------------------------------------------------------|-----|
| 11.1.2   | Todos .....                                              | 86  |
| 11.2     | Weekly Scrum and Sprint Retrospective / 06.03.2026 ..... | 86  |
| 11.2.1   | Agenda .....                                             | 86  |
| 11.2.2   | Sprint 1 Retrospective .....                             | 87  |
| 11.2.2.1 | What Went Well .....                                     | 87  |
| 11.2.2.2 | What Didn't Go Well .....                                | 87  |
| 11.2.2.3 | Improvements .....                                       | 87  |
| 11.2.3   | Todos .....                                              | 87  |
| 11.3     | Sprint Planning / 06.03.2026 .....                       | 88  |
| 11.3.1   | Sprint goal .....                                        | 88  |
| 11.4     | Sprint Review (Advisor) / 10.03.2026 .....               | 88  |
| 11.4.1   | Agenda .....                                             | 88  |
| 11.4.2   | Todos .....                                              | 89  |
| 11.4.3   | Next dates .....                                         | 89  |
| 11.5     | Weekly Scrum / 13.03.2026 .....                          | 89  |
| 11.5.1   | Agenda .....                                             | 89  |
| 11.5.2   | Todos .....                                              | 90  |
| 11.6     | Weekly Scrum and Sprint Retrospective / 20.03.2026 ..... | 91  |
| 11.6.1   | Agenda .....                                             | 91  |
| 11.6.2   | What went well .....                                     | 91  |
| 11.6.3   | What didn't .....                                        | 91  |
| 11.6.4   | Improvements .....                                       | 91  |
| 11.6.5   | Todos .....                                              | 92  |
| 11.7     | Sprint Planning / 20.03.2026 .....                       | 92  |
| 11.7.1   | Sprint goal .....                                        | 92  |
| 11.8     | Sprint Review (Advisor) / 24.03.2026 .....               | 92  |
| 11.8.1   | Agenda .....                                             | 93  |
| 11.8.2   | Todos .....                                              | 93  |
| 11.8.3   | Next dates .....                                         | 93  |
| 11.9     | Weekly Scrum / 27.03.2026 .....                          | 93  |
| 11.9.1   | Agenda .....                                             | 94  |
| 11.9.2   | Todos .....                                              | 94  |
| 11.10    | Weekly Scrum and Sprint Retrospective / 03.04.2026 ..... | 94  |
| 11.10.1  | Agenda .....                                             | 94  |
| 11.10.2  | What went well .....                                     | 95  |
| 11.10.3  | What didn't .....                                        | 95  |
| 11.10.4  | Todos .....                                              | 95  |
| 11.11    | Sprint Planning / 03.04.2026 .....                       | 95  |
| 11.11.1  | Sprint goal .....                                        | 95  |
| 11.12    | Sprint Review (Advisor) / 14.04.2026 .....               | 96  |
| 11.12.1  | Agenda .....                                             | 96  |
| 11.12.2  | Todos .....                                              | 96  |
| 11.12.3  | Next dates .....                                         | 97  |
| 11.13    | Weekly Scrum and Sprint Retrospective / 18.04.2026 ..... | 97  |
| 11.13.1  | Agenda .....                                             | 97  |
| 11.13.2  | What went well .....                                     | 97  |
| 11.13.3  | What didn't .....                                        | 97  |
| 11.13.4  | Todos .....                                              | 98  |
| 11.14    | Sprint Planning / 18.04.2026 .....                       | 98  |
| 11.14.1  | Sprint goal .....                                        | 98  |
| 11.15    | Weekly Scrum / 24.04.2026 .....                          | 98  |
| 11.15.1  | Agenda .....                                             | 99  |
| 11.15.2  | Todos .....                                              | 99  |
| 11.16    | Weekly Scrum and Sprint Retrospective / 30.04.2026 ..... | 99  |
| 11.16.1  | What went well .....                                     | 99  |
| 11.16.2  | What didn't .....                                        | 100 |
| 11.17    | Sprint Planning / 01.05.2026 .....                       | 100 |

|                                                                |     |
|----------------------------------------------------------------|-----|
| 11.17.1 Sprint goal .....                                      | 100 |
| 11.18 Sprint Review (Advisor) / 05.05.2026 .....               | 100 |
| 11.18.1 Agenda .....                                           | 101 |
| 11.18.2 Todos .....                                            | 101 |
| 11.18.3 Next dates .....                                       | 101 |
| 11.19 Weekly Scrum / 08.05.2026 .....                          | 101 |
| 11.19.1 Agenda .....                                           | 101 |
| 11.19.2 Todos .....                                            | 102 |
| 11.20 Weekly Scrum and Sprint Retrospective / 15.05.2026 ..... | 102 |
| 11.20.1 Agenda .....                                           | 102 |
| 11.20.2 What went well .....                                   | 102 |
| 11.20.3 What didn't .....                                      | 102 |
| 11.20.4 Todos .....                                            | 103 |
| 11.21 Sprint Planning / 15.05.2026 .....                       | 103 |
| 11.21.1 Sprint goal .....                                      | 103 |
| 11.22 Sprint Review (Advisor) / 19.05.2026 .....               | 103 |
| 11.22.1 Agenda .....                                           | 104 |
| 11.22.2 Todos .....                                            | 104 |
| 11.23 Weekly Scrum / 22.05.2026 .....                          | 104 |
| 11.23.1 Agenda .....                                           | 104 |
| 11.23.2 Todos .....                                            | 104 |
| 11.24 Weekly Scrum and Sprint Retrospective / 29.05.2026 ..... | 105 |
| 11.24.1 Agenda .....                                           | 105 |
| 11.24.2 What went well .....                                   | 105 |
| 11.24.3 What didn't .....                                      | 105 |
| 11.24.4 Todos .....                                            | 105 |

#### **IV List of Figures ..... 106**

#### **V List of Tables ..... 108**

# **Part I**

# **Management Summary**

# Management Summary

## Starting Point

GlowCore started as the idea of building a small singleplayer game where the player begins in a tiny lit area surrounded by darkness and has to expand the world by upgrading a central object called the GlowCore. The challenge for us as a team of four students had two sides. We had to design a game that is actually fun to play, and at the same time we had to build it cleanly enough that the code survives growing scope, four people working in parallel, and the usual deadline pressure of a Semester 4 project.

The other half of the goal was applying the software engineering process we learned about in lectures to a real codebase that we cared about. Scrum, CI/CD, automated testing, code reviews, branching, all of it had to work in practice and not just on paper.

## Our Approach

We picked Unity 6 with the Universal Render Pipeline as our engine. C# was a language everyone already knew and we used the time before the official project start to get up to speed on Unity, so we could be productive from day one. The code was split into two main layers, gameplay logic and presentation, with clear interfaces between them. ScriptableObjects are used for data like items, recipes and node definitions, which makes it possible to add new content without touching any code.

Work was organised in two-week sprints with a weekly Scrum meeting and a regular review with our advisor. Every change went through a pull request, was reviewed by a teammate, and had to pass our CI pipeline that runs linting, formatting checks and automated tests. Roles were split between the four of us so that everyone had a main area of responsibility but could still help out anywhere in the codebase.

One of the more important decisions we made during the project came after the first few sprints. The original plan was ambitious and included a combat system, multiple automation phases and additional biomes. After looking at our progress in April we decided to drop combat and the later automation phases from the Beta scope on purpose. The reasoning was simple, a smaller game that feels polished beats a bigger game that feels half finished. We documented the updated milestones and went through them with our advisor. Once the core of the game was solid, we picked automation back up and shipped it in the final sprint.

For the game itself we designed a core loop around gathering wood, crafting tools, upgrading the GlowCore and slowly automating the work with the axe machine. The intention was to give the player a steady sense of progress and to make automation feel like a reward rather than something the game hands out from the start.

## Result

GlowCore covers the full MVP we defined at the start of the project. The player can install the game, save and load progress, break and place nodes, expand the map by feeding the GlowCore, store items in chests, craft items at the workbench and the furnace, and change settings to their taste. All of this runs on Windows, Linux and in the browser through WebGL, with the WebGL build deployed automatically to GitHub Pages on every push to main or dev.

The numbers back this up. The game runs at around 120 FPS on WebGL during active play, well above the 60 FPS target we set ourselves. Worlds load in around 4 seconds for a fresh map, though large late-game saves can take up to 20 seconds, exceeding the original 3-second budget we set before the game grew this large, and save files come in at a few kilobytes despite a 5 MB ceiling. We ran two external user tests with players outside the team. Both testers reached the GlowCore upgrade loop on their own within a few minutes and confirmed that the core loop is fun to play. Their feedback on the parts that confused them or needed polish went straight into the next sprint.

On the engineering side the codebase stays consistent thanks to the automated checks, and the architecture proved its value when we added additional crafting stations late in the project without having to rewrite existing systems. The hours we tracked across the team came to 505, slightly above the 480 hour budget we planned at the start, which tells us that our storypoint estimates and sprint planning have been reasonably accurate.

## Looking Forward

GlowCore is in a good state to keep growing. The architecture was built with extensibility in mind, so adding new items, nodes, recipes or even entire crafting stations comes down to creating a new ScriptableObject asset and wiring it up in the inspector. Combat and potential future game systems such as farming are documented in the design and could be picked up either as a follow up project or as a continuation outside the semester. The foundation, the deployment pipelines, the testing setup and the documentation are all in place to support that, regardless of who continues the work.

## **Part II**

# **Product Documentation**

# 1 Glossary

| Term           | Definition                                                                                                                                                                                  |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grid           | Refers to the 2D Grid that divides the GlowCore world into rows and columns.                                                                                                                |
| Tile           | A Tile is a single unit of space in the grid and has the form of a square.                                                                                                                  |
| Node           | A Node refers to an object that is occupying a Tile. Not every Tile has a Node, but every Node belongs to a Tile.                                                                           |
| World Item     | World Items are items that can be picked up by the player and physically exist somewhere in the GlowCore world. In contrast to Nodes they do not belong to a Tile and do not have a hitbox. |
| Inventory Item | Inventory Items refer to the items that the player possesses. Certain Nodes such as Chests can also hold Inventory Items.                                                                   |
| Tool           | A Tool is an Inventory Item that can be used to break certain Nodes. For example an Axe can break Trees.                                                                                    |
| Block          | A Block is an Inventory Item that can be placed onto the Grid to build a Node.                                                                                                              |

**Table 1:** Glossary

# 2 Requirements

## 2.1 Functional Requirements

### 2.1.1 Actors / Personas

Since a game typically only has one primary actor - **The Player** - it makes sense to define Personas that reflect different types of players. The categorization into Casual- and Dedicated Players covers most player types. In the following, the pronoun **he** is used for simplicity, but the Personas are meant to include all players.

#### 2.1.1.1 Casual Player

- The Casual Player plays GlowCore infrequently and for short amounts of time. He wants to relax and unwind from his life. He prefers to play the game just for fun without putting effort into it.
- The Casual Player wants to experience the game from beginning to end with a story and explore the world. When he plays the game for the first time, he depends on the game to explain the rules of the game to him without being too overwhelming and tedious. As soon as he finishes the game, he will be done with it, having experienced everything he wanted.
- The Casual Player never changes settings in games, he never feels the need to customize anything. He likes to play the game as it is.

#### 2.1.1.2 Dedicated Player

- The Dedicated Player plays GlowCore in long sessions. He loves games where you build factories and automate everything. He searches for the feeling of progression and finds satisfaction in it. He always searches for a better way to progress and optimize his gameplay.
- The Dedicated Player has already played the game multiple times before, thus he doesn't want to be bored with a long tutorial every time he starts a new game. He wants to be able to dictate the pacing of the game.
- He doesn't care for the story and lore in games. He doesn't want to read through dialogues and doesn't like long cutscenes.



## 2.1.2 Use Case Diagram

The diagram shows a modified version of the Use Case Diagram. The Personas take the place of the Actors. An arrow indicates that a Persona uses a feature. A missing arrow means that the Persona is not interested in using a feature, although it remains accessible to him.

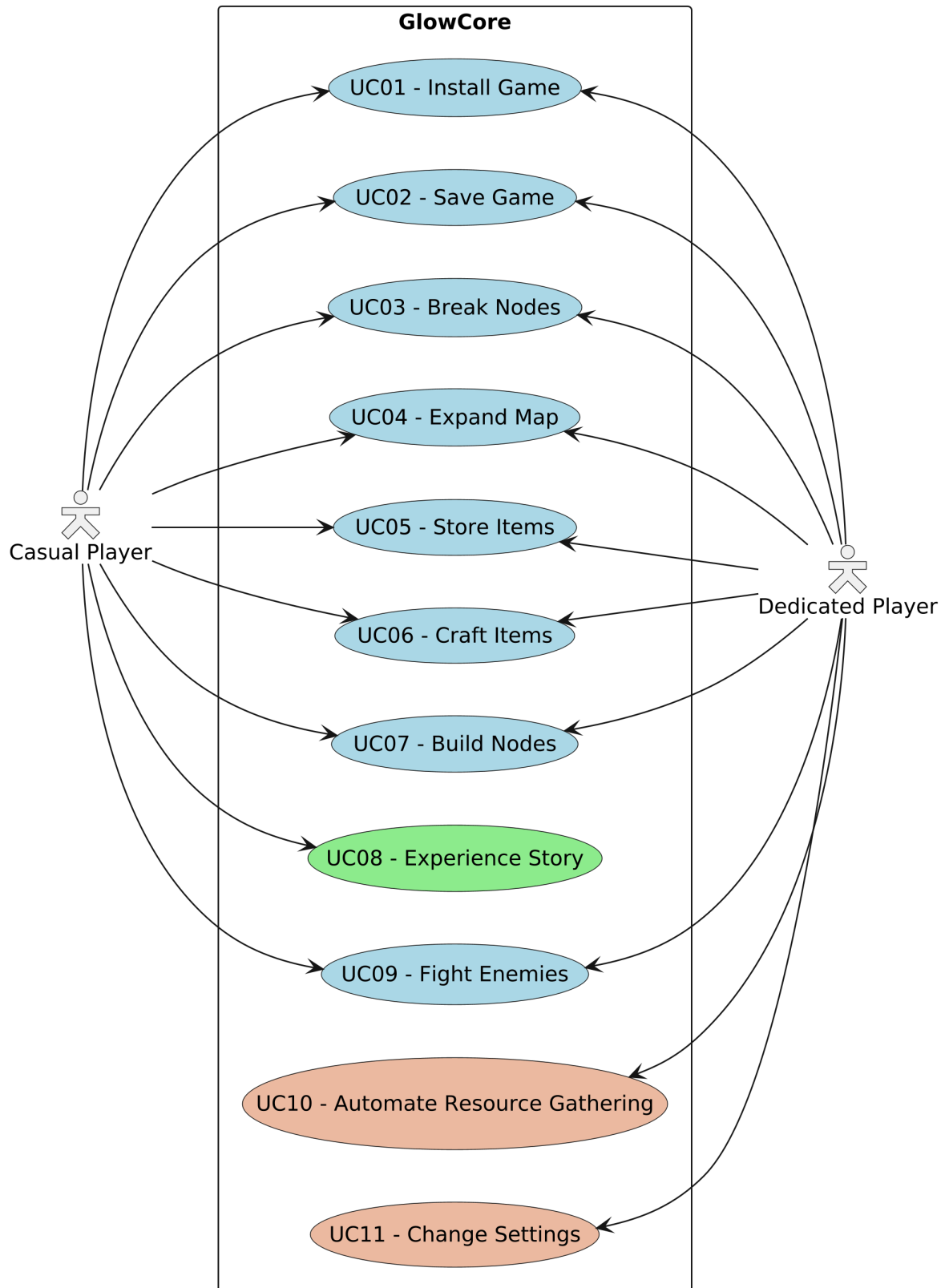


Image 1: Use Case Diagram

### 2.1.3 Use Cases

The Use Cases are described in the brief format, with each Use Case including an appropriately detailed description.

It is essential for the project that all Use Cases that define the **MVP** are implemented. The Use Cases marked with **Priority: Middle** are planned to be implemented, but can be reduced if the MVP takes more time than anticipated. The Use Cases marked with **Priority: Low** will be implemented if enough time is left.

Each Use Case table includes a colored status indicator in the top-right corner representing its current verification state:

| Color                                                                             | Status                                                          |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------|
|  | Not yet verified (default)                                      |
|  | Implemented – use case fully functional and verified            |
|  | Partially implemented – some aspects functional, others pending |
|  | Not implemented – use case not yet functional                   |
|  | N/A – decided not to implement                                  |

**Table 2:** UC Status Color Legend

### 2.1.4 UC Overview

The following table provides a compact overview of all Use Cases, their persona, priority, and current verification status.

| ID                   | Name                        | Persona                   | Priority   | Verify at   | Status                                                                                |
|----------------------|-----------------------------|---------------------------|------------|-------------|---------------------------------------------------------------------------------------|
| <a href="#">UC01</a> | Install Game                | Casual & Dedicated Player | High – MVP | M09 Beta    |  |
| <a href="#">UC02</a> | Save Game                   | Casual & Dedicated Player | High – MVP | M09 Beta    |  |
| <a href="#">UC03</a> | Break Nodes                 | Casual & Dedicated Player | High – MVP | M08 Alpha   |  |
| <a href="#">UC04</a> | Expand Map                  | Casual & Dedicated Player | High – MVP | M08 Alpha   |  |
| <a href="#">UC05</a> | Store Items                 | Casual & Dedicated Player | High – MVP | M08 Alpha   |  |
| <a href="#">UC06</a> | Craft Items                 | Casual & Dedicated Player | High – MVP | M08 Alpha   |  |
| <a href="#">UC07</a> | Build Nodes                 | Casual & Dedicated Player | High – MVP | M08 Alpha   |  |
| <a href="#">UC08</a> | Experience Story            | Casual Player             | Low        | M10 Release |  |
| <a href="#">UC09</a> | Fight Enemies               | Casual & Dedicated Player | Low        | M10 Release |  |
| <a href="#">UC10</a> | Automate Resource Gathering | Dedicated Player          | Middle     | M10 Release |  |
| <a href="#">UC11</a> | Change Settings             | Dedicated Player          | Middle     | M09 Beta    |  |

**Table 3:** UC Overview – All Use Cases at a Glance

|                    |                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC01</b>                                                                                                                                                       |
| <b>Name</b>        | Install Game                                                                                                                                                      |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                  |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                           |
| <b>Description</b> | The player can install the game. The installation requires no technical skills.                                                                                   |
| <b>Result</b>      | The game can be installed on Windows by downloading the zip from the GitHub release and launching the .exe directly. For that no technical knowledge is required. |

**Table 4:** UC01 - Install Game

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC02</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Name</b>        | Save Game                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Description</b> | When the player stops playing, the game saves his save state in a file and loads it when he resumes the game.                                                                                                                                                                                                                                                                                                                                                   |
| <b>Result</b>      | The game autosaves every 20 seconds and the save is loaded correctly the next time the player starts the game. Inventory contents, chest contents, the GlowCore level and any blocks the player built or broke are all restored. The player can also save at any time from the pause menu, and the game saves automatically when returning to the main menu. If the player just closes the window without saving, the most recent autosave is loaded next time. |

**Table 5:** UC02 - Save Game

|                    |                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC03</b>                                                                                                                                                                                                                                                      |
| <b>Name</b>        | Break Nodes                                                                                                                                                                                                                                                      |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                                                                 |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                                                                                          |
| <b>Description</b> | The player can break Nodes in the world with tools. When the Nodes break, they drop items depending on the type of Node. E.g. a Tree drops a Wood-Item. Certain Nodes that are built by the player themselves such as a workbench drop themselves upon breaking. |
| <b>Result</b>      | Verified manually. Trees drop wood when broken with an axe, stone nodes drop stone when broken with a pickaxe, and coal nodes drop both coal and stone. The crafting bench drops itself when broken with an axe. All acceptance criteria met.                    |

**Table 6:** UC03 - Break Nodes

|                    |                                                                                                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC04</b>                                                                                                                                                                                                                                                     |
| <b>Name</b>        | Expand Map                                                                                                                                                                                                                                                      |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                                                                |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                                                                                         |
| <b>Description</b> | The player is confined to only move on tiles that are unlocked at a given point. The player can drop off items at the GlowCore, whereupon the map expands. The player can then move into and explore the new area.                                              |
| <b>Result</b>      | Verified manually. The player is correctly confined to unlocked tiles and cannot move into locked areas. Feeding resources into the GlowCore triggers a map expansion, and the newly unlocked area becomes accessible immediately. All acceptance criteria met. |

**Table 7:** UC04 - Expand Map

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC05</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Name</b>        | Store Items                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Description</b> | Items that the player picks up are stored in his inventory. The player's inventory has a limited size. To organize items, the player can move items into chests and take them out again.                                                                                                                                                                                                                                                                                         |
| <b>Result</b>      | Verified manually. Items picked up by the player are stored in his inventory, and the inventory has a limited size and refuses additional items when it is full. The player can open a chest by interacting with it, move items between the chest and his inventory by dragging them, and the chest closes automatically when he walks away. Chest contents are saved and restored correctly on the next start, and breaking a chest drops its remaining contents on the ground. |

**Table 8:** UC05 - Store Items

|                    |                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC06</b>                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Name</b>        | Craft Items                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Description</b> | The player can craft items into other items with predefined recipes. Without a workbench the player is limited to a few simple recipes. The player can interact with a workstation such as a workbench to unlock more crafting recipes. Workstations such as an oven require additional items: e.g. Coal to smelt an Ingot.                                                                                  |
| <b>Result</b>      | Verified manually. Without a workbench the player has access to a few basic recipes that can be crafted directly from the inventory. Interacting with a crafting table opens its crafting UI and unlocks the full set of standard recipes. A furnace is also available as a second workstation with its own recipe list, where the player can smelt ores into ingots and has other furnace-specific recipes. |

**Table 9:** UC06 - Craft Items

|                    |                                                                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC07</b>                                                                                                                                                                                                                    |
| <b>Name</b>        | Build Nodes                                                                                                                                                                                                                    |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                               |
| <b>Priority</b>    | High - Required for MVP                                                                                                                                                                                                        |
| <b>Description</b> | The player can place certain items onto the world, where they become Nodes. Each Tile can hold only one Node, and the player is not allowed to build a Node on an already occupied Tile.                                       |
| <b>Result</b>      | Verified manually. The player can place items such as crafting tables and chests onto the world grid, where they become Nodes. Placing a Node on an already occupied tile is correctly prevented. All acceptance criteria met. |

**Table 10:** UC07 - Build Nodes

|                    |                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC08</b>                                                                                                                                                                                                                                                                                                                                                    |
| <b>Name</b>        | Experience Story                                                                                                                                                                                                                                                                                                                                               |
| <b>Persona</b>     | Casual Player                                                                                                                                                                                                                                                                                                                                                  |
| <b>Priority</b>    | Low                                                                                                                                                                                                                                                                                                                                                            |
| <b>Description</b> | The player can experience a story while playing the game. When the game starts there is an intro sequence. There is an end to the game, at which point a cutscene is played. The game contains small narrative hints that fit into the story conveyed by the GlowCore-World, including the acoustic, the visual art style and the overall gameplay experience. |
| <b>Result</b>      | Not implemented. Due to time constraints, no intro sequence, cutscene, or narrative elements were built. The feature was intentionally deprioritised as a Low-priority item, and the team's effort was directed at the MVP and automation instead.                                                                                                             |

**Table 11:** UC08 - Experience Story

|                    |                                                                                                                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC09</b>                                                                                                                                                                                                                                                              |
| <b>Name</b>        | Fight Enemies                                                                                                                                                                                                                                                            |
| <b>Persona</b>     | Casual Player & Dedicated Player                                                                                                                                                                                                                                         |
| <b>Priority</b>    | Low                                                                                                                                                                                                                                                                      |
| <b>Description</b> | The player can engage in combat with enemies. Defeated enemies drop items that are useful for the player. When the enemy hits the player, he loses hitpoints. When the player has no hitpoints left, he loses some of his items and respawns near the center of the map. |
| <b>Result</b>      | N/A — combat was intentionally removed from the project scope (see the Coverage Summary).                                                                                                                                                                                |

**Table 12:** UC09 - Fight Enemies

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC10</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Name</b>        | Automate Resource Gathering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Persona</b>     | Dedicated Player                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Priority</b>    | Middle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Description</b> | The player can automate tedious tasks, such as breaking Nodes. He can craft machines that are placed as Nodes and they automatically perform a task e.g. chopping down trees. There is a selection of machines that can be combined to automate more and more. Example: The player places multiple Replanter Machines in a row which automatically replant trees on the neighbouring tile. The player places an Axe Machine onto the start of the line, at which point the Axe Machine chops one tree after another and then moves onto the next tile. To automate it even further, the player places Rotators on both which cause the Axe Machine to turn around on contact.                                                                                                                                          |
| <b>Result</b>      | Verified manually. The player can craft three automation machines and place them as nodes. The Axe Machine faces a direction and automatically breaks the node in front of it using the appropriate tool. When the target tile is empty it drives forward to the next node. Drops are stored in the Axe Machine's built-in chest and the player can retrieve them at any time. The Replanter Machine plants saplings or other block items from its own chest onto the tile in front of it, automatically restocking resource nodes after the Axe Machine passes. Rotator nodes (Right, Down, Left) redirect the Axe Machine's heading when it drives into them, allowing the player to build looping or branching automation tracks. All three machines are craftable from the workbench. All acceptance criteria met. |

**Table 13:** UC10 - Automate Resource Gathering

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | <b>UC11</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Name</b>        | Change Settings                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Persona</b>     | Dedicated Player                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Priority</b>    | Middle                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Description</b> | The player can change settings of the game, such as the FOV and the Volume.                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Result</b>      | Verified manually. The player can open the settings from both the title screen and the in-game pause menu. The audio section has separate sliders for master, music and sound-effect volume. The display section has a fullscreen toggle, a resolution dropdown and separate camera sensitivity sliders for the X and Y axes. Changes are kept between sessions, and the player is asked to confirm before leaving the settings with unsaved changes. |

**Table 14:** UC11 - Change Settings

## 2.2 Non-Functional Requirements

This chapter documents the Non-Functional Requirements (NFRs) for the GlowCore project, categorized according to the ISO/IEC 25010:2011 quality model. The ISO/IEC 25010 standard defines a comprehensive set of software quality characteristics that serve as a framework for specifying and evaluating system quality. The following categories are used to structure our NFRs:

- **Performance Efficiency** – NFR1XX
- **Usability** – NFR2XX
- **Reliability** – NFR3XX
- **Maintainability** – NFR4XX
- **Portability** – NFR5XX

Each NFR is documented with a description, concrete acceptance criteria, a measurement method, a verification approach, and a priority. The priority consists of two parts:

- **MVP Relevance:**
  - **Required** – must be fulfilled for the MVP.
  - **Optional** – planned for a later stage, not required for MVP delivery.
- **Importance:**
  - **High** – critical for the product.
  - **Medium** – important but not blocking.
  - **Low** – nice to have.

Each NFR table includes a colored status indicator in the top-right corner representing its current verification state:

| Color                                                                               | Status                                               |
|-------------------------------------------------------------------------------------|------------------------------------------------------|
|  | Not yet tested (default)                             |
|  | Passed – all acceptance criteria met                 |
|  | Partially met – some criteria passed, others pending |
|  | Failed – acceptance criteria not met                 |
|  | N/A – decided not to implement                       |

**Table 15:** NFR Status Color Legend

## 2.2.1 NFR Overview

The following table provides a compact overview of all non-functional requirements, their category, priority, verification milestone, and current status. This serves as a quick reference before diving into the detailed NFR descriptions below.

| ID                     | Name                          | Category        | Priority          | Verify at   | Status                                                                                |
|------------------------|-------------------------------|-----------------|-------------------|-------------|---------------------------------------------------------------------------------------|
| <a href="#">NFR101</a> | World Load Time               | Performance     | Required / High   | M09 Beta    |    |
| <a href="#">NFR102</a> | In-Game Transition Time       | Performance     | Required / High   | M08 Alpha   |    |
| <a href="#">NFR103</a> | Frame Rate                    | Performance     | Required / High   | M08 Alpha   |    |
| <a href="#">NFR104</a> | Save File Size                | Performance     | Required / Medium | M09 Beta    |    |
| <a href="#">NFR105</a> | Minimal Hardware Req.         | Performance     | Optional / Medium | M09 Beta    |    |
| <a href="#">NFR201</a> | Player Restriction Visibility | Usability       | Required / High   | M08 Alpha   |    |
| <a href="#">NFR202</a> | Light Upgrade Progress        | Usability       | Required / High   | M08 Alpha   |    |
| <a href="#">NFR203</a> | New Player Learnability       | Usability       | Required / High   | M09 Beta    |    |
| <a href="#">NFR204</a> | Input Support                 | Usability       | Optional / Medium | M09 Beta    |    |
| <a href="#">NFR205</a> | Game Language                 | Usability       | Optional / Low    | M10 Release |    |
| <a href="#">NFR206</a> | Visual Style Consistency      | Usability       | Optional / Medium | M09 Beta    |    |
| <a href="#">NFR301</a> | Save File Portability         | Reliability     | Required / High   | M09 Beta    |  |
| <a href="#">NFR302</a> | Game Progress Persistence     | Reliability     | Required / High   | M09 Beta    |  |
| <a href="#">NFR303</a> | Save Data Integrity           | Reliability     | Required / High   | M09 Beta    |  |
| <a href="#">NFR401</a> | Extensibility of Game Systems | Maintainability | Optional / Medium | M09 Beta    |  |
| <a href="#">NFR402</a> | Automated Test Coverage       | Maintainability | Required / Medium | M09 Beta    |  |
| <a href="#">NFR501</a> | Multi-Platform Support        | Portability     | Required / Medium | M09 Beta    |  |
| <a href="#">NFR502</a> | Engine and Rendering Pipeline | Portability     | Optional / Low    | M10 Release |  |
| <a href="#">NFR503</a> | Display Resolution Support    | Portability     | Optional / Medium | M10 Release |  |

**Table 16:** NFR Overview – All Non-Functional Requirements at a Glance

Verification is tied to project milestones (see [Updated Milestones](#)) rather than fixed dates to ensure NFRs are checked at meaningful delivery checkpoints:

- **M08 – Alpha Release (10.04.2026):** The core Required / High NFRs scheduled for this milestone must pass. This is the MVP gate.
- **M09 – Beta Release (15.05.2026):** The remaining Required / High NFRs are verified, along with the Required / Medium and Optional / Medium NFRs. Performance tuning and playtesting rounds.
- **M10 – Official Release (05.06.2026):** Remaining Optional / Low NFRs are verified. Final polish and compliance check.

## 2.2.2 Priority Matrix

The priority matrix visualizes the distribution of NFRs across the two priority dimensions. This helps identify where the project's quality focus lies and whether the prioritization is balanced.

|               | Required (MVP)                                                             | Optional (Post-MVP)                      |
|---------------|----------------------------------------------------------------------------|------------------------------------------|
| <b>High</b>   | NFR101, NFR102, NFR103<br>NFR201, NFR202, NFR203<br>NFR301, NFR302, NFR303 | –                                        |
| <b>Medium</b> | NFR104, NFR402, NFR501                                                     | NFR105, NFR204, NFR206<br>NFR401, NFR503 |
| <b>Low</b>    | –                                                                          | NFR205, NFR502                           |

**Table 17:** NFR Priority Matrix – MVP Relevance vs. Importance

The matrix shows that the core quality effort is concentrated in the **Required / High** cell (9 NFRs), covering performance, usability, and reliability which are the three pillars that directly impact the player experience in the MVP. The **Optional / Low** cell contains only 2 NFRs that represent project constraints rather than testable quality goals. Priority is on delivering a stable, playable MVP before addressing optional requirements.

## 2.2.3 Performance Efficiency

Performance Efficiency addresses the amount of resources used under stated conditions. For GlowCore, this primarily concerns load times, frame rates, and resource consumption to ensure a smooth and responsive gameplay experience.

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR101</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Description</b>         | World Load Time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Acceptance Criteria</b> | Existing and newly created worlds must load within 3 seconds when the player presses “Start”.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Priority</b>            | Required / High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Measurement</b>         | Measure elapsed time from pressing “Start” until the world is fully loaded and interactive. Tested on minimum hardware specification.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Verification</b>        | Automated load-time benchmark test across 10 consecutive loads with varying world states (new and existing). All runs must complete within 3 seconds.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Result</b>              | Failed. Tested manually on the Windows build. A new world takes about 4 seconds, a small save about 6 seconds, and a late-game save up to 20 seconds. The world stays the same size, but the player fills it with placed and planted things, and all of that has to load. The more a save grows, the longer it takes. The 3-second target is not met. When we set this target, we did not expect to build a game of this size, so the original goal no longer fits what we ended up with. Bringing it back in range would need extra work on how the world is loaded. |

**Table 18:** NFR101 – World Load Time



|                            |                                                                                                                                                                                                        |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR102</b>                                                                                                                                                                                          |
| <b>Description</b>         | In-Game Transition Time                                                                                                                                                                                |
| <b>Acceptance Criteria</b> | In-game transitions (e.g. walking into a cave) shall not exceed 2 seconds.                                                                                                                             |
| <b>Priority</b>            | Required / High                                                                                                                                                                                        |
| <b>Measurement</b>         | Measure elapsed time from triggering the transition until the new scene is fully loaded and interactive.                                                                                               |
| <b>Verification</b>        | Trigger each scene transition during playtesting and record the load time. Automated scene-transition benchmark covering all transition points.                                                        |
| <b>Result</b>              | Not applicable. Due to the lack of time the team decided not to implement cave areas or any scene transitions during the project. This requirement no longer applies to the current scope of GlowCore. |

**Table 19:** NFR102 – In-Game Transition Time

|                            |                                                                                                                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR103</b>                                                                                                                                                                                                                                                                     |
| <b>Description</b>         | Frame Rate                                                                                                                                                                                                                                                                        |
| <b>Acceptance Criteria</b> | The game must run at a minimum of 60 FPS with 1% lows not dropping below 50 FPS on recommended hardware.                                                                                                                                                                          |
| <b>Priority</b>            | Required / High                                                                                                                                                                                                                                                                   |
| <b>Measurement</b>         | Use Unity Profiler or an external frame-time analysis tool to record FPS over a 5-minute gameplay session. Evaluate average FPS and 1% low values.                                                                                                                                |
| <b>Verification</b>        | Run a standardized gameplay scenario on minimum and recommended hardware. Capture FPS metrics and verify thresholds are met.                                                                                                                                                      |
| <b>Result</b>              | Manually tested on WebGL build using Chrome's Frame Rendering Stats overlay. The game ran at a stable 120 FPS during active gameplay with no significant drops observed. Exceeds the minimum threshold of 60 FPS and the 1% low threshold of 50 FPS. All acceptance criteria met. |

**Table 20:** NFR103 – Frame Rate

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR104</b>                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Description</b>         | Save File Size                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Acceptance Criteria</b> | Save files should remain compact and use a binary serialization format. Under the currently defined maximum expected game progression and world configuration, a save file should not exceed 5MB.                                                                                                                                                                                                                           |
| <b>Priority</b>            | Required / Medium                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Measurement</b>         | Measure the file size of the save file after a full gameplay session under the defined maximum expected game progression and world configuration.                                                                                                                                                                                                                                                                           |
| <b>Verification</b>        | Create save files at various progression stages and verify the file size remains within the defined limit for the current scope. Confirm that a binary serialization format is used.                                                                                                                                                                                                                                        |
| <b>Result</b>              | Tested by playing the game through several progression stages (fresh world, a few broken nodes, multiple chests filled with items, several built blocks) and measuring the on-disk size of the save file after each save. A fresh save was around 1KB, and a save taken after reaching the maximum GlowCore level peaked at 3KB, well under the 5MB limit. The save file is written in a custom binary format, as required. |

**Table 21:** NFR104 – Save File Size

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR105</b>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Description</b>         | Minimal Hardware Requirements                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Acceptance Criteria</b> | The game must run on minimal hardware with the following specifications: <ul style="list-style-type: none"> <li>• 2 GHz CPU</li> <li>• 8 GB RAM</li> <li>• NVIDIA GeForce GTX 1660</li> </ul>                                                                                                                                                                                                                         |
| <b>Priority</b>            | Optional / Medium                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Measurement</b>         | Test the game on a defined minimum hardware baseline and verify all performance NFRs (FPS, load times) are met.                                                                                                                                                                                                                                                                                                       |
| <b>Verification</b>        | Execute a full gameplay session on minimum-spec hardware. Record FPS, load times, and memory usage to confirm compliance.                                                                                                                                                                                                                                                                                             |
| <b>Result</b>              | Tested on a Microsoft Surface Pro 8 running on integrated Intel Iris Xe graphics with no dedicated GPU. The game ran smoothly throughout the entire session with no frame drops or lag. FPS stayed consistently high and load times were near instant. Memory usage remained stable with no noticeable spikes. For a game of this scope running on integrated graphics only, the minimal hardware requirement is met. |

**Table 22:** NFR105 – Minimal Hardware Requirements

## 2.2.4 Usability

Usability covers the degree to which the product can be used effectively, efficiently, and satisfactorily. For GlowCore, this includes visual clarity of game mechanics, learnability for new players, and input support.

|                            |                                                                                                                                                                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR201</b>                                                                                                                                                                                                                                                   |
| <b>Description</b>         | Player Restriction Visibility                                                                                                                                                                                                                                   |
| <b>Acceptance Criteria</b> | Player restrictions in the game must be clearly communicated through visual indicators such as greyed-out or red-tinted elements. Players must understand restrictions without additional explanation.                                                          |
| <b>Priority</b>            | Required / High                                                                                                                                                                                                                                                 |
| <b>Measurement</b>         | Conduct playtesting with at least 3 new players. Each player must correctly identify restricted actions without external guidance.                                                                                                                              |
| <b>Verification</b>        | During playtesting sessions, observe whether players attempt restricted actions and whether they understand the visual cues. Document success rate.                                                                                                             |
| <b>Result</b>              | User testing was conducted. The locked area border is visually communicated to the player. There is a progress indicator when breaking nodes and an interaction prompt when hovering over workstations such as the crafting table. All acceptance criteria met. |

**Table 23:** NFR201 – Player Restriction Visibility

|                            |                                                                                                                                                                                                                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR202</b>                                                                                                                                                                                                                                                                                                             |
| <b>Description</b>         | Light Upgrade Progress Indication                                                                                                                                                                                                                                                                                         |
| <b>Acceptance Criteria</b> | Light upgrade progress must be clearly indicated to the player at all times during gameplay.                                                                                                                                                                                                                              |
| <b>Priority</b>            | Required / High                                                                                                                                                                                                                                                                                                           |
| <b>Measurement</b>         | During playtesting, ask players to describe their current upgrade progress. At least 80% must answer correctly.                                                                                                                                                                                                           |
| <b>Verification</b>        | Conduct playtesting sessions and survey players about their perceived progress. Verify the UI communicates upgrade status unambiguously.                                                                                                                                                                                  |
| <b>Result</b>              | User Testing was conducted. The GlowCore upgrade UI clearly displays the current level, progress bar with percentage, required materials and amounts, a preview of the next level appearance, and the effect of the next upgrade. All information updates in real time as materials are fed. All acceptance criteria met. |

**Table 24: NFR202 – Light Upgrade Progress Indication**

|                            |                                                                                                                                                                                                           |  |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>ID</b>                  | <b>NFR203</b>                                                                                                                                                                                             |  |
| <b>Description</b>         | New Player Learnability                                                                                                                                                                                   |  |
| <b>Acceptance Criteria</b> | A new player shall understand core mechanics (movement, gathering, upgrading) within 5 minutes without external instructions.                                                                             |  |
| <b>Priority</b>            | Required / High                                                                                                                                                                                           |  |
| <b>Measurement</b>         | Time how long a new player takes to perform their first gather, move to a new area, and initiate an upgrade without any external help.                                                                    |  |
| <b>Verification</b>        | Conduct playtesting with at least 3 players who have never seen the game. Measure time to complete core actions. All players must achieve this within 5 minutes.                                          |  |
| <b>Result</b>              | User testing was conducted. All participants were able to upgrade the GlowCore multiple times in under 5 minutes. They learned the core mechanic quickly and were able to progress without external help. |  |

**Table 25: NFR203 – New Player Learnability**

|                            |                                                                                                                                                                                                                                                                              |  |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>ID</b>                  | <b>NFR204</b>                                                                                                                                                                                                                                                                |  |
| <b>Description</b>         | Input Support                                                                                                                                                                                                                                                                |  |
| <b>Acceptance Criteria</b> | The game must support both controller and keyboard input methods.                                                                                                                                                                                                            |  |
| <b>Priority</b>            | Optional / Medium                                                                                                                                                                                                                                                            |  |
| <b>Measurement</b>         | Test all core gameplay actions with both a standard controller (e.g. Xbox controller) and keyboard. All actions must be executable with both input methods.                                                                                                                  |  |
| <b>Verification</b>        | Perform a full gameplay session using only controller, then only keyboard. Verify all interactions, menus, and gameplay mechanics are fully accessible with each input method.                                                                                               |  |
| <b>Result</b>              | Keyboard input works for all core gameplay actions. Controller support was not implemented due to time constraints. Since the game uses Unity's new Input System, all input actions are already defined and adding controller bindings would not require significant effort. |  |

**Table 26: NFR204 – Input Support**

|                            |                                                                                                                                                                                                                                                                                                   |  |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>ID</b>                  | <b>NFR205</b>                                                                                                                                                                                                                                                                                     |  |
| <b>Description</b>         | Game Language                                                                                                                                                                                                                                                                                     |  |
| <b>Acceptance Criteria</b> | All in-game text, UI elements, and instructions must be in English.                                                                                                                                                                                                                               |  |
| <b>Priority</b>            | Optional / Low                                                                                                                                                                                                                                                                                    |  |
| <b>Measurement</b>         | Review all in-game text and UI elements for non-English content.                                                                                                                                                                                                                                  |  |
| <b>Verification</b>        | Manual review of all text-containing assets and UI screens to confirm English language throughout.                                                                                                                                                                                                |  |
| <b>Result</b>              | Verified by walking through every screen of the game (title screen, new game, pause menu, settings, inventory, crafting and chest UIs) and reviewing all visible text. All labels, buttons, item names, recipe names and tooltips are in English. No localized or translated strings are present. |  |

**Table 27: NFR205 – Game Language**

|                            |                                                                                                                                                                                                                                                                                                                                                                      |  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>ID</b>                  | <b>NFR206</b>                                                                                                                                                                                                                                                                                                                                                        |  |
| <b>Description</b>         | Visual Style Consistency                                                                                                                                                                                                                                                                                                                                             |  |
| <b>Acceptance Criteria</b> | All visual assets must follow the low-poly pastel art style consistently throughout the game.                                                                                                                                                                                                                                                                        |  |
| <b>Priority</b>            | Optional / Medium                                                                                                                                                                                                                                                                                                                                                    |  |
| <b>Measurement</b>         | Visual review of all assets against the defined art style guide. No asset shall deviate from the low-poly pastel aesthetic.                                                                                                                                                                                                                                          |  |
| <b>Verification</b>        | Conduct an art review session where all in-game assets are compared against reference material. Flag and correct any deviations.                                                                                                                                                                                                                                     |  |
| <b>Result</b>              | Went through all in-game assets during a gameplay session and compared them against our reference material. Tiles, trees, rocks, the player character, placed nodes, and UI elements all follow the low-poly pastel aesthetic consistently. No asset stood out as visually inconsistent. The color palette stays coherent throughout and nothing looks out of place. |  |

**Table 28:** NFR206 – Visual Style Consistency

## 2.2.5 Reliability

Reliability addresses the degree to which a system performs specified functions under specified conditions for a specified period of time. For GlowCore, this is critical for save data integrity and game stability.

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>ID</b>                  | <b>NFR301</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| <b>Description</b>         | Save File Portability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| <b>Acceptance Criteria</b> | Each world save must be stored as a single file that can be copied and moved. When placed into the game's save file folder, the save file must be recognized and function without issues.                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| <b>Priority</b>            | Required / High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <b>Measurement</b>         | Copy a world save file to a different machine or location, place it in the save folder, and verify the game loads it correctly with all progress intact.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
| <b>Verification</b>        | Create a world save file with significant progress. Copy it to a new installation. Load the game and verify all progress, items, and world state for that world are preserved.                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
| <b>Result</b>              | Tested manually by copying the save file (glowcore.bin) out of the save folder on one machine and dropping it into the save folder on a different installation. The save was checked after every scenario (moving the player, filling inventory and chests, breaking and building blocks, upgrading the GlowCore) and in each case the game recognized the file and restored everything correctly: player position, inventory contents, chest contents, GlowCore level and all blocks the player had built or broken. The save is a single self-contained file, so copying and moving it works without any additional steps. |  |

**Table 29:** NFR301 – Save File Portability

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR302</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Description</b>         | Game Progress Persistence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Acceptance Criteria</b> | Game progress shall not be lost during normal shutdown. Auto-save shall occur at least every 2 minutes or immediately after a light upgrade.                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Priority</b>            | Required / High                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Measurement</b>         | Verify auto-save triggers by monitoring save file timestamps during gameplay. Confirm no data loss after normal shutdown.                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Verification</b>        | Play for several minutes, perform a light upgrade, then shut down the game normally. Restart and verify all progress is retained. Additionally, verify auto-save timestamps occur within 2-minute intervals.                                                                                                                                                                                                                                                                                                   |
| <b>Result</b>              | Tested manually by playing several sessions and watching the save file timestamp. The autosave now fires every 20 seconds and keeps firing for the entire session, comfortably below the 2-minute limit required. After a normal shutdown and restart, the player position, inventory contents, chest contents, GlowCore level and all built or broken blocks were preserved. The worst case for unsaved progress is the 20 seconds since the last autosave, which is well within what the requirement allows. |

**Table 30:** NFR302 – Game Progress Persistence

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR303</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Description</b>         | Save Data Integrity                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Acceptance Criteria</b> | Save data must persist correctly across sessions without corruption.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Priority</b>            | Required / High                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Measurement</b>         | Perform several manual save/load cycles and verify that the game state after loading matches the state at the time of saving.                                                                                                                                                                                                                                                                                                                    |
| <b>Verification</b>        | Play the game, note the current state (inventory, player position, placed blocks), let the autosave trigger, restart the game, and verify the loaded state matches what was saved.                                                                                                                                                                                                                                                               |
| <b>Result</b>              | Tested manually by playing the game, noting the current state (player position, inventory contents, chest contents, GlowCore level and all placed or broken blocks), letting the autosave trigger, restarting the game, and comparing the loaded state to what was noted before. Across multiple save and load cycles the loaded state matched the saved state exactly in every case, with no missing items, misplaced blocks or corrupted data. |

**Table 31:** NFR303 – Save Data Integrity

## 2.2.6 Maintainability

Maintainability represents the degree of effectiveness and efficiency with which a product can be modified. For GlowCore, this covers extensibility of game systems and test coverage.

|                            |                                                                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR401</b>                                                                                                                                           |
| <b>Description</b>         | Extensibility of Game Systems                                                                                                                           |
| <b>Acceptance Criteria</b> | The code must allow adding new GlowCore upgrades and new items with their resource nodes without requiring code restructuring.                          |
| <b>Priority</b>            | Optional / Medium                                                                                                                                       |
| <b>Measurement</b>         | A developer unfamiliar with the codebase must be able to add a new upgrade or item by following existing patterns, without modifying core system code.  |
| <b>Verification</b>        | Task a team member with adding a new test upgrade and resource node. Measure the time required and verify no structural code changes were necessary.    |
| <b>Result</b>              | Verified manually by a team member adding a new resource node. The architecture supports adding new nodes and items without modifying core system code. |

**Table 32:** NFR401 – Extensibility of Game Systems

|                            |                                                                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR402</b>                                                                                                                                                   |
| <b>Description</b>         | Automated Test Coverage                                                                                                                                         |
| <b>Acceptance Criteria</b> | All systems must be covered by automated tests. The project targets a minimum of 40% code coverage. (This coverage target is relatively common for video games) |
| <b>Priority</b>            | Required / Medium                                                                                                                                               |
| <b>Measurement</b>         | Measure code coverage using the Unity Code Coverage package. Report overall and per-system coverage percentages.                                                |
| <b>Verification</b>        | Run the full automated test suite and generate a coverage report. Verify that overall coverage meets or exceeds 40%.                                            |
| <b>Result</b>              | Coverage was measured using the Unity Code Coverage package. The project achieves 43.1% method coverage, exceeding the 40% target. Line coverage sits at 38.9%. |

**Table 33:** NFR402 – Automated Test Coverage

## 2.2.7 Portability

Portability addresses the degree of effectiveness and efficiency with which a system can be transferred from one environment to another. For GlowCore, this covers platform support and display resolution compatibility.

|                            |                                                                                                                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR501</b>                                                                                                                                                                                                                                                                                              |
| <b>Description</b>         | Multi-Platform Support                                                                                                                                                                                                                                                                                     |
| <b>Acceptance Criteria</b> | The game must run on Windows, Linux, and Web platforms.                                                                                                                                                                                                                                                    |
| <b>Priority</b>            | Required / Medium                                                                                                                                                                                                                                                                                          |
| <b>Measurement</b>         | Build and deploy the game on all three platforms. Launch and complete a full gameplay session on each.                                                                                                                                                                                                     |
| <b>Verification</b>        | Execute the full test suite and a manual gameplay session on Windows, Linux, and a Web browser. Verify all features work identically across platforms.                                                                                                                                                     |
| <b>Result</b>              | Manually verified on WebGL (browser), Windows (native build via GitHub release zip), and Linux (native build tested in a Linux VM). The game launched and ran correctly on all three platforms. All three platform builds are generated automatically via the CI/CD pipeline. All acceptance criteria met. |

**Table 34:** NFR501 – Multi-Platform Support

|                            |                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>                  | <b>NFR502</b>                                                                                                                                                                                                                                                                                                                                 |
| <b>Description</b>         | Engine and Rendering Pipeline                                                                                                                                                                                                                                                                                                                 |
| <b>Acceptance Criteria</b> | The game must run on the Unity engine using the PhysX physics engine and the Universal Render Pipeline (URP).                                                                                                                                                                                                                                 |
| <b>Priority</b>            | Optional / Low                                                                                                                                                                                                                                                                                                                                |
| <b>Measurement</b>         | Verify project settings confirm Unity with PhysX and URP are configured as the active engine and rendering pipeline.                                                                                                                                                                                                                          |
| <b>Verification</b>        | Review Unity project settings and confirm PhysX is the active physics engine and URP is the active render pipeline. Verify no fallback to built-in pipeline occurs during gameplay.                                                                                                                                                           |
| <b>Result</b>              | Verified via Unity project settings and by running the game. The GraphicsSettings.asset confirms URP is the active render pipeline, and no fallback to the built-in pipeline was observed during gameplay. PhysX is Unity's default physics engine and is used throughout the project with no override configured. Both requirements are met. |

**Table 35:** NFR502 – Engine and Rendering Pipeline

|                            |                                                                                                                                                                                                                                                                                    |  |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>ID</b>                  | <b>NFR503</b>                                                                                                                                                                                                                                                                      |  |
| <b>Description</b>         | Display Resolution Support                                                                                                                                                                                                                                                         |  |
| <b>Acceptance Criteria</b> | The game must support common display resolutions with a 16:9 aspect ratio. Minimum: 1280×720. Recommended: 1920×1080, 2560×1440, and 3840×2160 (4K).                                                                                                                               |  |
| <b>Priority</b>            | Optional / Medium                                                                                                                                                                                                                                                                  |  |
| <b>Measurement</b>         | Launch the game at each supported resolution and verify the UI scales correctly, no elements are clipped, and the game renders properly.                                                                                                                                           |  |
| <b>Verification</b>        | Test the game at all specified resolutions. Take screenshots and verify correct rendering, UI scaling, and absence of visual artifacts.                                                                                                                                            |  |
| <b>Result</b>              | Tested manually in windowed mode. The settings menu provides a resolution dropdown and switching between resolutions correctly resizes the window and scales the UI. All tested resolutions from 1280x720 up to 1920x1080 rendered correctly with no clipping or visual artifacts. |  |

**Table 36:** NFR503 – Display Resolution Support

## 2.2.8 NFR-Architecture Traceability

The table below links NFRs to the specific technical solution in the architecture that is primarily responsible for satisfying it. The purpose of this table is to make the connection between quality requirements and design decisions explicit and verifiable.

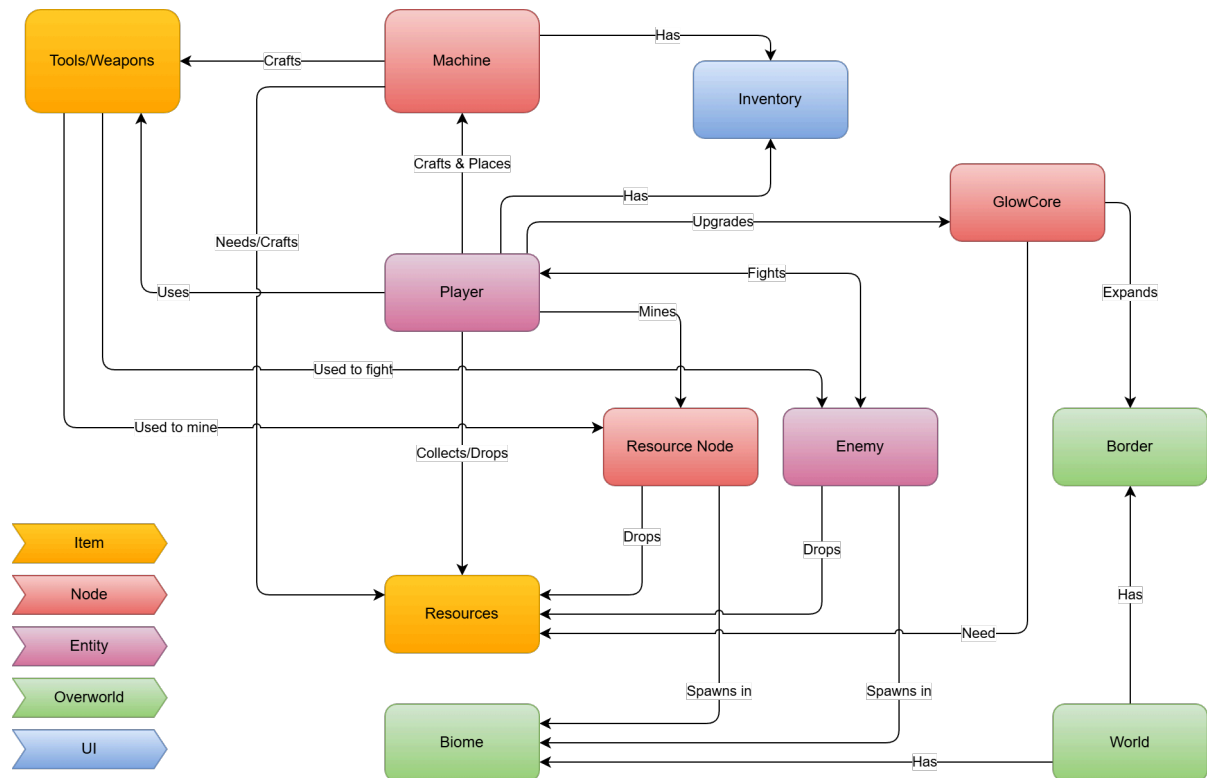
| NFR                                                                        | Quality Attribute                         | Technical Solution                                    | How the Architecture Addresses It                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">NFR401</a>                                                     | Maintainability<br>(Extensibility)        | IInteractable, IHandItem,<br>ScriptableObject         | New nodes implement IInteractable;<br>new usable items implement IHandItem;<br>new content is added as ScriptableObject<br>assets. No existing class needs to change.                                                                                                                                    |
| <a href="#">NFR402</a>                                                     | Maintainability<br>(Test Coverage)        | Plain C# classes;<br>interface-based design           | Inventory and CraftingSystem are plain<br>C# classes, not MonoBehaviours, so<br>NUnit can test them without a Unity<br>scene. The service interfaces enable<br>mock injection in integration tests.                                                                                                      |
| <a href="#">NFR501</a>                                                     | Portability<br>(Multi-Platform)           | Unity, URP,<br>New Input System                       | Unity builds natively to Windows, Linux,<br>and WebGL from one codebase. URP<br>renders consistently across all three. The<br>New Input System abstracts platform-<br>specific input, requiring no per-platform<br>code.                                                                                 |
| <a href="#">NFR502</a>                                                     | Portability<br>(Rendering Pipeline)       | Universal Render Pipeline (URP)                       | URP was chosen over the Built-<br>in Pipeline (deprecated) and HDRP<br>(too heavy). It supports cross-platform<br>shaders and custom post-processing (e.g.<br>LowResViewport) without HDRP's hard-<br>ware requirements.                                                                                 |
| <a href="#">NFR103</a>                                                     | Performance<br>(Frame Rate)               | URP; event-driven UI;<br>Component-Based Architecture | URP has a lower per-frame GPU cost<br>than HDRP. The event-driven UI only runs<br>when events like OnSlotChanged fire,<br>never polling each frame. Component-<br>Based Architecture avoids monolithic<br>Update() methods.                                                                              |
| <a href="#">NFR301</a><br><a href="#">NFR302</a><br><a href="#">NFR303</a> | Reliability<br>(Save System)              | Binary serialization;<br>ScriptableObject asset refs  | Binary serialization keeps file sizes small<br>(NFR104). Items are stored as Script-<br>ableObject references rather than copies,<br>ensuring consistency across sessions.<br>Auto-save triggers on key game events,<br>not only on shutdown.                                                            |
| <a href="#">NFR201</a><br><a href="#">NFR202</a><br><a href="#">NFR203</a> | Usability<br>(Feedback &<br>Learnability) | GetActionPromptText();<br>event-driven UI             | The Use/Interact abstraction reduces<br>controls to two input types, lower-<br>ing cognitive load (NFR203). Each<br>node provides its own prompt via<br>GetActionPromptText() (NFR201). The<br>event-driven UI updates instantly on state<br>changes, keeping upgrade indicators ac-<br>curate (NFR202). |

**Table 37:** NFR-to-Architecture Traceability



# 3 Domain Analysis

Below is the domain model diagram for the problem domain of our project. Due to the nature of this project, the domain is modeled using an entity-relationship diagram, which captures the key entities and their relationships within the system. The diagram provides a high-level overview of the main components and their interactions, which is essential for understanding the structure of the problem domain.

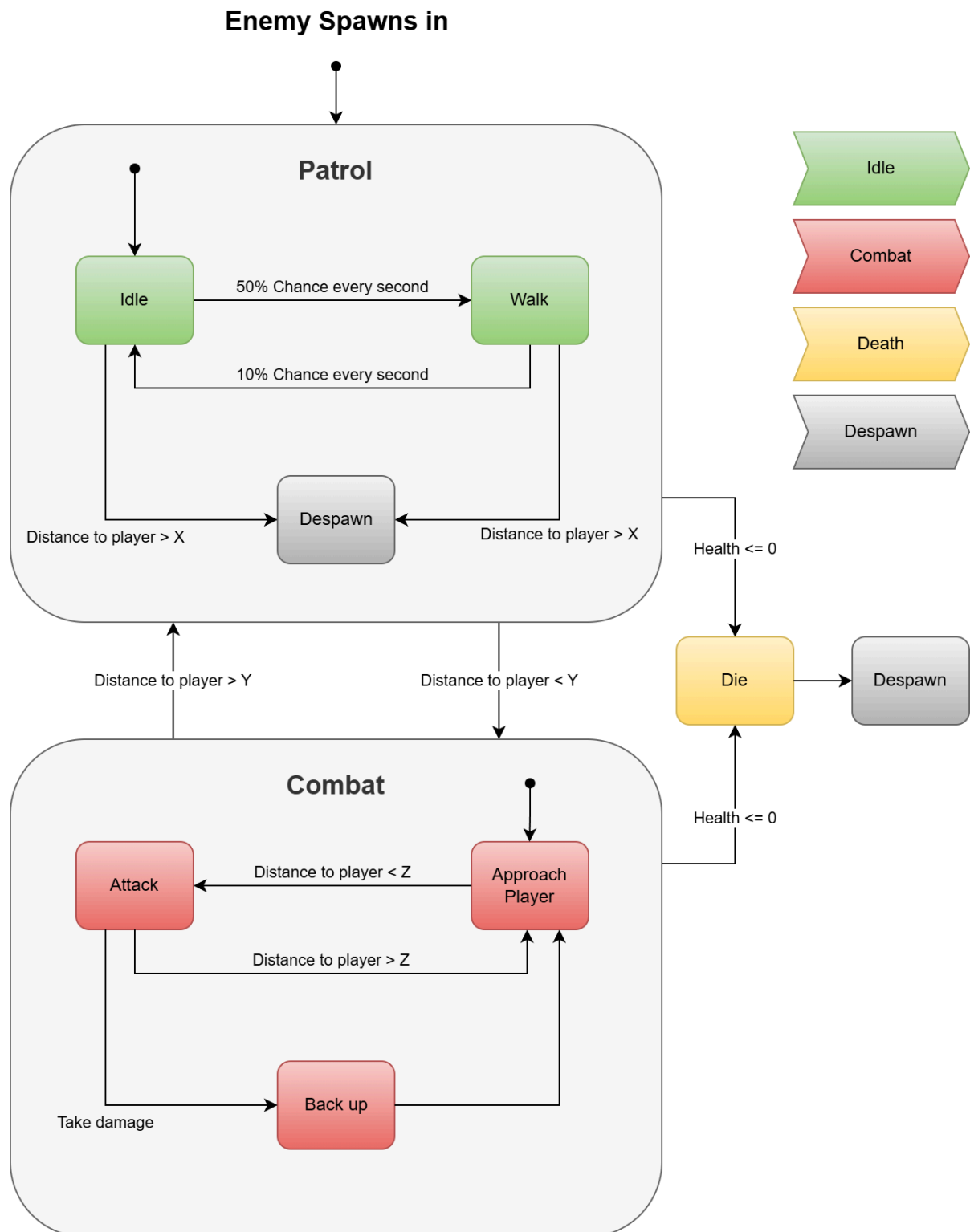


**Image 2: Domain Model Diagram**

One of the key aspects of the domain is the relationship between “Nodes” and “Items”. Some items can be placed in the world which turns them into Nodes. This means that the same concept can have different representations and behaviors depending on the context.

### 3.1 Enemy State Machine

The below diagram represents the state machine for the enemy behavior in our project. It captures the various states an enemy can be in as well as the transitions between those states and the conditions that trigger those transitions.



**Image 3:** Enemy State Machine Diagram

# 4 Architecture

## 4.1 Frontend Architecture

The user interface of GlowCore is designed to be self-explanatory. Therefore, only selected screens are explained in detail below.

### 4.1.1 Main Menu



**Image 4:** Main Menu

The Main Menu is the entry point of the game. The player can continue with an existing save of the game or create a new one. The player can only have one save at a time.

## 4.1.2 New Game



Image 5: New Game

## 4.1.3 Pause Menu



Image 6: Pause Menu

In the Pause Menu the player can save the game manually. The game is also automatically saved when the player exits to the Main Menu.

#### 4.1.4 Audio Settings

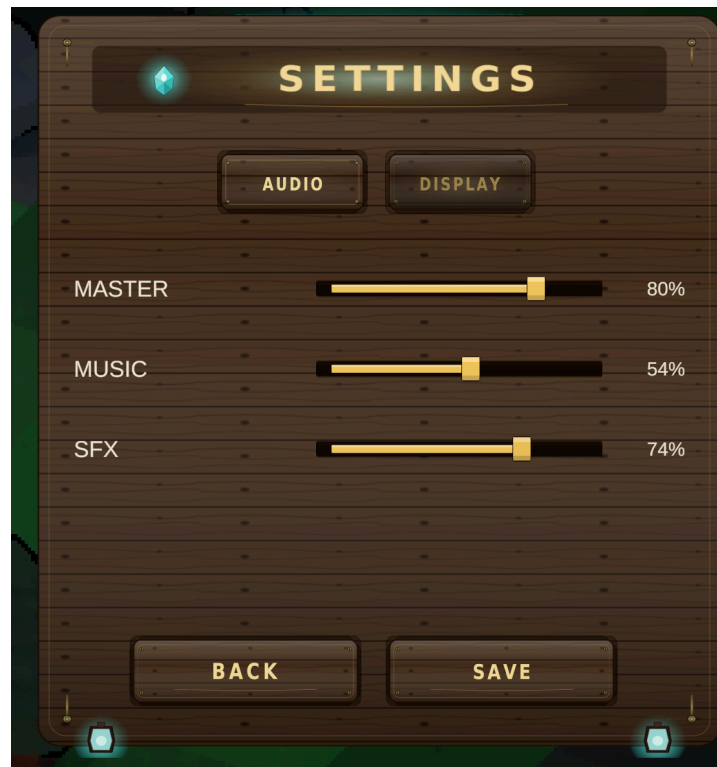


Image 7: Audio Settings

#### 4.1.5 Display Settings



Image 8: Display Settings

The appropriate screen size is automatically detected and selected when the game starts.

## 4.1.6 Unapplied Changes



Image 9: Unapplied Changes

When the player changes settings without applying them, this confirmation dialog is displayed. It prevents confusion by ensuring that unsaved changes are clearly communicated.

## 4.1.7 Inventory



Image 10: Inventory

The item stacks can be drag-and-dropped within the inventory. By dragging an item outside the inventory panel, the player drops it into the world. The player can also craft certain items in the inventory without the need for a crafting station.

## 4.1.8 Crafting Table



Image 11: Crafting Table

### 4.1.9 More Crafting Stations



Image 12: More Crafting Stations

Crafting Tables, Furnaces, Anvils etc. are abstracted into crafting stations that share a common UI but with different backgrounds and different available crafting recipes. This design follows our extensibility principle and allows new crafting stations to be added easily.

### 4.1.10 GlowCore Upgrade



Image 13: GlowCore Upgrade

### 4.1.11 Chest



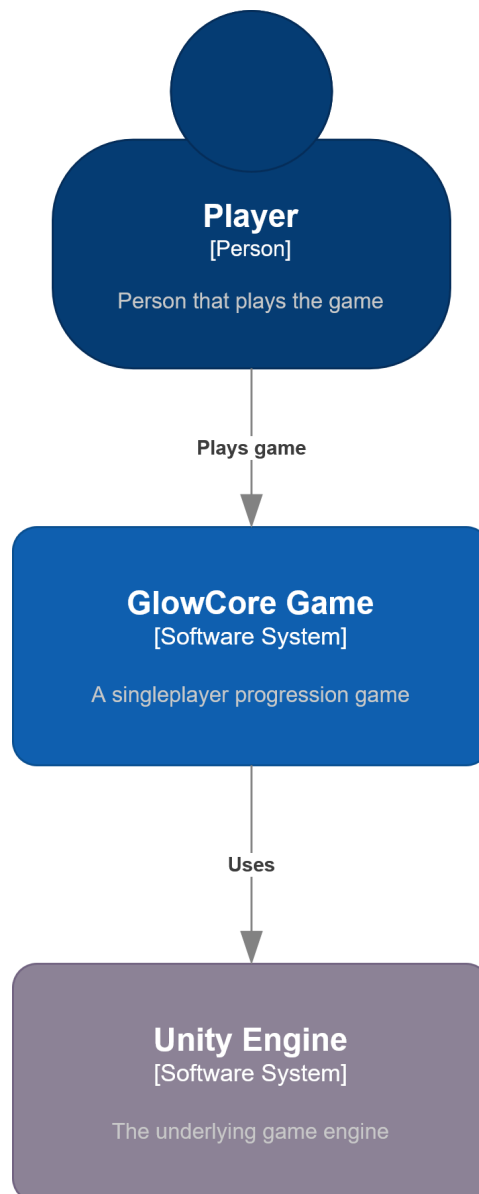
Image 14: Chest

## 4.2 C4 Model

To model the architecture of GlowCore we decided to use the C4 Model. The model is designed to provide stakeholders new to the project a high-level overview of GlowCore with different levels of abstraction. We decided against arc42 because it is too extensive for this project.

We intentionally omitted the Code Diagram (Layer 4 of C4) in this documentation because of how much complexity it would add without providing much value. It is common practice in project management to omit the Code Diagram.

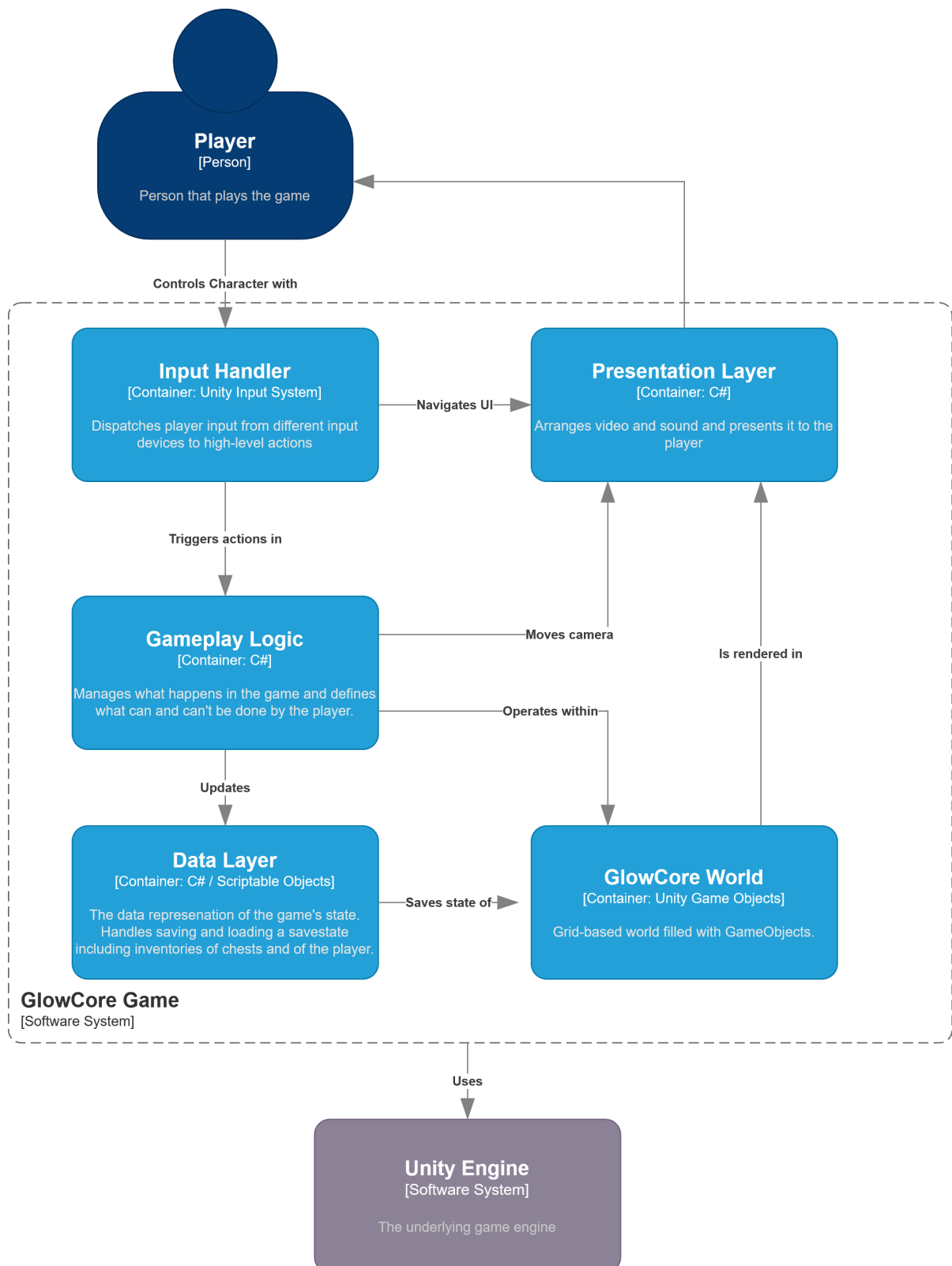
### 4.2.1 System Context Diagram



**Image 15:** C4 System Context Diagram



## 4.2.2 Container Diagram



**Image 16:** C4 Container Diagram

## 4.2.3 Component Diagram

The Component Diagram zooms into selected Containers to provide a deeper overview.

### 4.2.3.1 Gameplay Logic

A naive approach to implement features like Breaking Nodes, Building, Inventories and Crafting would be to just implement them separately, each with its own isolated input handling and execution logic. This would result in duplicated code and poorly maintainable code.

Instead, we analyzed the features and abstracted them into two groups: **Use Actions** and **Interact Actions**.

Use Actions are responsible for using the Item in the player's hand (The item that is selected in the hotbar and visually displayed in the player's hand). The player can only break Nodes when he holds a tool in his hand and he can only place Nodes if he holds a Block in his hand.

Interact Actions define the behaviour of hovered over Nodes. For example the player opens a chest by hovering over it with the cursor and pressing the corresponding button. The GlowCore and all crafting stations share the same input to avoid code duplication. This approach also makes our game extensible.

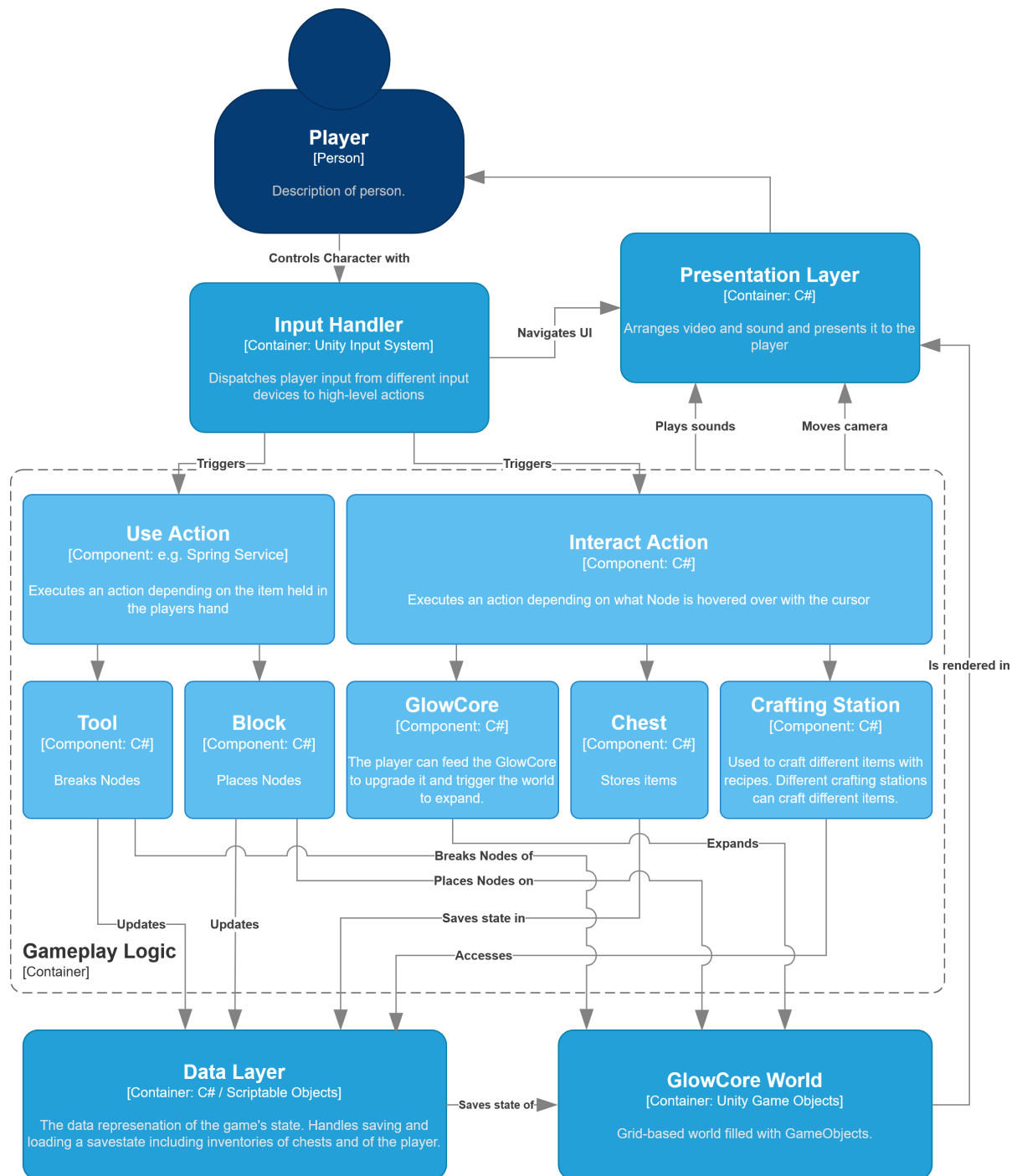


Image 17: C4 Component Diagram - Gameplay Logic

#### 4.2.3.2 Presentation Layer

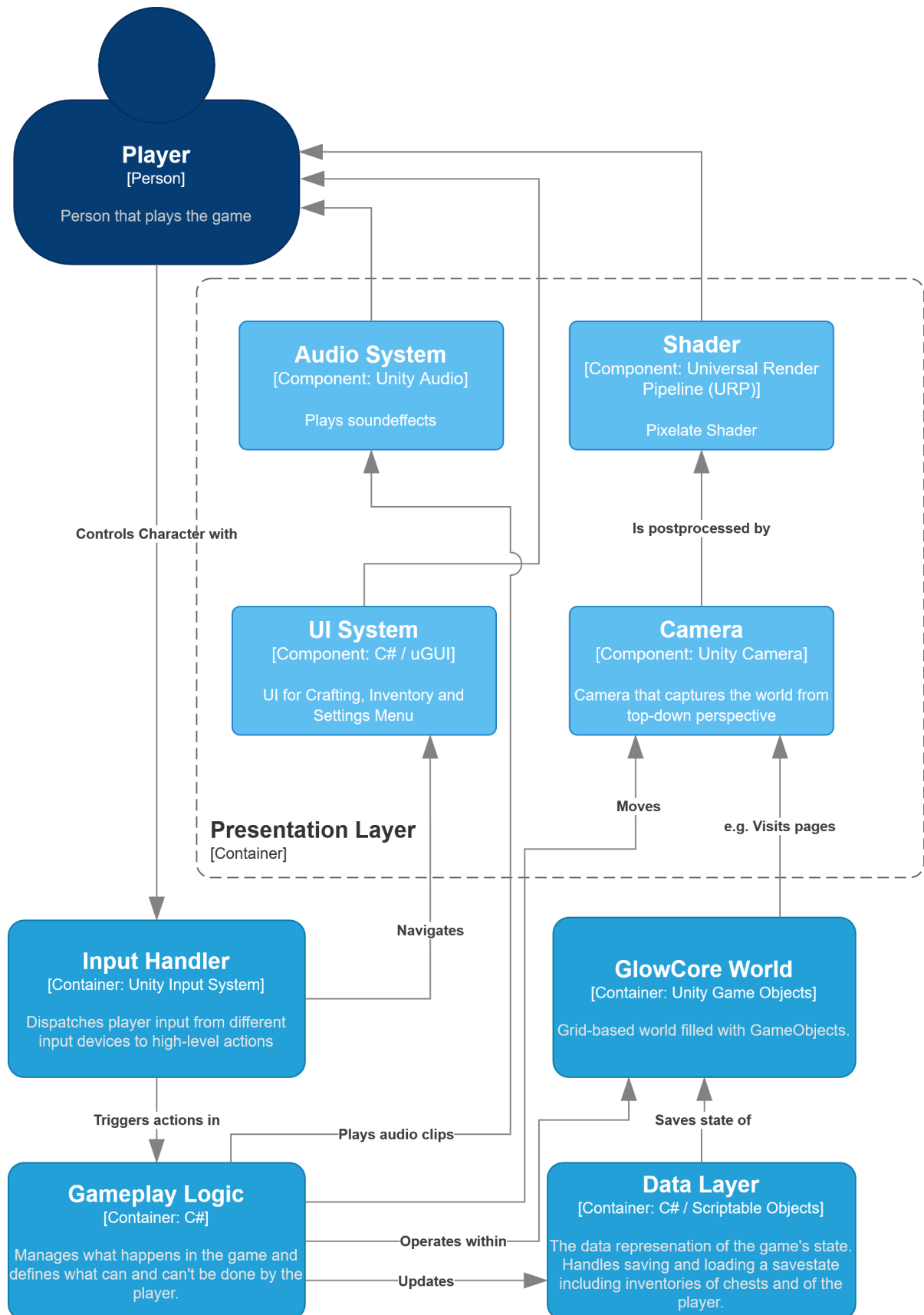
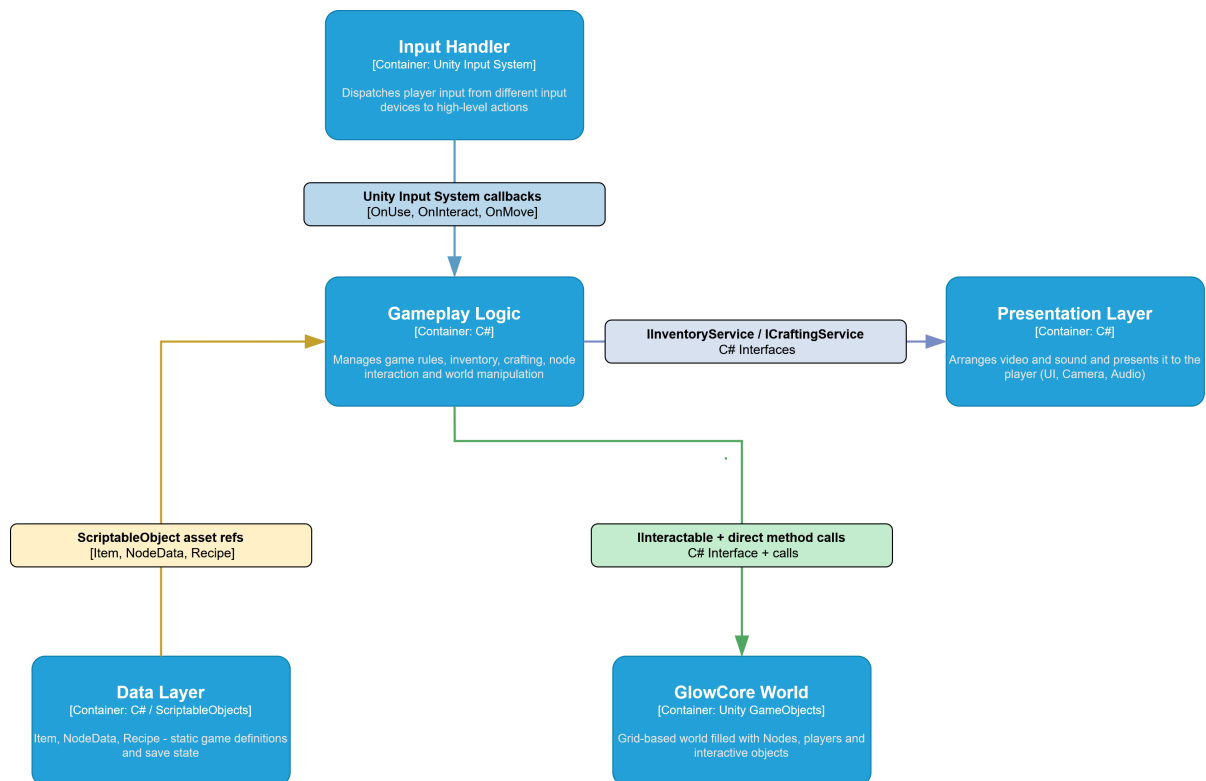


Image 18: C4 Component Diagram - Presentation Layer

## 4.3 Interfaces

GlowCore's container boundaries are expressed through either a C# interface or a ScriptableObject asset reference, with the exception of the input boundary where Unity's own message dispatch handles the connection directly.

### 4.3.1 Overview Diagram



**Image 19:** Interface Overview Diagram

### 4.3.2 Interface Contracts

| Interface<br><i>Boundary<br/>Mechanism</i>                                                                                               | Purpose                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Container Boundaries</b>                                                                                                              |                                                                                                                                                                                                                                                                                                                                         |
| <b>Unity Input callbacks</b><br><i>Input Handler → Gameplay Logic</i><br>Unity Input System (no C# interface)                            | Unity calls methods like <code>OnUse</code> , <code>OnInteract</code> and <code>OnMove</code> directly on <code>MonoBehaviour</code> components such as <code>PlayerHand</code> and <code>NodeActionSystem</code> . There is no C# interface at this boundary. Unity's own message dispatch handles the connection.                     |
| <b>IInteractable</b><br><i>Gameplay Logic → GlowCore World</i><br>C# interface                                                           | Nodes that do something when the player interacts with them implement this interface. A Chest opens its inventory, a <code>CraftingTableInteractable</code> shows the crafting UI, and <code>GlowCoreObject</code> runs its own logic, all through the same single-method contract.                                                     |
| <b>IInventoryService</b><br><i>Gameplay Logic → Presentation Layer</i><br>C# interface + events ( <code>Action&lt;T&gt;</code> )         | The main contract between the inventory system and the UI. Components like <code>InventoryUI</code> and <code>HotbarUI</code> subscribe to events such as <code>OnSlotChanged</code> and <code>OnHotbarSelectionChanged</code> to stay in sync with the game state, without ever referencing <code>PlayerInventory</code> directly.     |
| <b>ICraftingService</b><br><i>Gameplay Logic → Presentation Layer</i><br>C# interface + event ( <code>Action</code> )                    | Gives the crafting UI everything it needs: the current recipe list, availability checks, the <code>Craft()</code> command to actually execute a recipe, and the <code>OnRecipesRefreshed</code> event that triggers a refresh whenever the player's inventory changes.                                                                  |
| <b>ScriptableObject asset refs</b><br><i>Data Layer → Gameplay Logic</i><br>Unity serialised reference ( <code>[SerializeField]</code> ) | <code>Item</code> , <code>NodeData</code> and <code>Recipe</code> assets are set up in the Unity Inspector and read at runtime by the components that need them. There is no code dependency; the connection is entirely through serialised Unity references.                                                                           |
| <b>Gameplay Logic - Internal Component Boundaries</b>                                                                                    |                                                                                                                                                                                                                                                                                                                                         |
| <b>IHandItem</b><br><i>PlayerHand → Item Prefab (GL internal)</i><br>C# interface                                                        | Any item that can be used while held in the player's hand implements this interface. When the player presses the use button, <code>PlayerHand</code> calls <code>Use()</code> on whichever component is currently active, without caring whether it is a <code>ToolBehaviour</code> , a <code>BlockBehaviour</code> , or anything else. |
| <b>IPlayerInventoryAware</b><br><i>PlayerHand → Item Prefab (GL internal)</i><br>C# interface                                            | Some item prefab components need access to the player's inventory to function correctly. This interface lets <code>PlayerHand</code> pass them a reference when the item is equipped, avoiding any need for a singleton lookup.                                                                                                         |

Table 38: Interface Contracts

### 4.3.3 Extensibility through Stable Interfaces

These interfaces are what makes NFR401 achievable in practice. Each interface was designed so that adding something new only requires work in one place.

Adding a new usable item means implementing `IHandItem` on a prefab component. Adding a new interactive node means implementing `IInteractable`. In both cases nothing else in the system needs to change: the Input Handler, the Presentation Layer, and the Data Layer are completely unaware of the new addition.

Extending the item and crafting catalogue is just as simple. A new `Item` or `NodeData` asset can be created as a `ScriptableObject`, filled in through the Unity Inspector, and referenced wherever it is needed, without touching any C# code at all. The same applies to crafting recipes: a new `Recipe` asset is all that is needed to make a new combination available in the crafting UI.

The inventory and crafting UI are equally easy to extend. Because `InventoryUI` and `HotbarUI` depend only on `IInventoryService`, and `CraftingTableUI` only on `ICraftingService`, a new UI component for either system can be added by subscribing to the same events and use given methods, without modifying any existing class.

## 4.4 Technologies

### 4.4.1 Unity

The most important decision was the selection of the game engine. We decided to use Unity because we had the most experience with it and all members have already learned C#. Had our game been 2D and not 3D, Godot would have been our preferred choice.

### 4.4.2 Universal Render Pipeline (URP)

Unity provides 3 rendering pipelines:

- **Built-in Render Pipeline:** Legacy render pipeline
- **Universal Render Pipeline (URP):** Lightweight scriptable render pipeline.
- **High Definition Render Pipeline (HDRP):** For realistic surfaces.

We chose URP because the HDRP would be overkill for our project and the Built-in Render Pipeline is deprecated and new projects should use URP instead. URP is perfectly suitable to create custom shaders and viewports.

### 4.4.3 .NET & C#

Unity's default scripting language is C#, so this decision came bundled with the selection of the game engine. Although it is possible to use Plugins to run other languages such as C++, there are no significant benefits, so we settled on C#.

Unity uses Mono, a .NET runtime, to create cross-platform applications. We use NUnit via the Unity Testing Framework for automated testing.

### 4.4.4 GitHub

We use GitHub for version control. At the beginning of the project we used GitLab, but we switched to GitHub because we experienced problems setting up the CI/CD Pipeline to work. The documentation on GitLab was limited, since most tutorials and examples are focused on GitHub Actions.

### 4.4.5 Blender

To create our 3D Models we use Blender. Blender is free and open-source and offers all features of its competitors, so the choice was obvious. The workflow to import 3D Models from Blender into Unity is great and preserves all important information.

### 4.4.6 Audacity

We use Audacity to record and edit our sound effects. Audacity is free, easy to use and provides all necessary tools for simple sound design, so there was no need to choose a more sophisticated audio editing software.

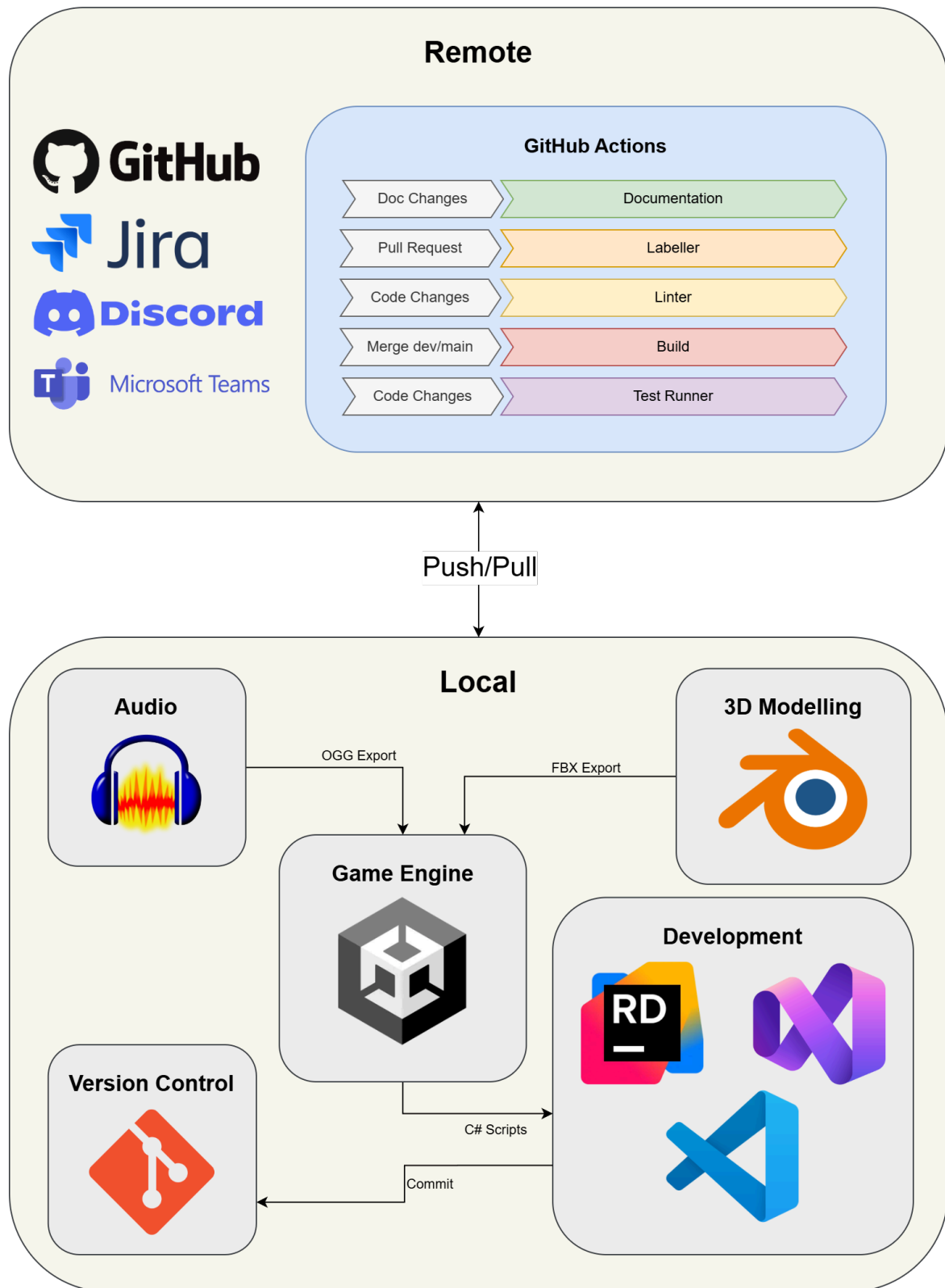


Image 20: Technologies Diagram



## 4.5 Used Design Patterns

### 4.5.1 Component-Based Architecture

We use Unity's Component-Based Architecture, where every `GameObject` has Components that define its behaviour. To implement custom Components, we use MonoBehaviour-Scripts to define the logic of a `GameObject`. This approach increases modularity because it is possible to split the logic into different Components and reuse them.

We don't have a real **Entity Component System (ECS)** because we don't separate the logic from the data. To support a real ECS we would need to use Unity's **Data-Oriented Technology Stack (DOTS)**. For our project the additional complexity of DOTS would outweigh its performance benefits.

### 4.5.2 Singleton

We use the Singleton Pattern where appropriate. In our game it definitely makes sense because we have systems that only exist once. For example there is only one Player and only one World-Grid.

### 4.5.3 Dependency Injection

Games often have tightly coupled systems. To counter this we use dependency injection wherever appropriate to decrease coupling.

### 4.5.4 Scriptable Objects

In our game we have to define and store data about Items and Nodes in an organized way. Meant are the Item/Node-Definitions, not the instances at runtime. For example for the Wood-Item we need to store its ID, 3D Model, Max Stack-Size etc.

Scriptable Objects solve this problem. Scriptable Objects are objects that are not bound to a `GameObject`. Instances of ScriptableObjects can be saved as **.asset** files. This improves Version Control for Items/Nodes since they are saved as individual files.

### 4.5.5 Prefabs

We use Prefabs to save specific `GameObjects` as **.prefab** files. This enables us to dynamically Instantiate Prefabs as `GameObjects` during runtime, which is necessary for e.g. Building.

## 4.6 Extensibility

Our game is designed to be extensible and exchangeable wherever feasible. Thanks to our ScriptableObject Architecture we can easily add new Items/Nodes just by creating a new Instance of the corresponding ScriptableObject and setting the parameters and referencing the 3D Model and Icon.

Thanks to the Component-Based Architecture we can easily add new Items that perform a specific action when used by the player. It would be easy to add for example Consumables. An additional benefit is that it is easy to add Nodes with custom behaviour for when the player interacts with it. For example you could just create a Signpost component and add it to a Node to create a Node where you can write something on it and read it.

We save our 3D Models separately from the ScriptableObjects in the .fbx format to improve exchangeability. It is possible to only exchange the 3D Model without touching the ScriptableObject.

A major limit of our architecture is that we are strongly tied to Unity. Switching to another game engine would require a significant effort due to how our systems are relying on Unity's systems.

## 4.7 External Dependencies

Our game is distributed via Steam, which is our only external dependency. Steam is responsible for installing, launching and updating the game on the end-user side. We also use Steam for marketing.

We create the store page for our game on Steamworks, where we can configure the description, upload marketing material etc. The executable is uploaded using Steamworks SteamPipe, which handles distribution and versioning.

## 4.8 Game Design

This chapter covers Gameplay Decisions rather than technical solutions, focusing on how the game is structured to create an entertaining experience. Game design can be understood as the application of user experience principles within the domain of game development.

### 4.8.1 Core Gameplay Loop

The core gameplay loop of GlowCore can be broken down into the following steps:

1. Gather resources
2. Craft items
3. Upgrade tools
4. Build Machines
5. Upgrade the GlowCore, which unlocks new areas and items
6. Repeat loop

### 4.8.2 Gameplay Progression

The following sections describe the gameplay progression in chronological order and explain the design decisions and their reasons.

#### 4.8.2.1 Early Game

The player starts with a basic wooden axe and gathers wood by manually cutting trees. Early trees can be chopped quickly but only give a small amount of wood. Over time, larger trees appear which require significantly more time to cut but also give more wood.

#### 4.8.2.2 Automation Phase 1

At this point, manual gathering becomes inefficient, encouraging the player to craft their first Axe Machine. This machine is placed in front of a tree and automatically cuts it over time. While the machine is operating, the player is free to perform other actions, such as gathering additional resources to craft more machines.

This creates a strong feeling of progress and a passive progression reward, since the player knows that progress is being made even while he is not directly interacting with the machine. This effect is reinforced as the player uses gathered resources to construct more machines, amplifying the loop.

When returning to the machine, the player collects the gathered resources and moves the machine to a new tree. This is acceptable at this stage because repositioning requires significantly less effort than manual tree cutting.

#### 4.8.2.3 Automation Phase 2

Over time, the player reaches a point where moving so many Axe Machines becomes tedious. At this stage, a new system is introduced: The Replanter block. The Replanter automatically places saplings. This allows trees to regrow automatically, removing the need to constantly reposition machines. However, this introduces a trade-off, as machines remain inactive while waiting for trees to regrow. This adds strategic depth to the system.

#### 4.8.2.4 Automation Phase 3

Later in progression, a new tree type is introduced that has short cut time and drops a relatively big amount of wood but has a very long growth time. To improve efficiency the player crafts the newly unlocked Axe Machine Tier 2, which moves automatically in the direction it's facing and harvests trees sequentially.

At this stage, multiple Replanter blocks can be used per machine, enabling more complex automated setups. A key design decision is that Replanter blocks must be significantly cheaper than Tier 2 machines, encouraging players to build efficient production lines.

#### 4.8.2.5 Full Automation

However, this introduces a new problem: machine orientation and placement still require manual adjustment. This is resolved by the introduction of Rotator blocks. When a machine collides with a Rotator block, it automatically rotates by 90, 180, or 270 degrees depending on the block type. This enables fully automated resource gathering systems and allows players to design complex production paths.

### 4.8.3 Automation Constraints

A core design principle in this system is that not everything should be fully automatable. If all actions were automated, the player would just watch the machines and become bored. Instead, the player must still actively gather certain resources in order to craft and expand automation systems.

### 4.8.4 Depth and Complexity

We followed the principle of providing as much depth as possible with as little complexity as possible. The goal of this design is to allow the player to create complex and interesting systems without introducing a large number of different mechanics and complicated rules.

In GlowCore, this is achieved by limiting the core automation system to only three main components: the Axe Machine, the Replanter, and the Rotator. When combined, these simple systems interact in ways that allow for emergent complexity, such as automation lines with different layouts.

## 4.9 Key Architectural Decisions

The following decisions had the greatest impact on the structure and long-term maintainability of the game. Each one is recorded together with the problem it solved and the reasoning behind the chosen approach.

### 4.9.1 ScriptableObjects as the Data Layer

**Decision:** All item, node, and recipe definitions are stored as ScriptableObject .asset files rather than in C# code or external data files (e.g. JSON).

**Why:** Game content, items, nodes, crafting recipes, must be defined once and consumed from many places without creating code dependencies. ScriptableObjects let non-programmers create and configure new content entirely through the Unity Inspector. Each asset is an individual file, which keeps version control diffs small and conflict-free. Crucially, the connection to gameplay code is a serialised Unity reference, so adding a new item, node, or recipe never requires touching any C# source file.

### 4.9.2 Interactable for World Interaction

**Decision:** Every node that responds to player interaction implements the IInteractable interface. NodeActionSystem calls Interact() on whichever node is currently hovered, without knowing its concrete type.

**Why:** Without a shared interface, NodeActionSystem would require an explicit branch for every interactive object type: chests, crafting stations, the GlowCore, signs, and anything added later. With IInteractable the input handler is permanently closed to modification on this axis, adding a new interactive node is purely additive. The GetActionPromptText() method on the same interface also lets each node control its own UI prompt label, removing another potential switch statement from the core input loop.

### 4.9.3 InventoryService and Event-Driven UI

**Decision:** The inventory system is exposed to all UI components exclusively through the IInventoryService interface, which carries both commands and Action<T> events (OnSlotChanged, OnHotbarSelectionChanged, etc.).

**Why:** Inventory state is modified from many independent sources: item collection, crafting, chest transfers, and GlowCore upgrades. A polling-based UI would either check every frame or miss updates; direct method calls from each source to each UI component would create a tightly coupled web of dependencies. Events mean every UI component reacts exactly when something relevant changes, at zero cost when idle. Because all UI depends only on the interface and not on PlayerInventory directly, the inventory implementation can change freely without any UI class breaking.

### 4.9.4 Use Actions and Interact Actions

**Decision:** Instead of implementing each player capability (breaking nodes, placing blocks, opening chests, crafting) as an isolated system with its own input handling, all player inputs are routed through exactly two abstractions: **Use Actions** (what the held item does, via IHandItem) and **Interact Actions** (what the hovered node does, via IInteractable).

**Why:** The two abstractions map directly to the two roles the player's cursor plays: the held item and the targeted node. Routing them separately through PlayerHand and NodeActionSystem keeps input handling in exactly one place per axis and avoids duplicating raycast logic, inventory checks, and UI-open guards. Adding a new usable item only requires implementing IHandItem on its prefab; adding a new interactive node only requires implementing IInteractable. Neither addition changes any existing class.

## 4.9.5 IPlayerInventoryAware via Explicit Injection

**Decision:** Item prefab components that need access to the player's inventory receive it through `IPlayerInventoryAware.SetInventory()`, called by `PlayerHand` at equip time, not through a singleton or a `FindObjectOfType` lookup.

**Why:** Tools and blocks need to read and modify the inventory at use-time (e.g. consuming the block stack when placing). Fetching `PlayerInventory.Instance` inside a prefab component hides the dependency, makes the coupling invisible to readers, and makes the component untestable in isolation. Passing the reference explicitly at equip time makes the dependency visible, eliminates hidden singleton coupling, and allows the component to be tested with any `PlayerInventory` instance, including a mock.

## 4.10 Deployment

The game has two deployment paths, both automated through GitHub Actions. Which path runs depends on what is pushed to the repository.

### 4.10.1 Continuous Deployment - WebGL (GitHub Pages)

Every push to `main` or `dev` triggers the WebGL build job. All jobs run on a self-hosted Linux runner with Unity 6000.3.9f1 installed directly on the machine. The Unity Editor is invoked in batch mode via a custom `BuildScript`, which avoids the overhead of spinning up a Docker container on every run. The license is activated per-run using `buildalot/activate-unity-license`. The resulting build artifact is deployed to the `pages` branch via `JamesIves/github-pages-deploy-action`. Builds from `main` land in `prod/webgl`; builds from `dev` land in `dev/webgl`. This gives stakeholders an always-up-to-date playable version of the game in a browser without any manual steps.

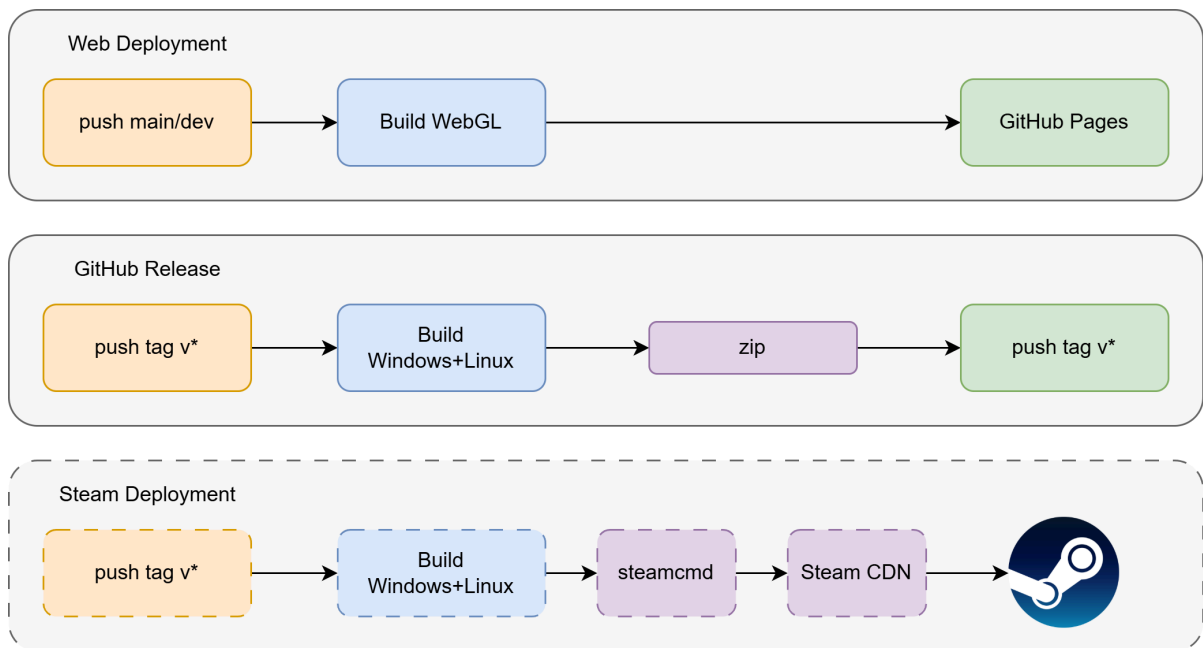
### 4.10.2 Release Deployment - Windows & Linux (GitHub Releases)

Pushing a version tag (e.g. `v1.0.0`) triggers the Windows and Linux build jobs in parallel, both running on the same self-hosted runner. Each job invokes the Unity Editor directly in batch mode, targeting `StandaloneWindows64` and `StandaloneLinux64` respectively. Once both builds complete, a release job downloads the artifacts, zips them, and publishes them as a GitHub Release with auto-generated release notes. Players can download the zipped executables directly from the GitHub Releases page.

### 4.10.3 Steam

Once the project is ready to be published to Steam, the release pipeline will be extended with a `SteamPipe` upload step after the Windows and Linux builds. The `steamcmd` CLI can authenticate using Steamworks credentials stored as GitHub Actions secrets and upload the build depot to Steam. Steam would then handle CDN distribution to all players who own the game. No structural change to the existing pipeline would be required.

## 4.10.4 Deployment Diagram



**Image 21:** Deployment Diagram

# 5 Quality Measures

## 5.1 Working Environment

### 5.1.1 CI/CD Pipeline

The project uses GitHub Actions for all build, test, and deployment automation. Five pipelines are defined, each with a distinct responsibility:

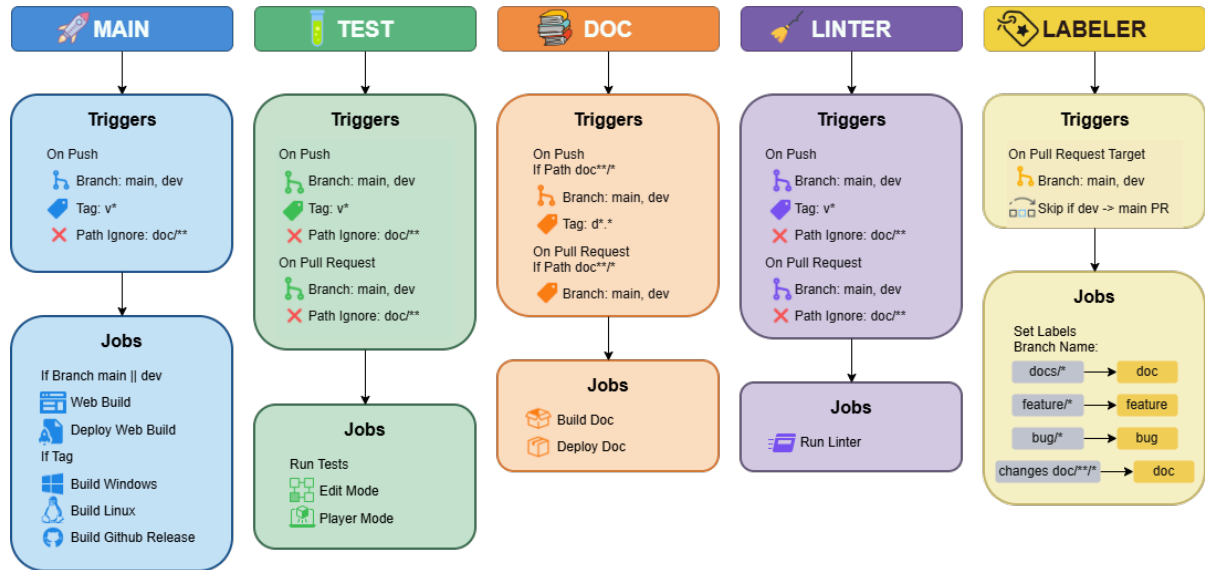
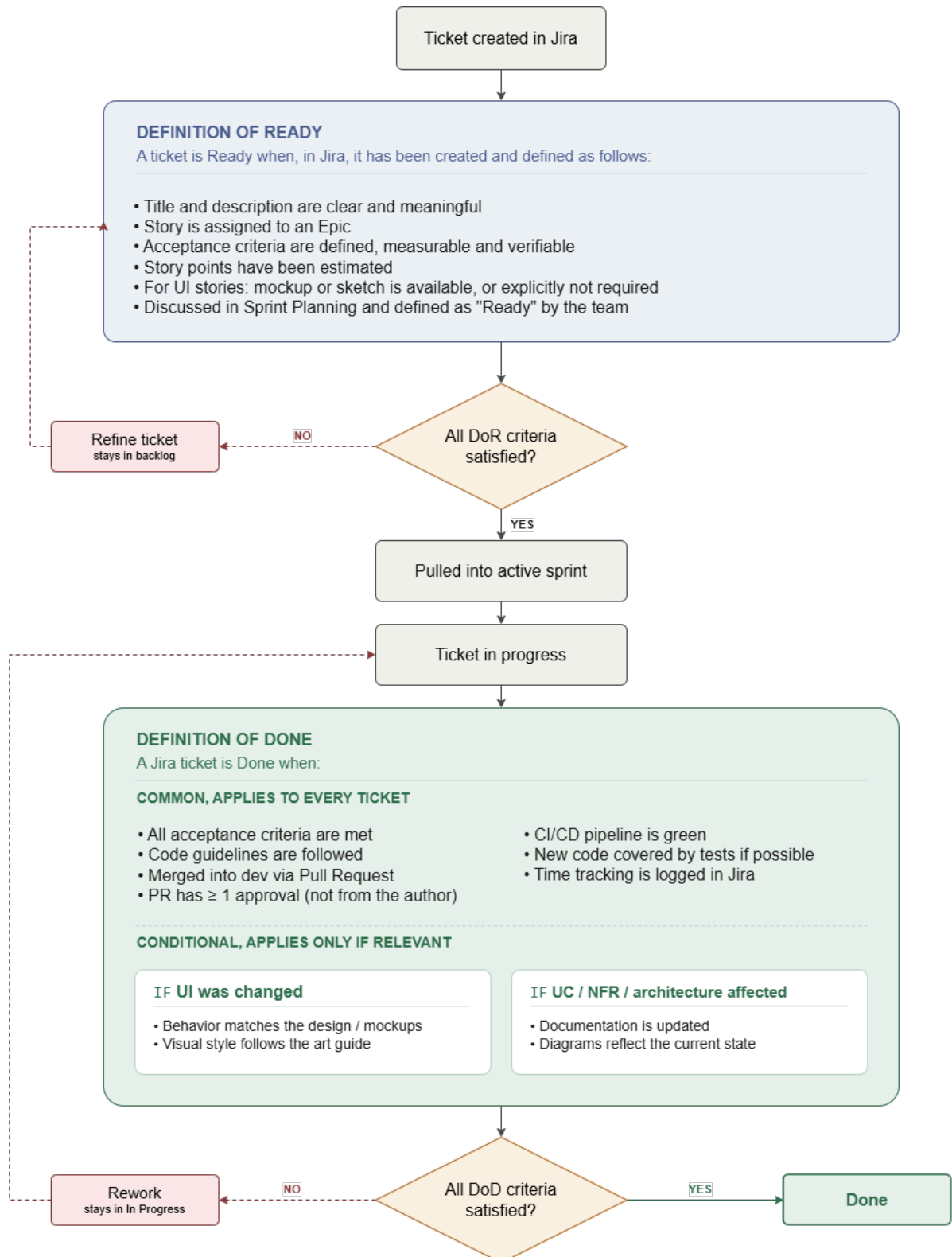


Image 22: CI/CD Pipeline Overview

- **MAIN:** runs on every push to main or dev (excluding doc/\*\* changes) and on version tags (v\*). Builds and deploys the WebGL build for pushes to the two main branches. On a version tag it additionally compiles Windows and Linux standalone builds and creates a GitHub Release.
- **TEST:** runs on push to main or dev and on pull requests targeting either branch (both excluding doc/\*\*). Executes the Unity Test Framework suite in both Edit Mode and Play Mode and generates a report on both results and coverage.
- **DOC:** triggered only when doc/\*\* files are modified: on push to main or dev, on doc tags (d\*. \*), and on pull requests that touch the documentation. Builds the Typst documentation and deploys it to GitHub Pages.
- **LINTER:** runs on the same push and pull-request triggers as TEST (excluding doc/\*\*). Runs the Roslyn code-analysis and dotnet format checks described in the Coding Guidelines section. A failed lint check blocks the merge.
- **LABELER:** fires on pull-request targets to main or dev (skipped for dev → main promotion PRs). Automatically applies labels to each PR based on branch prefix and changed paths (feature/\* → feature, bug/\* → bug, docs/\* or changes under doc/\*\*/\* → doc).

## 5.1.2 Definition of Ready and Definition of Done

Every Jira ticket follows a structured lifecycle governed by our Definition of Ready (DoR) and Definition of Done (DoD). A ticket may only be pulled into a sprint once all DoR criteria are met, including a clear description, defined acceptance criteria, story point estimate, epic assignment, and team sign-off during sprint planning. A ticket is only considered done once all DoD criteria are satisfied, covering code quality, CI/CD pipeline status, test coverage, PR approval and, conditionally, UI consistency and documentation updates.

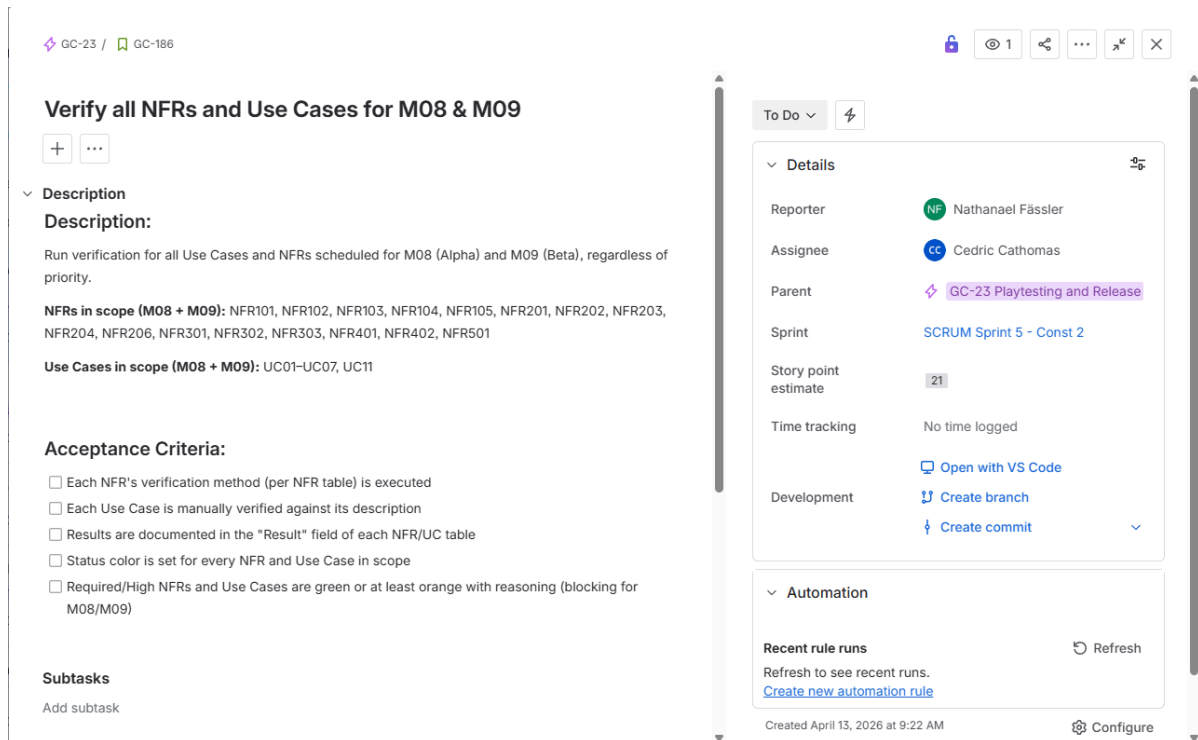


**Image 23:** Definition of Ready and Definition of Done - ticket lifecycle flowchart



### 5.1.2.1 DoR Example

The following example shows a real ticket (GC-186) that satisfies all DoR criteria and is ready to be worked on.



GC-23 / GC-186

## Verify all NFRs and Use Cases for M08 & M09

+ ...

**Description**

**Description:**

Run verification for all Use Cases and NFRs scheduled for M08 (Alpha) and M09 (Beta), regardless of priority.

**NFRs in scope (M08 + M09):** NFR101, NFR102, NFR103, NFR104, NFR105, NFR201, NFR202, NFR203, NFR204, NFR206, NFR301, NFR302, NFR303, NFR401, NFR402, NFR501

**Use Cases in scope (M08 + M09):** UC01-UC07, UC11

**Acceptance Criteria:**

- ☐ Each NFR's verification method (per NFR table) is executed
- ☐ Each Use Case is manually verified against its description
- ☐ Results are documented in the "Result" field of each NFR/UC table
- ☐ Status color is set for every NFR and Use Case in scope
- ☐ Required/High NFRs and Use Cases are green or at least orange with reasoning (blocking for M08/M09)

**Subtasks**

Add subtask

**To Do** ⚡

**Details**

Reporter: Nathanael Fässler

Assignee: Cedric Cathomas

Parent: GC-23 Playtesting and Release

Sprint: SCRUM Sprint 5 - Const 2

Story point estimate: 21

Time tracking: No time logged

Development: [Open with VS Code](#), [Create branch](#), [Create commit](#)

**Automation**

**Recent rule runs** Refresh

Refresh to see recent runs.  
[Create new automation rule](#)

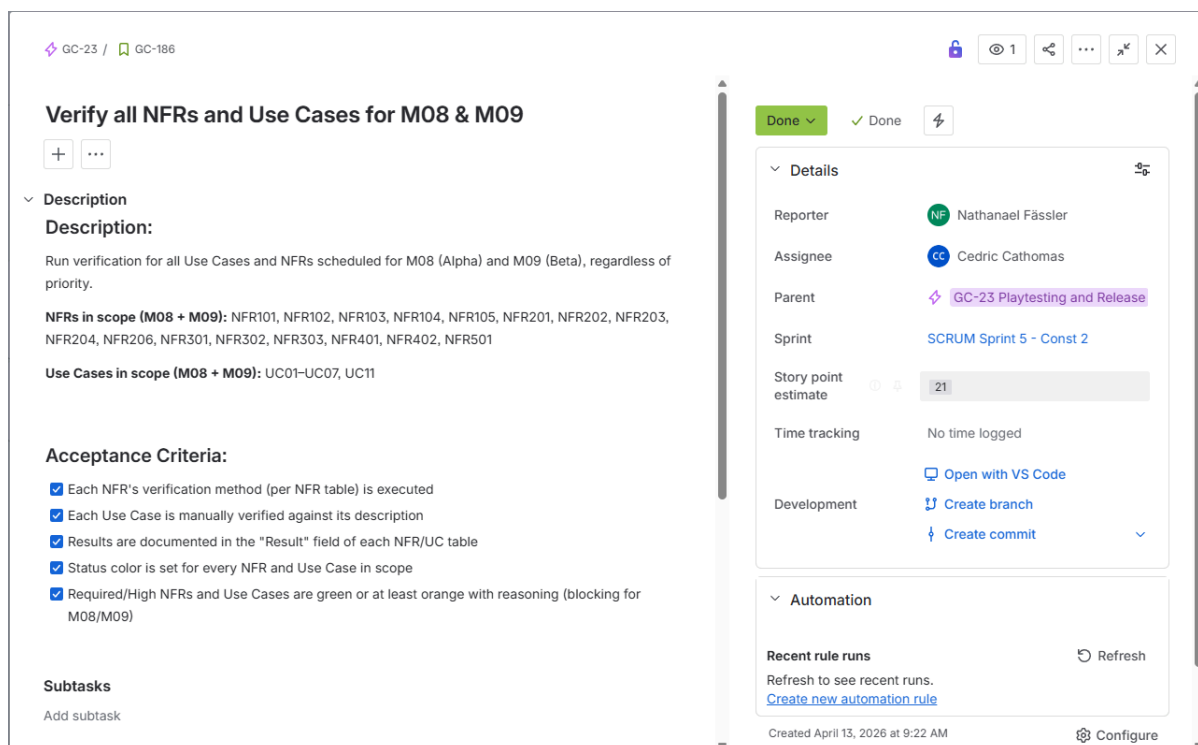
Created April 13, 2026 at 9:22 AM  
 Updated 13 seconds ago

Configure

Image 24: GC-186 - Example of a ticket satisfying all Definition of Ready criteria

### 5.1.2.2 DoD Example

The following example shows the same ticket (GC-186) with all acceptance criteria checked and status set to Done.



GC-23 / GC-186

## Verify all NFRs and Use Cases for M08 & M09

+ ...

**Description**

**Description:**

Run verification for all Use Cases and NFRs scheduled for M08 (Alpha) and M09 (Beta), regardless of priority.

**NFRs in scope (M08 + M09):** NFR101, NFR102, NFR103, NFR104, NFR105, NFR201, NFR202, NFR203, NFR204, NFR206, NFR301, NFR302, NFR303, NFR401, NFR402, NFR501

**Use Cases in scope (M08 + M09):** UC01-UC07, UC11

**Acceptance Criteria:**

- ☒ Each NFR's verification method (per NFR table) is executed
- ☒ Each Use Case is manually verified against its description
- ☒ Results are documented in the "Result" field of each NFR/UC table
- ☒ Status color is set for every NFR and Use Case in scope
- ☒ Required/High NFRs and Use Cases are green or at least orange with reasoning (blocking for M08/M09)

**Subtasks**

Add subtask

**Done** ✓ Done ⚡

**Details**

Reporter: Nathanael Fässler

Assignee: Cedric Cathomas

Parent: GC-23 Playtesting and Release

Sprint: SCRUM Sprint 5 - Const 2

Story point estimate: 21

Time tracking: No time logged

Development: [Open with VS Code](#), [Create branch](#), [Create commit](#)

**Automation**

**Recent rule runs** Refresh

Refresh to see recent runs.  
[Create new automation rule](#)

Created April 13, 2026 at 9:22 AM  
 Updated 13 seconds ago

Configure

Image 25: GC-186 - Example of a ticket satisfying all Definition of Done criteria

## 5.2 Test Concept

Our testing strategy for GlowCore is fundamentally shaped by the nature of the project. Since GlowCore is a standalone Unity game built for Windows, Linux, and WebGL, we don't have any server connections, a huge amount of data in a database or a frontend beyond simple game menus. This allows us to focus our entire testing effort directly on the client-side application, ensuring gameplay mechanics, performance, and user experience are as solid as possible.

The test concept is divided into automated testing for our codebase, manual system verification, extensive usability testing, and targeted checks for our non-functional requirements (NFRs).

### 5.2.1 Automated Testing (Unit & Integration)

At the code level, we rely heavily on the Unity Test Framework (UTF). We use it to write both unit tests and integration tests that verify whether individual functionalities and interconnected systems are working correctly.

A core part of our workflow is our CI/CD pipeline. Our UTF tests are set up to run automatically on the pipeline every time someone pushes new code. Because these tests run continuously on the latest codebase, they effectively act as our regression tests. They give us immediate feedback if a new feature or bug fix accidentally breaks an older, already working part of the game. We are aiming for a minimum of 40% automated test coverage across our systems, which is a pretty standard and realistic target for game development projects.

### 5.2.2 System Testing

We don't do automated system tests. Fully automating the entire game loop in Unity is just very difficult. It usually results in tests that break way too easily, for example, just because a small visual asset was moved slightly.

Instead, system testing is handled manually. For this, we regularly take the automated WebGL build generated from the current pipeline state and compare it directly against a known, stable release build. This manual side-by-side verification helps us ensure that the game as a whole still compiles, runs, and functions as expected before we consider a new version stable.

### 5.2.3 Usability Testing

There is a strict limit to what we can test automatically in a game project. A script might tell you if a player's health goes down correctly, but it can't tell you if the game is actually fun or if the controls make sense. Because of this, usability testing is a massive aspect of our QA process.

We will conduct regular playtesting sessions where we let several new users play the game without giving them any prior instructions. Our main goal here is to find out what feels good, what is frustrating, and if the core mechanics are actually understandable. For instance, we want to see if a brand-new player can figure out how to move, gather resources, and upgrade within the first 5 minutes. We will also observe if players naturally understand the visual cues for restricted actions (like red-tinted UI elements) or if they can accurately track their light upgrade progress.

## 5.2.4 Non-Functional Requirements (NFR) Verification

NFR tests are not done once at the end, but at appropriate points throughout the project. Each NFR is assigned to a milestone at which it must be verified. Additionally, our Definition of Done requires that any change which could affect an NFR is accompanied by a re-verification of that NFR and an updated result in the documentation. This ensures NFR results stay current and are not just a snapshot from a single test session.

A huge portion of our testing isn't just about finding bugs, but verifying our NFRs. We have defined a wide spectrum of NFRs covering different quality attributes. They are sorted by priority, meaning that while our ultimate goal is to fulfill all of them, the ones marked as "Required" are strict targets we have to hit for the final submission.

We cover NFR testing through a mix of automated benchmarks and manual playtesting protocols:

- **Performance Efficiency:** We regularly check our performance manually to make sure the game runs smoothly. We aim to ensure that the world loads quickly, in-game scene transitions are fast, and the game maintains a solid 60 FPS on recommended hardware.
- **Reliability:** Since saving the game correctly is critical, we manually test save and load cycles to ensure data doesn't corrupt. We also verify that save files remain small enough (under 5MB), that they are fully portable between machines, and that auto-saves trigger correctly.
- **Portability & Maintainability:** We manually verify that the game runs identically across Windows, Linux, and WebGL, and that it supports standard 16:9 resolutions. From a code perspective, we constantly test extensibility during normal development. We frequently add new elements like resource nodes, items, or recipes based on existing systems. If our internal architecture was too complicated to extend, we would notice it very quickly.
- **Usability & Visuals:** Beyond the playtesting mentioned earlier, we also manually review the game's language to ensure all text is in English. We test controller and keyboard inputs by simply playing with both, and we hold art reviews to make sure all new assets fit our low-poly pastel style guide.

By combining the automated safety net of our UTF pipeline with heavy manual playtesting and strict NFR benchmarking, we believe this concept will keep the game stable while still allowing us to iterate quickly on the gameplay.

## 5.3 Unity Testing

Testing a game is a bit different from testing a "standard" software project, like a web-based accounting tool or a simple CRUD application. In a typical software project, you are mostly concerned with whether a specific input leads to a specific output in a database or a UI. You can easily mock external dependencies and call functions in isolation.

In Unity, however, our code doesn't just sit there. It lives inside an engine that is constantly ticking. We have to deal with the "Game Loop," physics calculations, and the complex lifecycle of MonoBehaviours. To test a full cycle in the Unity engine, we are able to use Edit Mode and Play Mode.

### 5.3.1 Edit Mode Tests

Edit Mode tests are the closest thing we have to "traditional" unit tests. They run entirely within the Unity Editor and do not require the game to actually start. Since they don't have to load scenes or wait for the physics engine to initialize, they are fast.

Edit Mode tests can be used for our "pure" logic things like calculating resource costs, managing inventory math, or utility functions that don't depend on the game's frame rate. If a bug appears in our math, an Edit Mode test will catch it during our CI/CD run without us ever having to open a game window.

### 5.3.2 Play Mode Tests

Play Mode tests are where we handle the Unity Engine of the project. Unlike Edit Mode, these tests actually trigger the full game engine. They can load specific test scenes, instantiate Prefabs, and, most importantly, they can run over multiple frames.

This is essential for games because many mechanics are time-dependent. For example, if we want to test if the player really moves on input, we need the game clock to actually run. In Play Mode, we can use `yield return new WaitForSeconds(1);` to let the game simulate for a moment before checking the results. Play Mode tests can be used for:

- Physics interactions
- Component Lifecycle
- Player Input

The downside is that Play Mode tests are significantly slower because they have to “play” the game. However, they give us the confidence that our code isn’t just working on paper, but actually performing correctly within the simulated world of the engine.

### 5.3.3 Coverage Summary

At the final stage of the project (M10 Release reached), 9 out of 11 Use Cases are fully verified ( [UC01](#) , [UC02](#) , [UC03](#) , [UC04](#) , [UC05](#) , [UC06](#) , [UC07](#) , [UC10](#) , [UC11](#) ). [UC08](#) (Experience Story) was not implemented due to time constraints. [UC09](#) (Fight Enemies) is not applicable, since the team decided not to implement combat within the project scope.

For the NFRs, 16 requirements have fully passed ( [NFR103](#) , [NFR104](#) , [NFR105](#) , [NFR201](#) , [NFR202](#) , [NFR203](#) , [NFR205](#) , [NFR206](#) , [NFR301](#) , [NFR302](#) , [NFR303](#) , [NFR401](#) , [NFR402](#) , [NFR501](#) , [NFR502](#) , [NFR503](#) ), 1 is partially met ( [NFR204](#) ), and 1 failed ( [NFR101](#) ). [NFR102](#) is not applicable since in-game scene transitions were removed from scope. The partial NFR is input support: keyboard input works for all gameplay actions, but controller support was not implemented due to time constraints. [NFR101](#) (World Load Time) failed because load times grew with world content beyond what the 3-second target anticipated.

## 5.4 Code metrics

### 5.4.1 Lines of Code

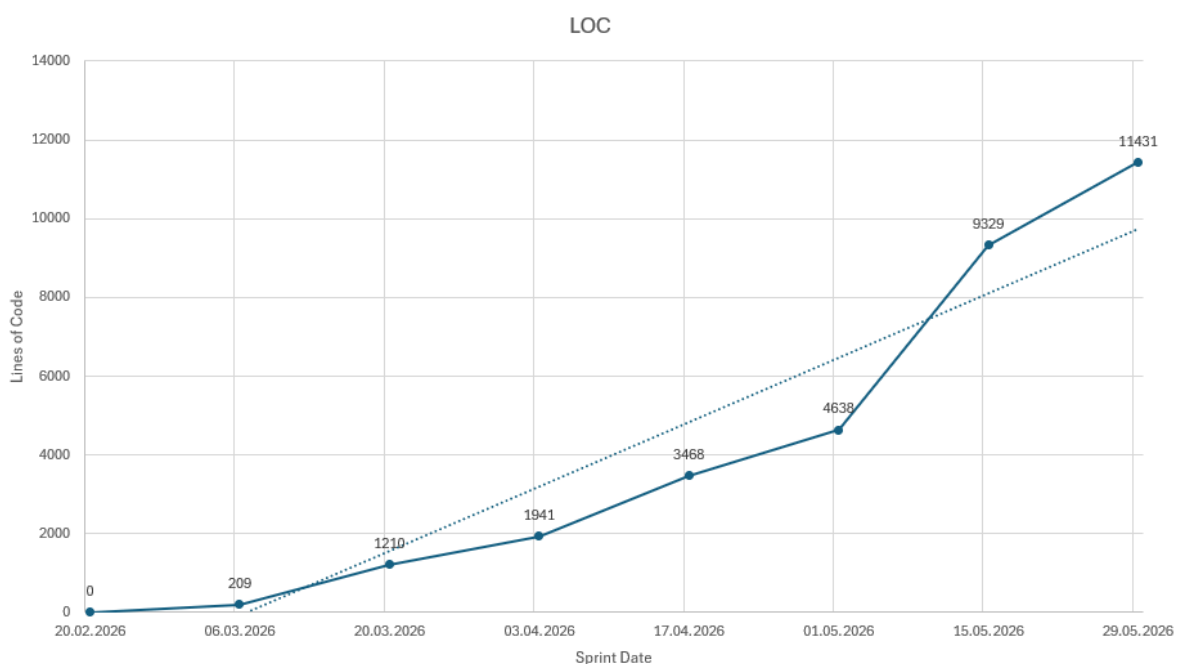


Image 26: Lines of Code over Time

| File                 | Blank | Comment | Code |
|----------------------|-------|---------|------|
| WorldGrid.cs         | 129   | 26      | 599  |
| InventoryUI.cs       | 88    | 20      | 393  |
| GlowCoreUpgradeUI.cs | 70    | 6       | 318  |
| SaveData.cs          | 65    | 6       | 280  |
| SceneTransiti...     | 39    | 4       | 264  |
| CraftingStationUI.cs | 45    | 11      | 242  |
| IconCaptureSe...     | 48    | 0       | 239  |
| ChestUI.cs           | 54    | 7       | 236  |
| Outline.cs           | 81    | 44      | 217  |
| Inventory.cs         | 44    | 0       | 212  |
| AxeMachine.cs        | 37    | 5       | 207  |
| GlowCore.cs          | 48    | 4       | 201  |
| ReadmeEditor.cs      | 46    | 1       | 195  |
| SettingsScreen.cs    | 36    | 9       | 189  |
| PlayerInventory.cs   | 41    | 12      | 169  |
| Node.cs              | 24    | 0       | 164  |
| RecipeRowUI.cs       | 36    | 2       | 158  |
| AudioManager.cs      | 37    | 0       | 157  |
| ActionPromptS...     | 27    | 3       | 155  |
| DisplaySettin...     | 30    | 1       | 149  |
| NodeActionSystem.cs  | 33    | 1       | 148  |
| ItemEditor.cs        | 31    | 1       | 141  |
| GameBootstrapper.cs  | 29    | 10      | 135  |
| SettingsScree...     | 33    | 5       | 134  |
| ItemSlotUI.cs        | 25    | 1       | 123  |
| UnsavedChange...     | 23    | 0       | 122  |
| ItemStackDrop.cs     | 29    | 1       | 119  |
| MenuManagerTests.cs  | 25    | 0       | 112  |
| GlowHoverEffect.cs   | 18    | 0       | 106  |
| BlockBehaviour.cs    | 24    | 0       | 105  |
| DisplayService.cs    | 21    | 6       | 104  |
| PlayerMovement.cs    | 16    | 0       | 103  |
| PlayerIntegra...     | 26    | 0       | 102  |
| BuildingInteg...     | 26    | 0       | 95   |
| PauseScreen.cs       | 19    | 2       | 93   |
| NewGameScreen.cs     | 19    | 0       | 93   |
| CraftingInteg...     | 26    | 0       | 92   |
| CraftingUI.cs        | 22    | 0       | 90   |
| HotbarUI.cs          | 13    | 0       | 88   |
| IconFraming.cs       | 19    | 3       | 87   |
| IconBatchMenu.cs     | 11    | 0       | 86   |
| IconRenderRig.cs     | 17    | 2       | 86   |
| ChestInteractable.cs | 19    | 3       | 84   |

| File                 | Blank | Comment | Code |
|----------------------|-------|---------|------|
| DropdownSetti...     | 10    | 0       | 40   |
| AutosaveService.cs   | 6     | 0       | 38   |
| NodeData.cs          | 6     | 0       | 38   |
| IInventoryService.cs | 4     | 3       | 37   |
| PauseEscapeCo...     | 9     | 1       | 37   |
| GameStateCont...     | 7     | 0       | 37   |
| ToggleSetting...     | 9     | 0       | 36   |
| WeirdPotion.cs       | 6     | 0       | 36   |
| Item.cs              | 5     | 2       | 34   |
| FakeDisplaySe...     | 8     | 0       | 32   |
| PauseButton.cs       | 7     | 0       | 32   |
| ExampleDamage...     | 7     | 0       | 31   |
| AudioSettings...     | 4     | 0       | 31   |
| InventoryEsca...     | 7     | 4       | 30   |
| FakeMenuScreen.cs    | 8     | 0       | 29   |
| EscapeRouter.cs      | 4     | 1       | 29   |
| SceneTransiti...     | 4     | 0       | 29   |
| TitleCameraDrift.cs  | 4     | 0       | 29   |
| UIColors.cs          | 7     | 6       | 29   |
| VisionPotion.cs      | 6     | 0       | 29   |
| IconCaptureSe...     | 5     | 0       | 27   |
| ItemIconHelper.cs    | 4     | 0       | 27   |
| PauseInputHandler.cs | 7     | 1       | 27   |
| IconImportPos...     | 3     | 0       | 25   |
| Sapling.cs           | 4     | 0       | 25   |
| NewGameService.cs    | 3     | 0       | 25   |
| ItemRegistryE...     | 3     | 0       | 24   |
| MiningPotion.cs      | 6     | 0       | 24   |
| SpeedPotion.cs       | 6     | 0       | 24   |
| FakeMenuManager.cs   | 7     | 0       | 24   |
| SettingsEscap...     | 6     | 0       | 24   |
| SettingsRowFa...     | 4     | 0       | 23   |
| Sign.cs              | 5     | 0       | 22   |
| MenuScreenLocator.cs | 4     | 0       | 22   |
| Fire.cs              | 4     | 0       | 21   |
| LightFlicker.cs      | 3     | 0       | 21   |
| IPlayerPrefsB...     | 1     | 0       | 21   |
| IDisplayService.cs   | 1     | 0       | 20   |
| IGlowCoreObject.cs   | 4     | 4       | 19   |
| Recipe.cs            | 2     | 0       | 19   |
| FakeGameLaunc...     | 4     | 0       | 19   |
| ProgressBarUI.cs     | 2     | 0       | 19   |
| GlowCoreLevel...     | 2     | 0       | 18   |

| File                 | Blank | Comment | Code |
|----------------------|-------|---------|------|
| InventoryTests.cs    | 20    | 0       | 83   |
| UnsavedChange...     | 15    | 0       | 79   |
| DescriptorRou...     | 19    | 0       | 78   |
| CraftingStati...     | 15    | 3       | 77   |
| AudioBridgeTests.cs  | 15    | 0       | 73   |
| FeedMaterialRowUI.cs | 14    | 0       | 72   |
| AmountStepperUI.cs   | 13    | 0       | 71   |
| HotbarSlotUI.cs      | 12    | 0       | 71   |
| PlayerCamera.cs      | 17    | 17      | 70   |
| NamedMutexSin...     | 13    | 2       | 70   |
| TooltipUI.cs         | 16    | 4       | 70   |
| MenuManager.cs       | 15    | 0       | 70   |
| AxeMachineAni...     | 14    | 0       | 69   |
| TitleScreen.cs       | 15    | 0       | 69   |
| SaveDataTests.cs     | 11    | 9       | 67   |
| MenuBootstrapper.cs  | 17    | 10      | 67   |
| LowResViewport.cs    | 18    | 0       | 66   |
| SettingsEscap...     | 11    | 1       | 64   |
| SettingsRepos...     | 10    | 0       | 62   |
| PlayerHand.cs        | 9     | 0       | 61   |
| Chest.cs             | 23    | 12      | 59   |
| ConfirmDialog...     | 13    | 0       | 59   |
| ToolBehaviour.cs     | 9     | 1       | 58   |
| NewGameServic...     | 8     | 0       | 56   |
| DisplaySettin...     | 10    | 0       | 56   |
| SettingDescriptor.cs | 5     | 0       | 56   |
| ContainerOps.cs      | 9     | 0       | 54   |
| ItemRegistry.cs      | 13    | 0       | 54   |
| FogFollow.cs         | 20    | 0       | 50   |
| Replanter.cs         | 12    | 0       | 49   |
| MachineProgre...     | 8     | 0       | 49   |
| AudioMixerVol...     | 15    | 0       | 49   |
| ItemStack.cs         | 13    | 0       | 48   |
| NamedMutexSin...     | 11    | 1       | 48   |
| SliderSetting...     | 10    | 0       | 47   |
| BuildScript.cs       | 5     | 0       | 46   |
| CraftingSystem.cs    | 15    | 0       | 46   |
| SlotData.cs          | 6     | 0       | 44   |
| GameLaunchContext.cs | 10    | 0       | 44   |
| AutosaveServi...     | 17    | 0       | 43   |
| ProcessGuardTests.cs | 12    | 0       | 43   |
| MapUI.cs             | 12    | 0       | 42   |
| ProcessGuard.cs      | 7     | 0       | 42   |

| File                 | Blank | Comment | Code |
|----------------------|-------|---------|------|
| PlayerPrefsSe...     | 8     | 0       | 18   |
| Consumable.cs        | 3     | 2       | 17   |
| RecipeRowTheme.cs    | 2     | 0       | 16   |
| ItemContainer.cs     | 4     | 3       | 15   |
| Map.cs               | 3     | 0       | 15   |
| FakeSingleIns...     | 4     | 0       | 15   |
| FakePlayerPre...     | 8     | 0       | 14   |
| Readme.cs            | 2     | 0       | 14   |
| CraftingStation.cs   | 1     | 0       | 13   |
| IMenuManager.cs      | 1     | 0       | 13   |
| MenuScreenId.cs      | 0     | 0       | 13   |
| AudioBridge.cs       | 4     | 0       | 13   |
| IAudioBridge.cs      | 1     | 0       | 13   |
| ISaveDataGateway.cs  | 1     | 0       | 13   |
| ISettingsRepo...     | 0     | 0       | 13   |
| Block.cs             | 1     | 0       | 12   |
| FakeSaveDataG...     | 4     | 0       | 12   |
| ICraftingService.cs  | 3     | 3       | 11   |
| SpinningObject.cs    | 2     | 0       | 11   |
| DamageCalculator.cs  | 2     | 0       | 11   |
| IGameStateCon...     | 0     | 0       | 11   |
| RecipeList.cs        | 2     | 0       | 10   |
| IMenuScreen.cs       | 0     | 0       | 10   |
| IGameLaunchCo...     | 0     | 0       | 10   |
| Tool.cs              | 1     | 0       | 9    |
| IMenuScreenLo...     | 0     | 0       | 9    |
| ISceneTransition.cs  | 0     | 0       | 9    |
| ISingleInstan...     | 1     | 0       | 9    |
| NullSingleIns...     | 0     | 0       | 9    |
| ISettingsCategory.cs | 1     | 0       | 9    |
| SettingContro...     | 0     | 0       | 9    |
| GameLaunchMode.cs    | 0     | 0       | 8    |
| IEscapeConsumer.cs   | 0     | 0       | 8    |
| INewGameService.cs   | 0     | 0       | 8    |
| ISettingControl.cs   | 0     | 0       | 8    |
| TreeNode.cs          | 1     | 0       | 7    |
| IMenuArgsReceiver.cs | 0     | 0       | 7    |
| NodeActionChild.cs   | 1     | 0       | 6    |
| SaveService.cs       | 1     | 0       | 6    |
| IHandleItem.cs       | 1     | 0       | 5    |
| ISaveService.cs      | 0     | 0       | 5    |
| IWeapon.cs           | 0     | 0       | 5    |
| IPlayerInvent...     | 0     | 0       | 4    |

## 5.4.2 Test Coverage

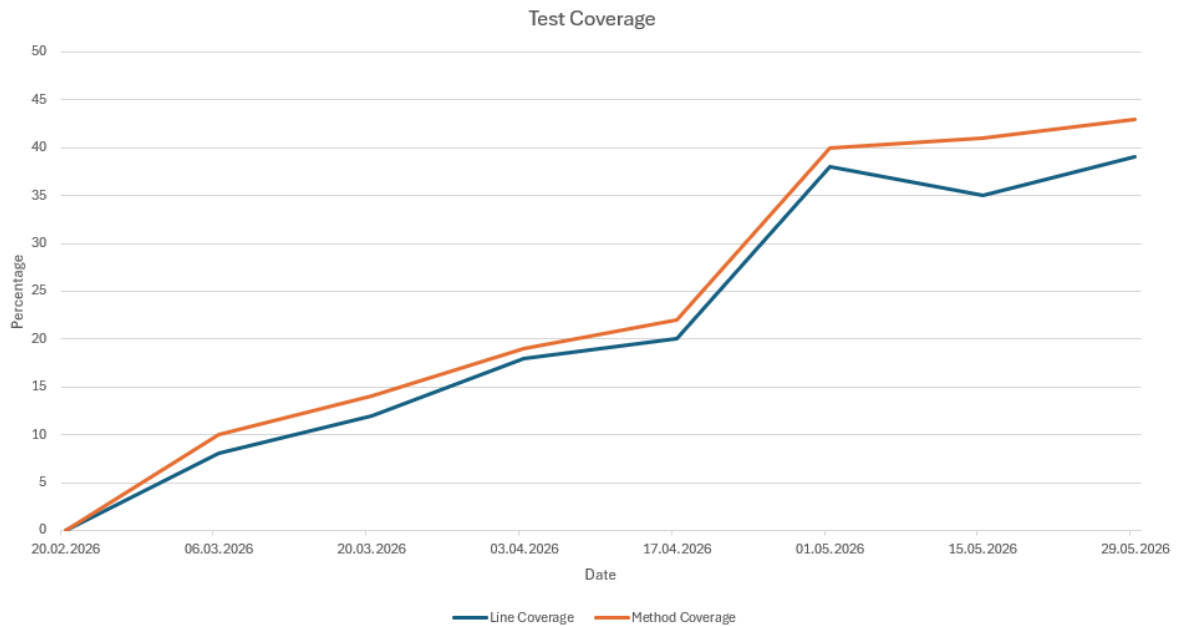


Image 27: Test Coverage Chart

## 5.5 User Testing

In game development, the most important aspect of the product is that it is entertaining. As developers, we are not representative users for playtesting, since we have spent many hours developing the features and already know how everything works. A first-time player may not immediately understand game rules that are obvious to us. This strongly relates to our usability NFRs, which require the game to be intuitive which is a prerequisite for it to be enjoyable. This is the main purpose of the user tests of GlowCore.

During testing, the participant is given the game on a device with a keyboard and mouse, where no further instructions are given beyond what is provided in the actual game. They are then asked to play the game while commenting on what they are currently doing. At the end, the participant answers a set of questions to evaluate the corresponding NFRs. The results of the NFRs are documented in their chapter.

### 5.5.1 User Test 1 (UT01)

- **Participant:** omega-800
- **Gaming Experience / Similar Games:** Has played Factorio
- **Date:** 28.04.2026
- **Tested release:** v0.2.0

| Field                                                                                              | Result                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR201</b> Player Restriction Visibility. Did you understand the visual clues?                  | Yes, I especially liked the red and green clues when building.                                                                                          |
| <b>NFR202</b> Light Upgrade Progress Indication. Did you understand what causes the map to expand? | Yes, when I feed wood into the fire it grows, and the darkness goes away.                                                                               |
| <b>NFR203</b> New Player Learnability. How long did it take for the first upgrade?                 | 2 minutes.                                                                                                                                              |
| What was confusing?                                                                                | I couldn't move the camera vertically.                                                                                                                  |
| Did you have fun? What did you like?                                                               | Yes, I enjoyed playing it. I liked the visual art style with the outlines.                                                                              |
| Was the pacing good, too slow or too fast?                                                         | The pacing was good. I had fun and felt motivated to progress. I got bored at level 11 because there were no better tools to craft, nothing to aim for. |

**Table 39:** UT01

#### 5.5.1.1 Additional Feedback

| Topic         | Explanation                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GlowCore UI   | He didn't notice that he had to click "Upgrade To Level 2" to progress. At level 14 it was not obvious that he had to scroll down to add coal.              |
| Signs         | At first he did not read the second sign, assuming that all signs display the same text.                                                                    |
| Node hovering | It should be possible to select a Node located behind the player (the raycast should go through the player).                                                |
| Hotbar        | He suggested allowing the player to scroll through the hotbar using the mouse wheel.                                                                        |
| Building      | He stopped the main task of progressing to build something. This is considered positive, because the game is intended to be open-world to a certain degree. |

**Table 40:** UT01 - Additional Feedback

#### 5.5.1.2 Conclusion

The participant was mainly motivated to progress because he wanted to craft better tools in order to break things faster to progress faster creating a continuous progression loop. This loop is called the core game loop of a game and this is exactly what we wanted to accomplish and what makes the game entertaining ultimately. This user test confirms that the core game loop is engaging.

Based on the feedback we will make improvements to the UI, in particular we will change the GlowCore upgrade UI to be more intuitive.



## 5.5.2 User Test 2 (UT02)

- **Participant:** Anonymous
- **Gaming Experience / Similar Games:** Has played various game genres including automation games.
- **Date:** 30.04.2026
- **Tested release:** v0.2.0

| Field                                                                                              | Result                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR201</b> Player Restriction Visibility. Did you understand the visual clues?                  | Yes, the outlines were obvious.                                                                                                                           |
| <b>NFR202</b> Light Upgrade Progress Indication. Did you understand what causes the map to expand? | Yes, when I upgrade the fire.                                                                                                                             |
| <b>NFR203</b> New Player Learnability. How long did it take for the first upgrade?                 | 3 minutes.                                                                                                                                                |
| What was confusing?                                                                                | The border visual is strange.                                                                                                                             |
| Did you have fun? What did you like?                                                               | It got boring fast, because it's very repetitive. I liked how the crafting table looked. The proportions of the stone-fire and the player were a bit off. |
| Was the pacing good, too slow or too fast?                                                         | The pacing was okay, but I think you should get more wood out of trees.                                                                                   |

**Table 41:** UT02

### 5.5.2.1 Additional Feedback

| Topic             | Explanation                                                                               |
|-------------------|-------------------------------------------------------------------------------------------|
| Gameplay          | The game has potential but there are not enough things to do, not enough content.         |
| Node break prompt | The break prompt should only be displayed for the first few Nodes that the player breaks. |
| Camera            | He suggested that you should be able to zoom in and out a bit with the mouse wheel.       |
| Building bug      | When you hold a building block and click somewhere in the UI it still builds.             |

**Table 42:** UT02 - Additional Feedback

### 5.5.2.2 Conclusion

This user test provided valuable feedback on what the gameplay is missing and what it is already doing right. In Sprint 6 we will refine the UI and controls accordingly. We will also add more Items, Nodes and Crafting Recipes. Thanks to our extensible architecture which follows the Open-Closed Principle (OCP), new content can be added without modifying existing code.

## 5.6 Coding Guidelines

Our coding guidelines are based on the Unity C# scripting conventions and are enforced through Roslyn analyzers and EditorConfig rules. Every C# file must conform to these rules, which are checked automatically on every push to main or dev via the CI pipeline.

### 5.6.1 Naming Conventions

| Element                      | Convention                             | Example                        |
|------------------------------|----------------------------------------|--------------------------------|
| Classes / Interfaces         | PascalCase; interfaces prefixed with I | PlayerController, ICollectible |
| Methods                      | PascalCase                             | MovePlayer(), TakeDamage()     |
| Bool methods                 | Question-word prefix                   | IsAlive(), CanCollect()        |
| Parameters / Local variables | camelCase                              | playerSpeed, itemCount         |
| Constants (const)            | k prefix + PascalCase                  | kMaxHealth, kGravity           |
| Private fields               | m_ prefix + camelCase                  | m_playerHealth, m_enemyCount   |
| Public fields                | PascalCase (avoid in classes)          | PlayerHealth                   |
| Properties                   | PascalCase                             | Health, Score                  |
| Static fields                | s_ prefix + camelCase                  | s_instanceCount                |
| Events                       | On prefix + PascalCase                 | OnPlayerDeath, OnItemCollected |
| Enums / Values               | PascalCase                             | PlayerState.Idle               |
| Namespaces                   | PascalCase, project-based              | GlowCore.Player                |
| Generic type parameters      | T prefix + PascalCase                  | TItem, TState                  |

### 5.6.2 Class Member Ordering

Members within a class must appear in the following order:

1. Constants
2. Static fields
3. Instance fields
4. Constructors
5. Properties
6. Public methods
7. Private methods

### 5.6.3 General Guidelines

**Properties:** For get-only properties, prefer expression-bodied members (e.g. `public int Health => m_health;`). Simple get/set properties may also use expression bodies when the logic fits on one line.

**Method parameters:** Avoid `in` parameters. The compiler may silently create defensive copies, which is a hidden performance cost.

**Access modifiers:** Always declare access modifiers explicitly; never rely on C# defaults.

**Public fields:** Avoid public fields in classes, use properties instead. Public fields in structs are acceptable.

## 5.6.4 Formatting Rules

These rules are enforced by `.editorconfig` and verified by `dotnet format` in CI. A build with formatting violations is treated as a failure.

**Line endings:** LF (`\n`) only. CRLF is not permitted in any file.

**Indentation:** 4 spaces for C# files; 2 spaces for XML/JSON/config files. Tabs are never used.

**Brace style (Allman):** Opening braces always appear on their own line for all constructs (`if`, `for`, method bodies, class bodies, etc.).

**Single-line bodies:** `if`, `for`, `while`, and similar statements with a single-line body omit braces; the body is placed on the next indented line.

**this. qualifier:** Do not qualify field or member access with `this.` unless it is required to resolve an ambiguity.

**Language keywords over BCL types:** Use `int`, `string`, `bool`, etc. instead of `Int32`, `String`, `Boolean`.

**var keyword:** Use `var` for built-in types and whenever the type is apparent from the right-hand side of an assignment.

**using directives:** Placed outside the namespace declaration. `System.*` namespaces are sorted first.

**Modifier order:** `public/private/protected/internal > static > extern/new/virtual/abstract/sealed/override > readonly > unsafe/volatile/async`.

**Null checks:** Prefer null-coalescing (`??`) and null-conditional (`?.`) operators over explicit null comparisons where possible.

## 5.7 Version Control Workflow

The project uses Git with a two-tier branching model:

- **main:** stable production branch. Only receives merges from `dev` after a release is considered stable.
- **dev:** integration branch. All feature work is merged here first and must pass CI before it is promoted to `main`.

### 5.7.0.1 Branch Naming

Every branch must be linked to a Jira issue. Branch names follow the pattern:

`<type>/gc-<issue-id>-short-description`

Where `<type>` is one of:

| Type    | When to use                                        |
|---------|----------------------------------------------------|
| feature | New features or significant enhancements           |
| bug     | Bug fixes                                          |
| docs    | Documentation updates                              |
| task    | General tasks that do not fit the above categories |

### 5.7.0.2 Rebasing Workflow

We follow a **rebasing workflow**: before opening a pull request, the feature branch is rebased onto the current tip of `dev`. This keeps the project history linear and avoids unnecessary merge commits.

If a branch accumulates many small or intermediate commits that do not add meaningful history (e.g. “fix typo”, “WIP”), the branch is squashed before or during merge to keep the log clean.

### 5.7.0.3 Pull Requests

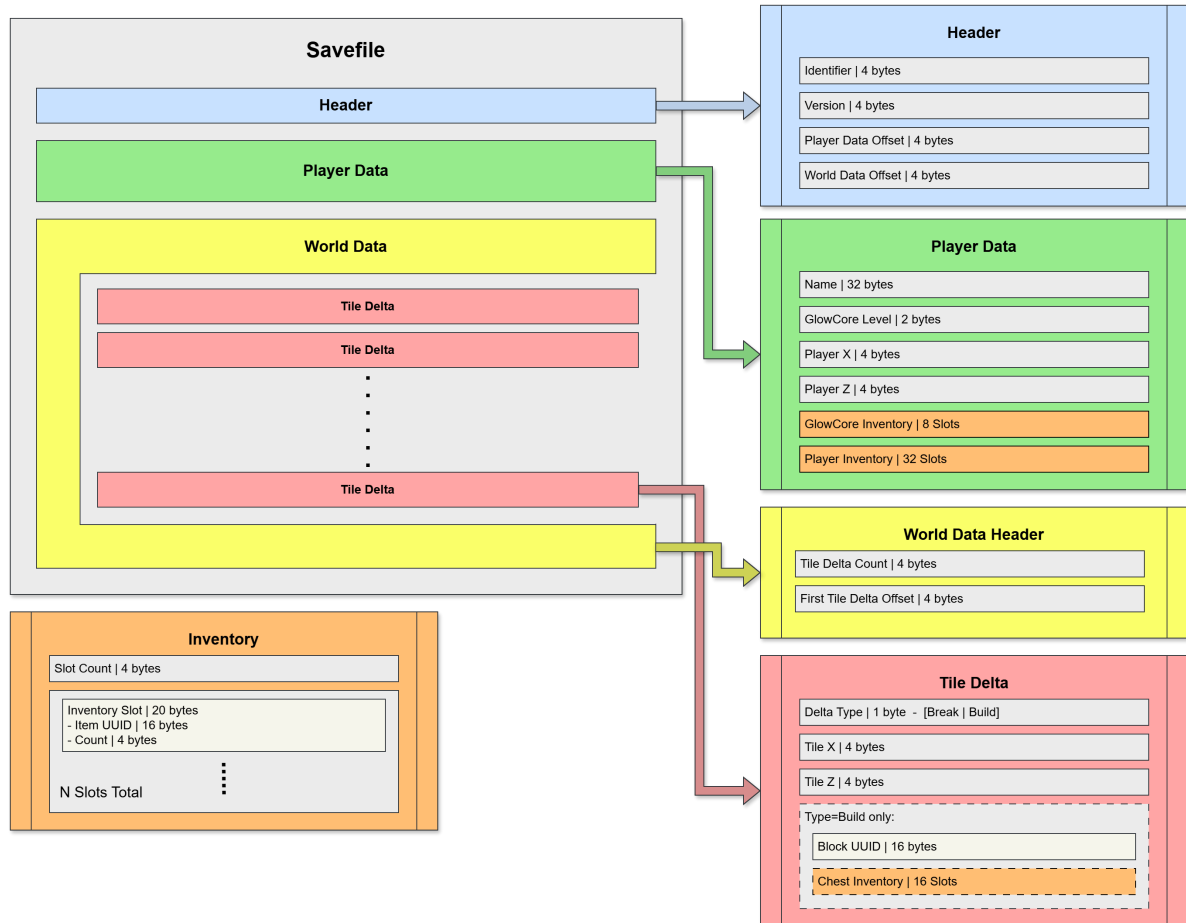
Pull requests always target `dev`. Each PR must:

- Reference the corresponding Jira issue in its description.
- Receive **at least one approving review** from a team member before it can be merged.
- Pass all CI checks (linting, formatting, automated tests, and build).

No direct pushes to `dev` or `main` are permitted.

# 6 Savefile

The game periodically saves the player's progress in a savefile. This file contains all the necessary information to restore the game state when the player loads it again. It is stored as a binary file on the player's device. The following diagram illustrates the structure of the savefile:



**Image 28:** Savefile Structure

The savefile is organized into several sections, each containing specific types of data:

- **Header:** Contains metadata about the savefile, such as an identifier, version number, and offsets to subsequent sections.
- **Player Data:** Stores information about the player itself, including name, GlowCore level, inventory, etc.
- **World Data:** Contains information about the game world itself, such as where nodes have been placed or destroyed.

The world data is stored as a list of deltas, instead of a full snapshot of the world state. Each delta represents a change to the world from its initial state, such as placing or destroying a node. This approach allows for more efficient storage and faster loading times, as only the changes need to be recorded rather than the entire world state.

## **Part III**

# **Project Documentation**

# 7 Initial Project Proposal

**Project name:** GlowCore

**Date:** 19.12.2025

## 7.1 Team Members

1. Cedric Cathomas ( [Cedric.Cathomas@ost.ch](mailto:Cedric.Cathomas@ost.ch) )
2. Dominik Wyss ( [Dominik.Wyss1@ost.ch](mailto:Dominik.Wyss1@ost.ch) )
3. Nathanael Fässler ( [Nathanael.Faessler@ost.ch](mailto:Nathanael.Faessler@ost.ch) )
4. Yoris Kucera ( [Yoris.Kucera@ost.ch](mailto:Yoris.Kucera@ost.ch) )

## 7.2 Availabilities

| Time slot   | Mon  | Tue  | Wed | Thu | Fri  |
|-------------|------|------|-----|-----|------|
| 08h00–09h00 | -    | (XR) | -   | -   | (XO) |
| 09h00–10h00 | -    | (XR) | -   | -   | (XO) |
| 10h00–11h00 | (XR) | -    | -   | -   | (XO) |
| 11h00–12h00 | (XR) | -    | -   | -   | (XO) |
| 12h00–13h00 | -    | -    | -   | -   | (XO) |
| 13h00–14h00 | XR   | -    | -   | -   | (XO) |
| 14h00–15h00 | XR   | -    | -   | -   | (XO) |
| 15h00–16h00 | (XR) | XR   | -   | -   | (XO) |
| 16h00–17h00 | (XR) | XR   | -   | -   | (XO) |
| 17h00–18h00 | -    | (XR) | -   | -   | (XO) |
| 18h00–19h00 | -    | (XR) | -   | -   | -    |

**Table 43:** Team availability for review meetings

## 7.3 Vision

GlowCore is a singleplayer progression game focused on resource gathering and gradual world expansion. At the start, the player can only access a small, limited area of the game world, while the rest remains hidden and unreachable. At the center of the world is a special entity that can be upgraded using collected materials. Each upgrade expands the accessible area, revealing new parts of the world and enabling further progression. The core gameplay loop is built around exploring newly unlocked areas, gathering resources, and using them to continuously improve the central entity and push the boundaries of the world outward.

As the world expands, additional gameplay systems deepen the experience. The player can obtain and use different materials, interact with tools, craft items, and face enemies within the expanding world. More advanced mechanics such as machines for processing resources, farming systems, and specialized equipment can be introduced over time to increase complexity and strategic depth.

## 7.4 Proposed Realisation

The project will be implemented using Unity as the primary game engine, using its integrated systems for rendering, physics, input handling, animation, audio, and cross-platform deployment. Development will be done in C# using Unity's component-based architecture (GameObjects, MonoBehaviours, and ScriptableObjects) to structure gameplay logic and systems. Unity's built-in physics engine and animation system (Animator) will be used to realise core gameplay and user interaction. Visual assets will be rendered using Unity's Universal Render Pipeline (URP). Individual systems will be tested for robustness using unit tests by means of Unity's own testing framework, based on NUnit. Version control will be handled with Git. The project will be developed within the Unity Editor.

# 8 Project Plan

## 8.1 Resources

### 8.1.1 People

The project team consists of 4 students. Each member contributes approximately 8.6 hours per week and is responsible for tasks of their defined roles.

#### 8.1.1.1 Project Management Roles

| Role            | Description                                                                                                                                                          | Names             |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Product Owner   | <ul style="list-style-type: none"> <li>• Define product vision</li> <li>• Manage backlog</li> <li>• Prioritize features</li> <li>• Validate requirements</li> </ul>  | Nathanael Fässler |
| Scrum Master    | <ul style="list-style-type: none"> <li>• Facilitate meetings</li> <li>• Remove blockers</li> <li>• Ensure Scrum process</li> <li>• Support team workflow</li> </ul>  | Dominik Wyss      |
| Project Manager | <ul style="list-style-type: none"> <li>• Schedule planning</li> <li>• Resource management</li> <li>• Risk control</li> <li>• Progress reporting</li> </ul>           | Cedric Cathomas   |
| DevOps Engineer | <ul style="list-style-type: none"> <li>• CI/CD maintenance</li> <li>• Build automation</li> <li>• Deployment support</li> <li>• Infrastructure management</li> </ul> | Yoris Kucera      |

**Table 44:** Project Management Roles



### 8.1.1.2 Developer Roles

| Role                     | Description                                                                                                                                                                                  | Names                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Lead Developer           | <ul style="list-style-type: none"> <li>• Overall technical coordination</li> <li>• Code quality control</li> <li>• Architecture decisions</li> <li>• Integration management</li> </ul>       | Yoris Kucera                                                         |
| Developer                | <ul style="list-style-type: none"> <li>• Gameplay implementation</li> <li>• Bug fixing</li> <li>• Feature development</li> <li>• System integration</li> <li>• Writing Test Cases</li> </ul> | Dominik Wyss<br>Yoris Kucera<br>Nathanael Fässler<br>Cedric Cathomas |
| Art Director / 3D Artist | <ul style="list-style-type: none"> <li>• Asset creation</li> <li>• Scene design</li> <li>• Visual consistency</li> <li>• Asset optimization</li> </ul>                                       | Nathanael Fässler                                                    |
| UI Architect             | <ul style="list-style-type: none"> <li>• UI/UX design</li> <li>• Menu systems</li> <li>• HUD implementation</li> <li>• User interaction flow</li> </ul>                                      | Cedric Cathomas                                                      |
| Graphics Programmer      | <ul style="list-style-type: none"> <li>• Shader development</li> <li>• Particle effects</li> <li>• Performance optimization</li> <li>• Visual effects integration</li> </ul>                 | Yoris Kucera                                                         |
| Audio Engineer           | <ul style="list-style-type: none"> <li>• Sound design</li> <li>• Audio integration</li> <li>• Music management</li> <li>• Audio balancing</li> </ul>                                         | Dominik Wyss                                                         |
| Level Designer           | <ul style="list-style-type: none"> <li>• Level layout</li> <li>• Gameplay balancing</li> <li>• Difficulty tuning</li> <li>• Playtesting feedback</li> </ul>                                  | Nathanael Fässler<br>Cedric Cathomas                                 |

**Table 45:** Developer Roles

## 8.1.2 Time

The project runs from 16.02.2026 to 12.06.2026 with an estimated effort of 120 hours per person, resulting in a total workload of 480 hours, structured in milestones and sprints.

## 8.1.3 Cost

Most required tools and platforms are freely available through free or student licenses. We plan on publishing GlowCore on Steam for which Steam charges a 100 US-Dollar publishing fee. The cost will be shared by the members.

## 8.2 Organization

### 8.2.1 Meetings

The project meetings follow mainly the Scrum standard, with weekly Scrum meetings replacing daily stand-ups and biweekly sprint planning including retrospectives. We also hold regular advisor meetings to present and discuss our results so far.

| Type                    | Todos                                                                                                                                                                                                       | Frequency                                              |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Sprint Review (Advisor) | <ul style="list-style-type: none"> <li>• Present sprint results</li> <li>• Demonstrate features</li> <li>• Discuss open issues</li> <li>• Collect advisor feedback</li> <li>• Document decisions</li> </ul> | Every 2 weeks<br>Tuesday 15:00 - 16:00                 |
| Weekly Scrum            | <ul style="list-style-type: none"> <li>• Report progress</li> <li>• Define next steps</li> <li>• Identify blockers</li> <li>• Update tasks</li> <li>• Support team members</li> </ul>                       | Weekly<br>Friday 14:00 - 15:00                         |
| Sprint Retrospective    | <ul style="list-style-type: none"> <li>• Analyze sprint process</li> <li>• Identify problems</li> <li>• Define improvements</li> <li>• Document lessons learned</li> </ul>                                  | Every 2 weeks<br>Friday 15:00 - 16:00                  |
| Sprint Planning         | <ul style="list-style-type: none"> <li>• Define sprint goals</li> <li>• Select backlog items</li> <li>• Estimate effort</li> <li>• Assign tasks</li> <li>• Update schedule</li> </ul>                       | Every 2 weeks<br>Friday 16:00 - 17:00<br>(after retro) |

**Table 46:** Planned Meetings Overview

## 8.2.2 Planning Tools

| Tool            | Use Case                                                                                                                                                                      |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GitHub          | <ul style="list-style-type: none"> <li>• Version control</li> <li>• GitHub Actions (CI/CD)</li> <li>• Pull requests</li> <li>• GitHub notifications via Discord</li> </ul>    |
| Typst           | <ul style="list-style-type: none"> <li>• Project documentation</li> <li>• PDF generation</li> </ul>                                                                           |
| Microsoft Teams | <ul style="list-style-type: none"> <li>• File sharing</li> <li>• Communication</li> </ul>                                                                                     |
| Jira            | <ul style="list-style-type: none"> <li>• Task management</li> <li>• Sprint planning</li> <li>• Backlog tracking</li> <li>• Issue tracking</li> <li>• Time tracking</li> </ul> |
| Miro            | <ul style="list-style-type: none"> <li>• Brainstorming</li> <li>• Diagrams</li> <li>• Project planning</li> </ul>                                                             |
| Discord         | <ul style="list-style-type: none"> <li>• Online meetings</li> <li>• Informal communication</li> <li>• Quick coordination</li> <li>• GitHub activity notifications</li> </ul>  |

**Table 47:** Planning Tools

## 8.3 Schedule

The project follows the OST internal SCRUM+ framework for its overall organization. SCRUM+ combines elements of Scrum with the Rational Unified Process (RUP). Long-term planning is based on the RUP approach, while short-term development is organized using Scrum with two-week iterations.

At the start of each iteration, we estimate all tickets with storypoints. The story **GC-54 Make Player** is the reference story which is worth 5 storypoints.

### 8.3.1 Work Items

We work with 5 types of Work Items

| Name          | Description                                                            |
|---------------|------------------------------------------------------------------------|
| Epic          | A group of features that we want to implement.                         |
| Story         | Describes a feature that is directly related to the game               |
| Task          | Everything that isn't directly related to the game                     |
| Bug           | Bugs are created when a bug is found and fixed in an unplanned matter. |
| Documentation | Work related to the documentation                                      |

**Table 48:** Work Items

## 8.3.2 Long Term Plan

The Long Term Planning is separated into two categories:

- Milestones: Serve as checkpoint for releases and review meetings.
- Epics: Provide the structure to organize the stories.

### 8.3.2.1 Milestones

The milestones for the review meetings and for the releases are interleaved and independent.

In agile project management it is common to reevaluate the Long Term Plan during the project. For this reason there are two Milestone Tables. The first table shows the Initial Milestones that were defined at the beginning of the project and the second table contains updated Milestones as of 07.04.2026.

#### 8.3.2.1.1 Initial Milestones

| Nr  | Milestone Name                        | Date       | Description                                                                                                                                                                                                            |
|-----|---------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M01 | Review 1                              | 10.03.2026 | Initial Project Setup                                                                                                                                                                                                  |
| M02 | Review 2                              | 24.03.2026 | Requirements                                                                                                                                                                                                           |
| M03 | Review 3                              | 14.04.2026 | End of Elaboration                                                                                                                                                                                                     |
| M04 | Review 4                              | 28.04.2026 | Quality                                                                                                                                                                                                                |
| M05 | Review 5                              | 12.05.2026 | Architecture                                                                                                                                                                                                           |
| M06 | Project Presentation                  | 12.06.2026 | Final Presentation to SEProject-Advisors and fellow students. Completion of SEProject.                                                                                                                                 |
| M07 | Pre-Alpha Release                     | 20.03.2026 | First playable prototype is ready. Core gameplay loops are implemented. The player can chop trees, pick up wood and expand the GlowCore.                                                                               |
| M08 | Alpha Release                         | 10.04.2026 | MVP is done and working. Core gameplay Mechanics are implemented. Player can craft, build, expand the map and he has an inventory. The player can save his savestate.                                                  |
| M09 | Beta Release                          | 15.05.2026 | More Biomes and Materials are added. There is a combat system with enemies and peaceful mobs.<br>There is a title screen and a settings menu.<br>Automation mechanics are implemented. There are more GlowCore Phases. |
| M10 | Official Release and Final Submission | 05.06.2026 | The game is polished and has no major bugs. The game has a beginning and an ending.                                                                                                                                    |

**Table 49:** Initial Milestones

### 8.3.2.1.2 Updated Milestones as of 07.04.2026

Milestone M09 - Beta Release was updated and the combat system and the automation mechanics were removed. The reduction in scope allows the team to deliver a stable and more polished game at the end. The savestate system was moved from Alpha to Beta. We prioritize quality over quantity. If there is enough time at the end of the project, we can still implement the features as a bonus.

| Nr  | Milestone Name                        | Date       | Description                                                                                                                                          |
|-----|---------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| M01 | Review 1                              | 10.03.2026 | Initial Project Setup                                                                                                                                |
| M02 | Review 2                              | 24.03.2026 | Requirements                                                                                                                                         |
| M03 | Review 3                              | 14.04.2026 | End of Elaboration                                                                                                                                   |
| M04 | Review 4                              | 28.04.2026 | Quality                                                                                                                                              |
| M05 | Review 5                              | 12.05.2026 | Architecture                                                                                                                                         |
| M06 | Project Presentation                  | 12.06.2026 | Final Presentation to SEProject-Advisors and fellow students. Completion of SEProject.                                                               |
| M07 | Pre-Alpha Release                     | 20.03.2026 | First playable prototype is ready. Core gameplay loops are implemented. The player can chop trees, pick up wood and expand the GlowCore.             |
| M08 | Alpha Release                         | 10.04.2026 | MVP is done and working. Core gameplay Mechanics are implemented. Player can craft, build, expand the map and he has an inventory.                   |
| M09 | Beta Release                          | 15.05.2026 | More Biomes and Materials are added. There is a title screen and a settings menu. There are more GlowCore Phases. The player can save his savestate. |
| M10 | Official Release and Final Submission | 05.06.2026 | The game is polished and has no major bugs. The game has a beginning and an ending.                                                                  |

**Table 50:** Updated Milestones

### 8.3.2.2 Epics

| Epic Name                     | Description/Requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documentation                 | This epic includes all documentation related work.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Project Setup                 | This epic contains all work that is done in the Inception-Phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Core Gameplay Loop Mechanics  | <ul style="list-style-type: none"> <li>The player can chop down trees, collect wood and feed it into the GlowCore.</li> <li>The GlowCore levels up and expands the map.</li> <li>The player is restricted to the unlocked map. The grid system is implemented.</li> </ul>                                                                                                                                                                                                                                                                                              |
| Inventory, Crafting, Building | <ul style="list-style-type: none"> <li>Inventory: The player has an inventory and can store a limited amount of items. He can have items in his hands and in his backpack(or similar).</li> <li>Crafting: The player can craft items to other items on different workstations.</li> <li>Building: The player can place objects onto the world-grid.</li> </ul>                                                                                                                                                                                                         |
| Graphics                      | <ul style="list-style-type: none"> <li>The Artstyle of the game is set.</li> <li>The 3D Models follow a consistent Artstyle.</li> <li>A Shader is implemented.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              |
| Game Framework & Completeness | <p>This epic contains everything that makes a game complete and feel like a game.</p> <ul style="list-style-type: none"> <li>Intro: When the Game starts the GlowCore Logo is displayed. After the title screen a Cutscene is played that leads into the game.</li> <li>Ending: An ending of the game is defined. When the player finishes the game, a Cutscene is played and the credits roll.</li> <li>UI: A Title Screen is available. In-Game there is a Debug Screen, and a Settings Menu.</li> <li>Gamedata Saving: The player can save his progress.</li> </ul> |
| Big World & GlowCore Phases   | <ul style="list-style-type: none"> <li>World building: The game has a complete world to play in.</li> <li>GlowCore Phases: The GlowCore has multiple phases, all of which need different materials to level up. The GlowCore changes visually with every Level.</li> </ul>                                                                                                                                                                                                                                                                                             |
| Biomes & Materials            | <ul style="list-style-type: none"> <li>The world has multiple biomes: e.g. Forest, Grassland, Mountain, Volcano, Jungle etc.</li> <li>Each biome has unique resources.</li> <li>There are Caves which you can enter.</li> </ul>                                                                                                                                                                                                                                                                                                                                        |
| Combat                        | <ul style="list-style-type: none"> <li>Enemies can spawn and fight with the player.</li> <li>The player can take damage and can die.</li> <li>A respawn mechanic is set up.</li> <li>There are peaceful mobs.</li> </ul>                                                                                                                                                                                                                                                                                                                                               |
| Resource Automation           | <ul style="list-style-type: none"> <li>The Player can replant trees.</li> <li>The Player can craft machines that automatically gather resources.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                            |
| Playtesting & Release         | <ul style="list-style-type: none"> <li>Playtesting: The game is tested by players that play it for the first time (blind test).</li> <li>Release: The game is released on Steam on its own Steam page and has a trailer, screenshots and a description.</li> </ul>                                                                                                                                                                                                                                                                                                     |

**Table 51: Epics**

The following image shows the Timeline in Jira as of 04.03.2026.

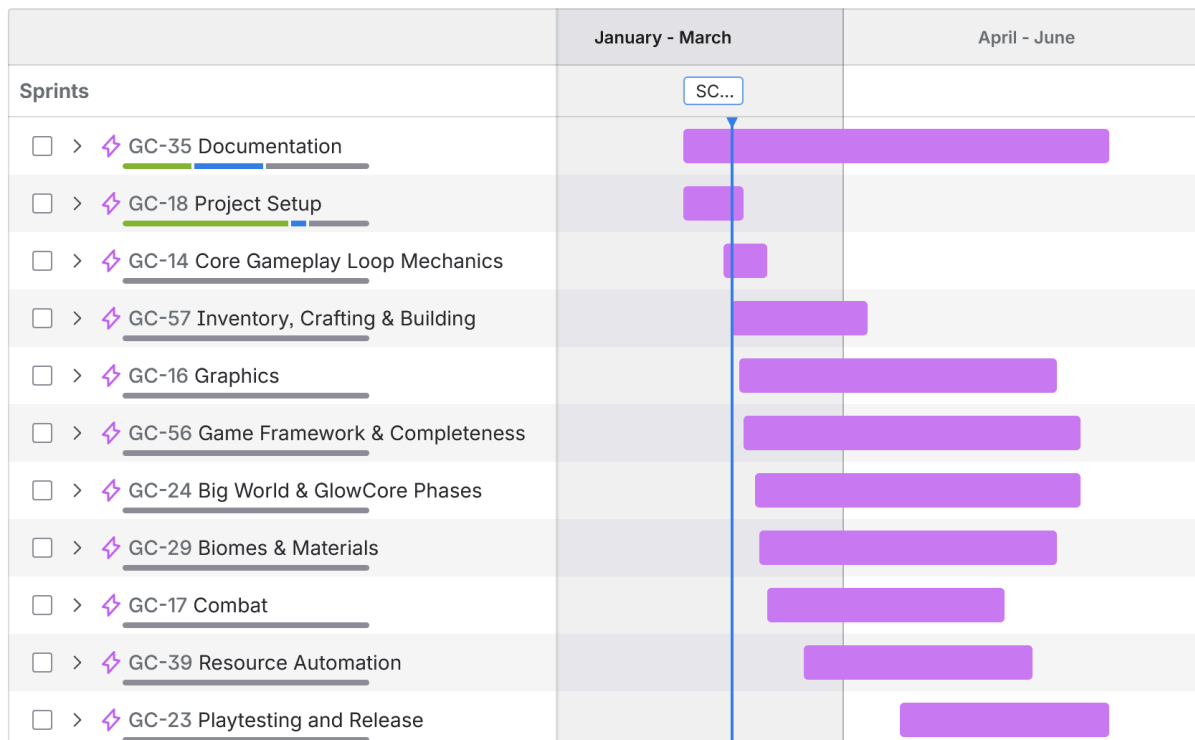


Image 29: Initial Jira Timeline

The Long Term Plan was updated on 07.04.2026. The following image shows the new Timeline in Jira as of 07.04.2026.

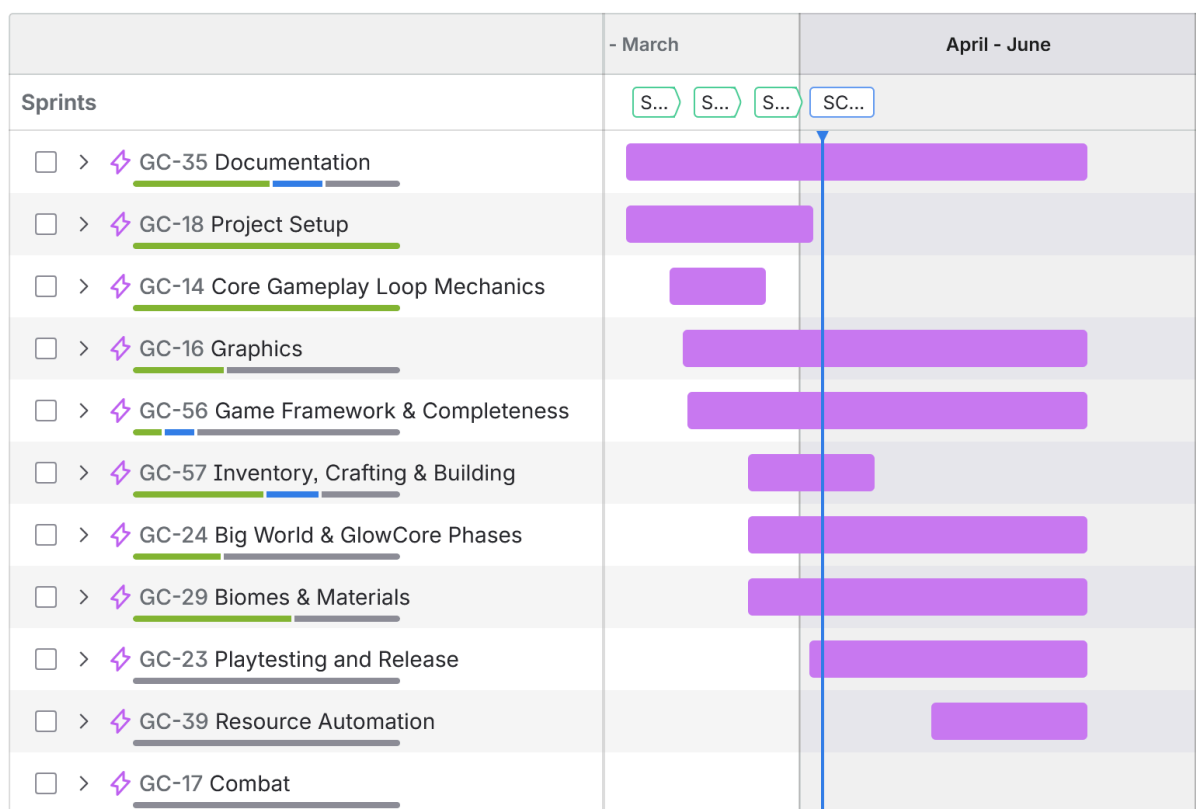


Image 30: Updated Jira Timeline

### 8.3.2.3 Initial Backlog & MVP

The priority of Stories is defined by the ordering in the backlog. As we progress in Agile, we may divide Stories further and add new Stories. The MVP is reached when every Story up to and including **GC-101 Build World** is completed.

The following is the Initial Backlog as of 04.03.2026.

▼

Backlog

(44 work items)

0

0

0

Create sprint

|  |        |                                |                      |         |   |  |
|--|--------|--------------------------------|----------------------|---------|---|--|
|  | GC-54  | There is a player              | CORE GAMEPLAY LO...  | TO DO ▼ | - |  |
|  | GC-78  | Create Tree-Node 3D Model      | GRAPHICS             | TO DO ▼ | - |  |
|  | GC-95  | Create Wood-Item 3D Model      | GRAPHICS             | TO DO ▼ | - |  |
|  | GC-47  | Trees can be chopped           | CORE GAMEPLAY LO...  | TO DO ▼ | - |  |
|  | GC-109 | Make GlowCore Phase 1 3D Model | GRAPHICS             | TO DO ▼ | - |  |
|  | GC-46  | GlowCore expands World         | CORE GAMEPLAY LO...  | TO DO ▼ | - |  |
|  | GC-69  | Make Inventory-System          | INVENTORY, CRAFTI... | TO DO ▼ | - |  |
|  | GC-74  | Save Game-Data                 | GAME FRAMEWORK &...  | TO DO ▼ | - |  |
|  | GC-108 | Create Axe 3D Model            | GRAPHICS             | TO DO ▼ | - |  |
|  | GC-97  | Create Coal 3D Model           | GRAPHICS             | TO DO ▼ | - |  |
|  | GC-106 | Add Coal Node and Item         | BIOMES & MATERIALS   | TO DO ▼ | - |  |
|  | GC-96  | Create Stone 3D Model          | GRAPHICS             | TO DO ▼ | - |  |
|  | GC-93  | Add Stone Node and Item        | BIOMES & MATERIALS   | TO DO ▼ | - |  |
|  | GC-107 | Make Pickaxe 3D Model          | GRAPHICS             | TO DO ▼ | - |  |
|  | GC-102 | Add Pickaxe                    | BIOMES & MATERIALS   | TO DO ▼ | - |  |
|  | GC-79  | Add Grassland Biome            | BIOMES & MATERIALS   | TO DO ▼ | - |  |
|  | GC-111 | Make GlowCore Phase 2 3D Model | GRAPHICS             | TO DO ▼ | - |  |
|  | GC-100 | Add GlowCore Phase 2           | BIG WORLD & GLOWC... | TO DO ▼ | - |  |
|  | GC-71  | Make Crafting-System           | INVENTORY, CRAFTI... | TO DO ▼ | - |  |
|  | GC-70  | Make Building-System           | INVENTORY, CRAFTI... | TO DO ▼ | - |  |
|  | GC-77  | Implement Pixelize-Shader      | GRAPHICS             | TO DO ▼ | - |  |
|  | GC-103 | Add Chest                      | INVENTORY, CRAFTI... | TO DO ▼ | - |  |
|  | GC-101 | Build World                    | BIG WORLD & GLOWC... | TO DO ▼ | - |  |
|  | GC-99  | Add Rock Biome                 | BIOMES & MATERIALS   | TO DO ▼ | - |  |
|  | GC-72  | Game Intro                     | GAME FRAMEWORK &...  | TO DO ▼ | - |  |

GC-73

Game Ending

GAME FRAMEWORK &...

TO DO ▼

-

GC-105

Add Combat Logic

COMBAT

TO DO ▼

-

GC-124

Create Enemy 3D Model

GRAPHICS

TO DO ▼

-

GC-81

Add Enemy

COMBAT

TO DO ▼

-

GC-75

Ingame Debug Settings

GAME FRAMEWORK &...

TO DO ▼

-

GC-76

Player Settings-Menu

GAME FRAMEWORK &...

TO DO ▼

-

GC-125

Create Peaceful-Mob 3D Model

GRAPHICS

TO DO ▼

-

GC-82

Add peaceful Mob

COMBAT

TO DO ▼

-

GC-123

Make Sword 3D Model

GRAPHICS

TO DO ▼

-

GC-83

Add Sword

COMBAT

TO DO ▼

-

GC-120

Make Axe-Machine 3D Model

GRAPHICS

TO DO ▼

-

GC-114

Add Axe-Machine

RESOURCE AUTOMAT...

TO DO ▼

-

GC-121

Make Replant-Machine 3D Model

GRAPHICS

TO DO ▼

-

GC-115

Add Replant-Machine

RESOURCE AUTOMAT...

TO DO ▼

-

GC-122

Make Rotater-Block 3D Model

GRAPHICS

TO DO ▼

-

GC-116

Add Rotater-Block

RESOURCE AUTOMAT...

TO DO ▼

-

GC-117

Conduct Playtesting with User

PLAYTESTING AND R...

TO DO ▼

-

GC-118

Release Game on Steam

PLAYTESTING AND R...

TO DO ▼

-

GC-119

Make Marketing Material for Steam Page

PLAYTESTING AND R...

TO DO ▼

-

Image 31: Initial Backlog



### 8.3.3 Short Term Plan

In a Sprint the priority of the tasks and stories is defined by the ordering.

#### 8.3.3.1 Sprint 1 (Sprint Zero) 20.02.2026 - 06.03.2026

The following shows the Jira sprint backlog of the first sprint. Since this was Sprint Zero, not all guidelines were followed initially because they had not yet been established.

| Work                                                                                                                                                | Assignee                                                                                              | Status |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------|
|  <del>GG-88</del> Document Long and Short Term Plan                |  Nathanael Fässler   | DONE ▾ |
|  <del>GG-135</del> Document Sprint Planning Meeting Sprint 2       |  Nathanael Fässler   | DONE ▾ |
|  <del>GG-92</del> Setup Contract with OST about Ownership          |  Nathanael Fässler   | DONE ▾ |
|  <del>GG-89</del> Document Meeting Weekly Scrum Sprint 1           |  Nathanael Fässler   | DONE ▾ |
|  <del>GG-4</del> Concept for Code Quality, Linter etc.             |  Yoris Kucera        | DONE ▾ |
|  <del>GG-3</del> Unity Setup                                       |  Yoris Kucera        | DONE ▾ |
|  <del>GG-126</del> Documentation list of tables                    |  Cedric Cathomas     | DONE ▾ |
|  <del>GG-1</del> Risk Specification                               |  Yoris Kucera       | DONE ▾ |
|  <del>GG-91</del> Document Guidelines for code and documentation |  Yoris Kucera      | DONE ▾ |
|  <del>GG-6</del> GitLab Timetracking Research                    |  Dominik Wyss      | DONE ▾ |
|  <del>GG-5</del> GitLab Pipeline Setup                           |  Dominik Wyss      | DONE ▾ |
|  <del>GG-2</del> GitHub Setup                                    |  Yoris Kucera      | DONE ▾ |
|  <del>GG-87</del> Document Meetings Structure                    |  Cedric Cathomas   | DONE ▾ |
|  <del>GG-84</del> Typst documentation changes                    |  Cedric Cathomas   | DONE ▾ |
|  <del>GG-86</del> Document Collaboration as a team               |  Cedric Cathomas   | DONE ▾ |
|  <del>GG-7</del> Typst Documentation Setup                       |  Cedric Cathomas   | DONE ▾ |
|  <del>GG-8</del> People, Cost, Time Documentation                |  Cedric Cathomas   | DONE ▾ |
|  <del>GG-85</del> Roles Documentation                            |  Cedric Cathomas   | DONE ▾ |
|  <del>GG-12</del> Longtermplan & Milestones                      |  Nathanael Fässler | DONE ▾ |
|  <del>GG-9</del> Jira Setup                                      |  Nathanael Fässler | DONE ▾ |
|  <del>GG-112</del> Unity Folder Structure                        |  Nathanael Fässler | DONE ▾ |
|  <del>GG-11</del> Steamworks Registration                        |  Nathanael Fässler | DONE ▾ |
|  <del>GG-13</del> Shorttermplan for first sprint                 |  Nathanael Fässler | DONE ▾ |

**Image 32:** Sprint 1 Backlog

### 8.3.3.2 Sprint 2 06.03.2026 - 20.03.2026

In this sprint the first actual work on the game was carried out in Unity. The focus was to implement an initial working prototype.

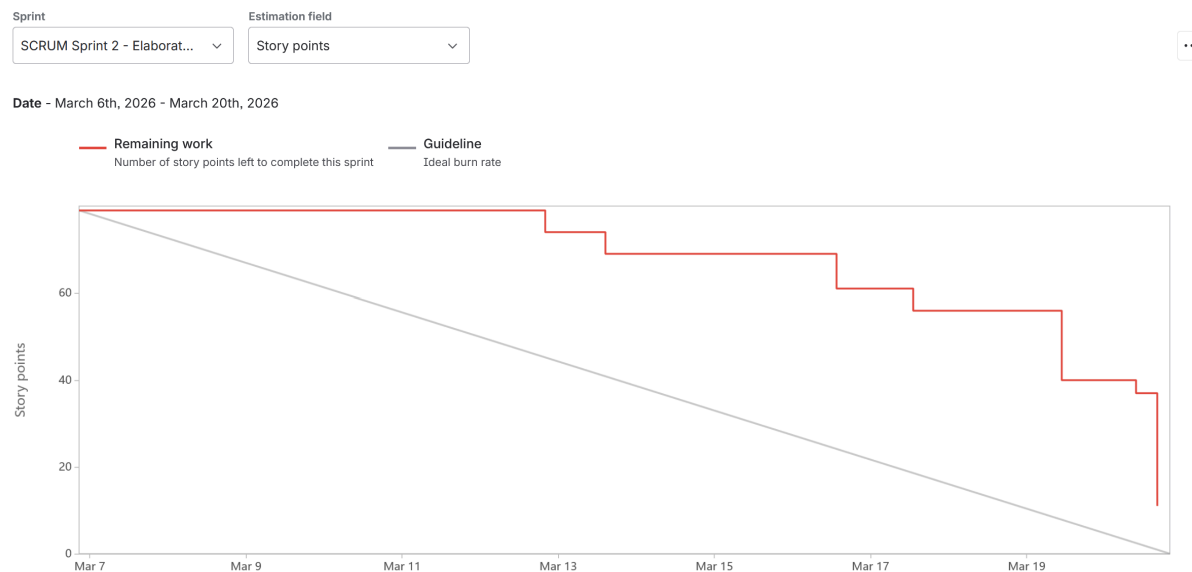


Image 33: Sprint 2 Burndown Chart

Report: SCRUM Sprint 2 - Elaboration 1

Scope changes log

| Date       | Key                    | Summary                                | Work type | Epic | Details of scope change   | Change in estimation |
|------------|------------------------|----------------------------------------|-----------|------|---------------------------|----------------------|
| 2026-03-13 | <a href="#">GC-138</a> | GitHub Runner Disk Space usage to high | Bug       |      | Work item added to sprint | -                    |

Incomplete work items

| Key                    | Summary                                    | Work type | Epic                          | Status      | Assignee | Story points |
|------------------------|--------------------------------------------|-----------|-------------------------------|-------------|----------|--------------|
| <a href="#">GC-130</a> | Unity Testing Setup and configure Pipeline | Task      | PROJECT SETUP                 | TESTING     | NY       | 3            |
| <a href="#">GC-133</a> | Implement Basic inventory system           | Story     | INVENTORY, CRAFTING & BUIL... | IN PROGRESS | NY       | 8            |

Completed work items

| Key                    | Summary                                | Work type     | Epic                        | Status | Assignee | Story points |
|------------------------|----------------------------------------|---------------|-----------------------------|--------|----------|--------------|
| <a href="#">GC-95</a>  | Create Wood-Item 3D Model              | Story         | GRAPHICS                    | DONE   | NY       | 1            |
| <a href="#">GC-28</a>  | Create Tree-Node 3D Model              | Story         | GRAPHICS                    | DONE   | NY       | 1            |
| <a href="#">GC-34</a>  | Make Player                            | Story         | CORE GAMEPLAY LOOP MECHA... | DONE   | NY       | 5            |
| <a href="#">GC-109</a> | Make GlowCore Phase 1 3D Model         | Story         | GRAPHICS                    | DONE   | NY       | 2            |
| <a href="#">GC-41</a>  | Trees can be chopped                   | Story         | CORE GAMEPLAY LOOP MECHA... | DONE   | NY       | 5            |
| <a href="#">GC-48</a>  | GlowCore expands World                 | Story         | CORE GAMEPLAY LOOP MECHA... | DONE   | NY       | 8            |
| <a href="#">GC-108</a> | Create Axe 3D Model                    | Story         | GRAPHICS                    | DONE   | NY       | 1            |
| <a href="#">GC-106</a> | Implement Node Hover Picking           | Story         | CORE GAMEPLAY LOOP MECHA... | DONE   | NY       | 8            |
| <a href="#">GC-130</a> | Document Functional Requirements       | Documentation | DOCUMENTATION               | DONE   | NY       | 8            |
| <a href="#">GC-131</a> | Document Non-Functional Requirements   | Documentation | DOCUMENTATION               | DONE   | NY       | 8            |
| <a href="#">GC-132</a> | Document Domain-Model                  | Documentation | DOCUMENTATION               | DONE   | NY       | 13           |
| <a href="#">GC-134</a> | Pipeline for doc build                 | Task          | PROJECT SETUP               | DONE   | NY       | 5            |
| <a href="#">GC-136</a> | Make UI Sketches                       | Task          | GAME FRAMEWORK & COMPLET... | DONE   | NY       | 3            |
| <a href="#">GC-138</a> | GitHub Runner Disk Space usage to high | Bug           |                             | DONE   | NY       | -            |

Image 34: Sprint 2 Backlog

### 8.3.3.3 Sprint 3 20.03.2026 - 03.04.2026

In Sprint 3 we focused on implementing the MVP. The planned amount of storypoints was too ambitious so we had to move some Tasks into Sprint 4. From this experience we gained a better understanding of our team's capacity.

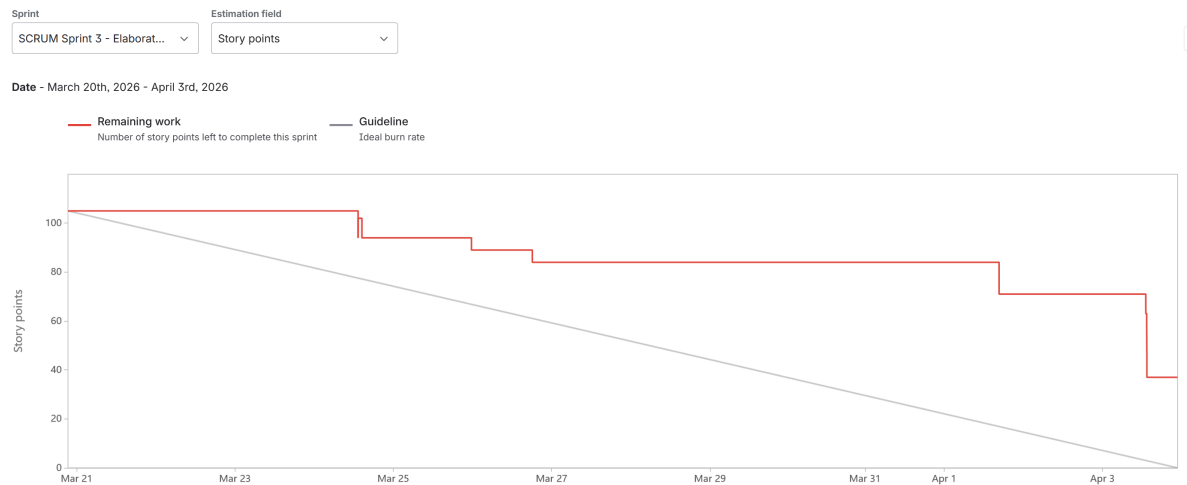


Image 35: Sprint 3 Burndown Chart

Report: SCRUM Sprint 3 - Elaboration 2

Scope changes log

| Date       | Key                    | Summary                                       | Work type | Epic | Details of scope change   | Change in estimation |
|------------|------------------------|-----------------------------------------------|-----------|------|---------------------------|----------------------|
| 2026-04-03 | <a href="#">GC-170</a> | Building decent work and pickaxe doesnt exist | Bug       |      | Work item added to sprint | 0                    |

Incomplete work items

| Key                    | Summary                 | Work type | Epic                          | Status      | Assignee | Story points |
|------------------------|-------------------------|-----------|-------------------------------|-------------|----------|--------------|
| <a href="#">GC-171</a> | Make Crafting-System    | Story     | INVENTORY, CRAFTING & BUIL... | IN PROGRESS | AS       | 8            |
| <a href="#">GC-174</a> | Save Game-Data          | Story     | GAME FRAMEWORK & COMPLET...   | IN PROGRESS | AS       | 13           |
| <a href="#">GC-142</a> | Make Crafting UI        | Story     | INVENTORY, CRAFTING & BUIL... | IN PROGRESS | AS       | 8            |
| <a href="#">GC-144</a> | Add GlowCore Upgrade UI | Story     | INVENTORY, CRAFTING & BUIL... | TO DO       | AS       | 8            |

Completed work items

| Key                    | Summary                                       | Work type | Epic                          | Status | Assignee | Story points |
|------------------------|-----------------------------------------------|-----------|-------------------------------|--------|----------|--------------|
| <a href="#">GC-352</a> | Add Pickaxe                                   | Story     | ITEMS & MATERIALS             | DONE   | AS       | 3            |
| <a href="#">GC-357</a> | Create Coal 3D Model                          | Story     | GRAPHICS                      | DONE   | AS       | 1            |
| <a href="#">GC-357</a> | Make Player 3D Model                          | Story     | GRAPHICS                      | DONE   | AS       | 1            |
| <a href="#">GC-29</a>  | Implement Building-System                     | Story     | INVENTORY, CRAFTING & BUIL... | DONE   | AS       | 13           |
| <a href="#">GC-348</a> | Make Inventory UI                             | Story     | INVENTORY, CRAFTING & BUIL... | DONE   | AS       | 13           |
| <a href="#">GC-498</a> | Make Inventory-System                         | Story     | INVENTORY, CRAFTING & BUIL... | DONE   | AS       | 8            |
| <a href="#">GC-348</a> | Create Stone 3D Model                         | Story     | GRAPHICS                      | DONE   | AS       | 1            |
| <a href="#">GC-302</a> | Make Pickaxe 3D Model                         | Story     | GRAPHICS                      | DONE   | AS       | 1            |
| <a href="#">GC-308</a> | Add Coal Node and Item                        | Story     | ITEMS & MATERIALS             | DONE   | AS       | 1            |
| <a href="#">GC-333</a> | Add Stone Node and Item                       | Story     | ITEMS & MATERIALS             | DONE   | AS       | 1            |
| <a href="#">GC-373</a> | Implement Pixelator-Shader Prototype          | Story     | GRAPHICS                      | DONE   | AS       | 8            |
| <a href="#">GC-383</a> | Build World for Alpha Release                 | Story     | 3D WORLD & GLSLCORE PHAS...   | DONE   | AS       | 3            |
| <a href="#">GC-343</a> | Implement Tool Logic                          | Story     | INVENTORY, CRAFTING & BUIL... | DONE   | AS       | 5            |
| <a href="#">GC-348</a> | Add Crafting Table 3D Model                   | Story     | GRAPHICS                      | DONE   | AS       | 1            |
| <a href="#">GC-333</a> | Implement Basic inventory system              | Story     | INVENTORY, CRAFTING & BUIL... | DONE   | AS       | 8            |
| <a href="#">GC-38</a>  | Unity Testing Setup and configure Pipeline    | Task      | PROJECT SETUP                 | DONE   | AS       | 3            |
| <a href="#">GC-170</a> | Building decent work and pickaxe doesnt exist | Bug       |                               | DONE   | AS       | 0            |

Image 36: Sprint 3 Backlog

### 8.3.3.4 Sprint 4 03.04.2026 - 17.04.2026

Because the week of 06.04.2026 to 12.04.2026 was the spring break, we had a sprint with reduced work.

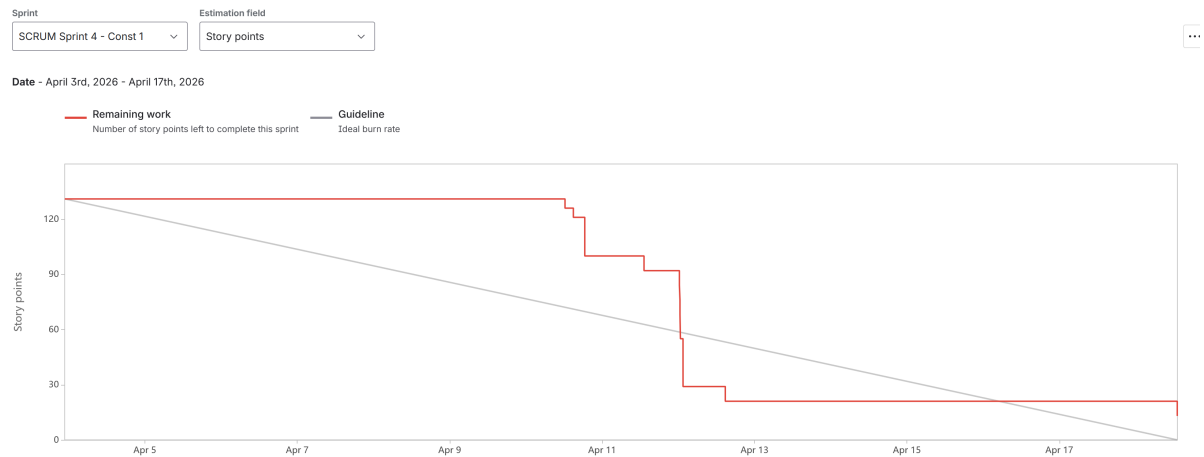


Image 37: Sprint 4 Burndown Chart

Report: SCRUM Sprint 4 - Const 1

Scope changes log

| Date       | Key                    | Summary                                 | Work type | Epic                    | Details of scope change       | Change in estimation |
|------------|------------------------|-----------------------------------------|-----------|-------------------------|-------------------------------|----------------------|
| 2026-04-06 | <a href="#">GC-176</a> | GitHub Action failed, free disk space   | Bug       |                         | Work item added to sprint     | -                    |
| 2026-04-08 | <a href="#">GC-177</a> | Test Pipeline run when only doc changes | Bug       |                         | Work item added to sprint     | -                    |
| 2026-04-13 | <a href="#">GC-186</a> | Test NFIs                               | Story     | PLAYTESTING AND RELEASE | Work item added to sprint     | -                    |
| 2026-04-13 | <a href="#">GC-186</a> | Test NFIs                               | Story     | PLAYTESTING AND RELEASE | Work item removed from sprint | -                    |

Incomplete work items

| Key                   | Summary        | Work type | Epic                        | Status      | Assignee | Story points |
|-----------------------|----------------|-----------|-----------------------------|-------------|----------|--------------|
| <a href="#">GC-74</a> | Save Game-Data | Story     | GAME FRAMEWORK & COMPLET... | IN PROGRESS | NP       | 13           |

Completed work items

| Key                    | Summary                                          | Work type     | Epic                          | Status | Assignee | Story points |
|------------------------|--------------------------------------------------|---------------|-------------------------------|--------|----------|--------------|
| <a href="#">GC-193</a> | Make Risk-Analysis                               | Documentation | DOCUMENTATION                 | DONE   | NP       | 5            |
| <a href="#">GC-188</a> | Update Project Plan                              | Documentation | DOCUMENTATION                 | DONE   | NP       | 5            |
| <a href="#">GC-187</a> | Create Test Concept                              | Documentation | DOCUMENTATION                 | DONE   | NP       | 3            |
| <a href="#">GC-188</a> | Document Architecture                            | Documentation | DOCUMENTATION                 | DONE   | NP       | 13           |
| <a href="#">GC-147</a> | Register Steampage                               | Story         | PLAYTESTING AND RELEASE       | DONE   | NP       | 5            |
| <a href="#">GC-171</a> | Dev environment setup diagram                    | Documentation | DOCUMENTATION                 | DONE   | NP       | 13           |
| <a href="#">GC-172</a> | Document Technologies and patterns               | Documentation | DOCUMENTATION                 | DONE   | NP       | 21           |
| <a href="#">GC-173</a> | AB Document interfaces between system components | Documentation | DOCUMENTATION                 | DONE   | NP       | 13           |
| <a href="#">GC-174</a> | P1-P3                                            | Documentation | DOCUMENTATION                 | DONE   | NP       | 8            |
| <a href="#">GC-175</a> | Document Savefile structure                      | Documentation | DOCUMENTATION                 | DONE   | NP       | 8            |
| <a href="#">GC-71</a>  | Make Crafting-System                             | Story         | INVENTORY, CRAFTING & BUIL... | DONE   | NP       | 8            |
| <a href="#">GC-143</a> | Make Crafting UI                                 | Story         | INVENTORY, CRAFTING & BUIL... | DONE   | NP       | 8            |
| <a href="#">GC-144</a> | Add GlowCore Upgrade UI                          | Story         | INVENTORY, CRAFTING & BUIL... | DONE   | NP       | 8            |
| <a href="#">GC-176</a> | GitHub Action failed, free disk space            | Bug           |                               | DONE   | NP       | -            |
| <a href="#">GC-177</a> | Test Pipeline run when only doc changes          | Bug           |                               | DONE   | NP       | -            |

Image 38: Sprint 4 Backlog

### 8.3.3.5 Sprint 5 17.04.2026 - 01.05.2026



Image 39: Sprint 5 Burndown Chart

Report: SCRUM Sprint 5 - Const 2

Scope changes log

| Date       | Key       | Summary                                     | Work type | Open          | Details of scope change   | Change in estimation |
|------------|-----------|---------------------------------------------|-----------|---------------|---------------------------|----------------------|
| 2026-04-25 | SC-392-17 | Create DOD and DoD example                  | Task      | DOCUMENTATION | Work item added to sprint | 0                    |
| 2026-04-25 | SC-392-17 | Q1 UDC and Test Coverage diagrams over time | Task      |               | Work item added to sprint | 0                    |
| 2026-04-25 | SC-392-17 | Java and GraphQL overview contribution      | Task      |               | Work item added to sprint | 0                    |
| 2026-04-25 | SC-392-17 | T2 TA                                       | Task      |               | Work item added to sprint | 0                    |

Incomplete work items

| Key       | Summary                | Work type | Open    | Status | Assignee | Story points |
|-----------|------------------------|-----------|---------|--------|----------|--------------|
| SC-392-17 | Improve Canvas         | Story     | BACKLOG | TO DO  |          | 8            |
| SC-392-17 | Anti-Corrupted Module  | Story     | BACKLOG | TO DO  |          | 10           |
| SC-392-17 | Save Game Data         | Story     | BACKLOG | TO DO  |          | 10           |
| SC-392-17 | Refined GraphQL Router | Task      | BACKLOG | TO DO  |          | 10           |
| SC-392-17 | T2 TA                  | Task      | BACKLOG | TO DO  |          | 0            |

Completed work items

| Key       | Summary                                     | Work type     | Open                    | Status | Assignee | Story points |
|-----------|---------------------------------------------|---------------|-------------------------|--------|----------|--------------|
| SC-392-17 | Document Quality Measures                   | Documentation | DOCUMENTATION           | DONE   |          | 8            |
| SC-392-17 | Document Testing                            | Documentation | DOCUMENTATION           | DONE   |          | 5            |
| SC-392-17 | Make it Playable                            | Story         | PLAYTESTING AND RELEASE | DONE   |          | 20           |
| SC-392-17 | Verify all MBs and Use Cases for MGS & MGS  | Story         | PLAYTESTING AND RELEASE | DONE   |          | 20           |
| SC-392-17 | Make GameCore Logic                         | Story         | PLAYTESTING AND RELEASE | DONE   |          | 3            |
| SC-392-17 | Write Integration Tests                     | Task          |                         | DONE   |          | 20           |
| SC-392-17 | Conduct Field Player Tests                  | Story         | PLAYTESTING AND RELEASE | DONE   |          | 5            |
| SC-392-17 | Create DOD and DoD example                  | Task          | DOCUMENTATION           | DONE   |          | 0            |
| SC-392-17 | Q1 UDC and Test Coverage diagrams over time | Task          |                         | DONE   |          | 0            |
| SC-392-17 | Java and GraphQL overview contribution      | Task          |                         | DONE   |          | 0            |

Image 40: Sprint 5 Backlog

### 8.3.3.6 Sprint 6 01.05.2026 - 15.05.2026

Sprint: **SCRUM Sprint 6 - Const 3** Estimation field: **Story points**

Date - May 2nd, 2026 - May 15th, 2026

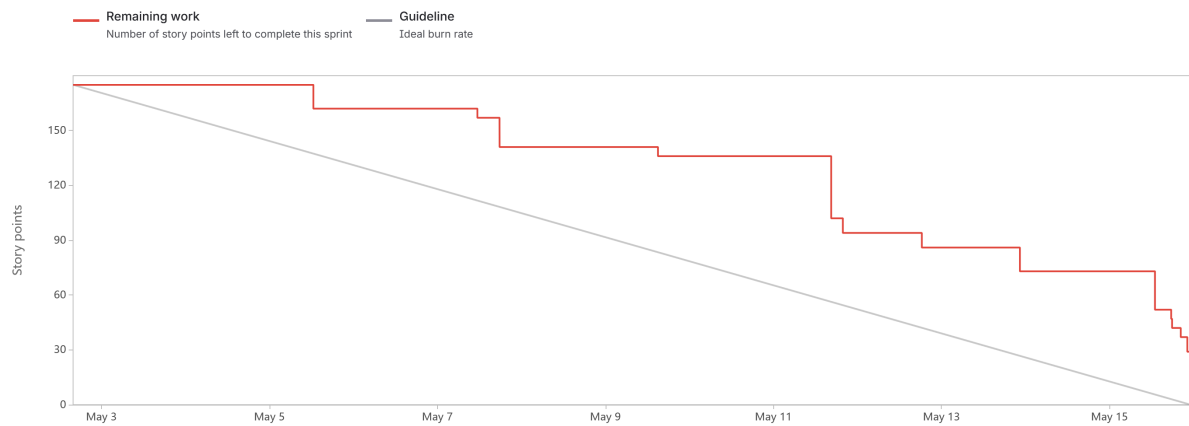


Image 41: Sprint 6 Burndown Chart

Report: SCRUM Sprint 6 - Const 3

Scope changes log

| Date       | Key    | Summary              | Work type     | Epic          | Details of scope change   | Change in estimation |
|------------|--------|----------------------|---------------|---------------|---------------------------|----------------------|
| 2026-05-14 | GC-214 | Document Game Design | Documentation | DOCUMENTATION | Work item added to sprint | -                    |

Incomplete work items

| Key    | Summary          | Work type | Epic                        | Status      | Assignee | Story points |
|--------|------------------|-----------|-----------------------------|-------------|----------|--------------|
| GC-212 | Game Ending      | Story     | GAME FRAMEWORK & COMPLET... | TO DO       | JP       | 5            |
| GC-165 | Add SoundEffects | Story     | GAME FRAMEWORK & COMPLET... | IN PROGRESS | JP       | 20           |
| GC-88  | Add Rock Biome   | Story     | BIOMES & MATERIALS          | IN REVIEW   | JP       | 3            |

Completed work items

| Key    | Summary                                     | Work type     | Epic                          | Status | Assignee | Story points |
|--------|---------------------------------------------|---------------|-------------------------------|--------|----------|--------------|
| GC-161 | Add Chest UI                                | Story         | BEST WORLD & GLOWCORE PHAS... | DONE   | JP       | 5            |
| GC-79  | Add Game Menus                              | Story         | GAME FRAMEWORK & COMPLET...   | DONE   | JP       | 20           |
| GC-168 | Extend Crafting Stations                    | Story         | BIOMES & MATERIALS            | DONE   | JP       | 5            |
| GC-166 | Improve Border Visuals                      | Story         | GRAPHICS                      | DONE   | JP       | 8            |
| GC-169 | Improve GlowCore UI                         | Story         | BEST WORLD & GLOWCORE PHAS... | DONE   | JP       | 5            |
| GC-167 | Make new Nodes and Items                    | Story         | GRAPHICS                      | DONE   | JP       | 34           |
| GC-164 | Make better Player 3D Model                 | Story         | GRAPHICS                      | DONE   | JP       | 3            |
| GC-169 | Document Deployment Diagram                 | Documentation | DOCUMENTATION                 | DONE   | JP       | 5            |
| GC-208 | Document MFR linking to technical solutions | Documentation | DOCUMENTATION                 | DONE   | JP       | 5            |
| GC-207 | Document Design Decisions                   | Documentation | DOCUMENTATION                 | DONE   | JP       | 3            |
| GC-202 | Document Frontend Architecture              | Documentation | DOCUMENTATION                 | DONE   | JP       | 5            |
| GC-169 | Selfhosted Github Runner                    | Task          |                               | DONE   | JP       | 15           |
| GC-74  | Save Game-Data                              | Story         | GAME FRAMEWORK & COMPLET...   | DONE   | JP       | 15           |
| GC-167 | Improve Camera                              | Story         | GRAPHICS                      | DONE   | JP       | 8            |
| GC-163 | T2-16                                       | Task          |                               | DONE   | JP       | -            |
| GC-79  | Add Grassland Biome                         | Story         | BIOMES & MATERIALS            | DONE   | JP       | 15           |
| GC-214 | Document Game Design                        | Documentation | DOCUMENTATION                 | DONE   | JP       | -            |

Image 42: Sprint 6 Backlog

8.3.3.7 Sprint 7 15.05.2026 - 29.05.2026

Sprint

SCRUM Sprint 7 - Transition

Estimation field

Story points

Date - May 15th, 2026 - May 29th, 2026

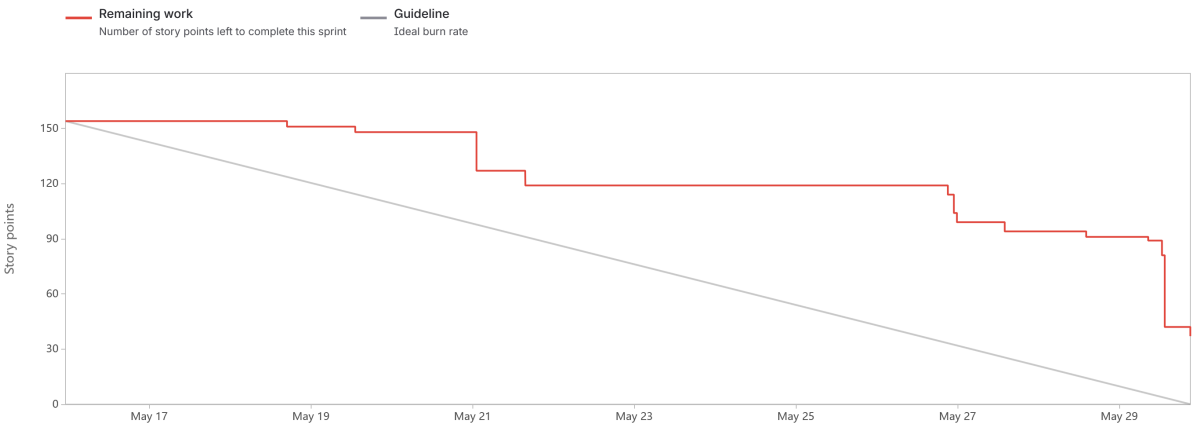


Image 43: Sprint 7 Burndown Chart

Report: SCRUM Sprint 7 - Transition

• Work item added after sprint start

View in All work navigator

Scope changes log

| Date       | Key                    | Summary                                          | Work type | Epic     | Details of scope change   | Change in estimation |
|------------|------------------------|--------------------------------------------------|-----------|----------|---------------------------|----------------------|
| 2026-05-16 | <a href="#">GC-220</a> | webgl                                            | Bug       |          | Work item added to sprint | -1                   |
| 2026-05-17 | <a href="#">GC-220</a> | Player stuck inside GlowCore                     | Bug       |          | Work item added to sprint | -1                   |
| 2026-05-17 | <a href="#">GC-220</a> | Interaction Range to short for high lvl Glowcore | Bug       |          | Work item added to sprint | -1                   |
| 2026-05-18 | <a href="#">GC-222</a> | GlowCore Inventory is not saved                  | Bug       |          | Work item added to sprint | -1                   |
| 2026-05-19 | <a href="#">GC-223</a> | Linux issue: does not start                      | Bug       |          | Work item added to sprint | -1                   |
| 2026-05-21 | <a href="#">GC-224</a> | Make Inventory Stacks Splittable                 | Story     |          | Work item added to sprint | -1                   |
| 2026-05-21 | <a href="#">GC-225</a> | Saved Sound Settings are not loaded              | Bug       |          | Work item added to sprint | -1                   |
| 2026-05-21 | <a href="#">GC-226</a> | Add Crafting Station Themes                      | Story     | GRAPHICS | Work item added to sprint | -1                   |
| 2026-05-21 | <a href="#">GC-227</a> | Crafting Recipes with long names are truncated   | Bug       |          | Work item added to sprint | -1                   |

Incomplete work items

View in All work navigator

| Key                    | Summary                                          | Work type     | Epic                        | Status | Assignee | Story points |
|------------------------|--------------------------------------------------|---------------|-----------------------------|--------|----------|--------------|
| <a href="#">GC-195</a> | Refine Graphic Settings                          | Story         | GRAPHICS                    | TO DO  |          | 5            |
| <a href="#">GC-22</a>  | Game Intro                                       | Story         | GAME FRAMEWORK & COMPLET... | TO DO  |          | 5            |
| <a href="#">GC-118</a> | Release Game on Steam                            | Story         | PLAYTESTING AND RELEASE     | TO DO  | NP       | 2            |
| <a href="#">GC-198</a> | Configure Build Settings                         | Story         | PLAYTESTING AND RELEASE     | TO DO  | YS       | 3            |
| <a href="#">GC-158</a> | Make Animation for breaking Nodes                | Story         | GRAPHICS                    | TO DO  |          | 8            |
| <a href="#">GC-163</a> | Animate Collecting Items                         | Story         | GRAPHICS                    | TO DO  |          | 3            |
| <a href="#">GC-169</a> | Make Animation for Building                      | Story         | GRAPHICS                    | TO DO  |          | 3            |
| <a href="#">GC-159</a> | Make Animation for GlowCore Upgrade              | Story         | GRAPHICS                    | TO DO  |          | 5            |
| <a href="#">GC-181</a> | Add Sound Effects to UI                          | Story         | GRAPHICS                    | TO DO  |          | 3            |
| <a href="#">GC-148</a> | Make Inputs Controller Compatible                | Story         | GAME FRAMEWORK & COMPLET... | TO DO  |          | -1           |
| <a href="#">GC-128</a> | General documentation changes                    | Documentation | DOCUMENTATION               | TO DO  |          | -1           |
| <a href="#">GC-221</a> | Interaction Range to short for high lvl Glowcore | Bug           |                             | TO DO  | NP       | -1           |
| <a href="#">GC-223</a> | Linux issue: does not start                      | Bug           |                             | TO DO  | YS       | -1           |
| <a href="#">GC-227</a> | Crafting Recipes with long names are truncated   | Bug           |                             | TO DO  | NP       | -1           |

Completed work items

View in All work navigator

| Key                    | Summary                                                | Work type     | Epic                         | Status | Assignee | Story points |
|------------------------|--------------------------------------------------------|---------------|------------------------------|--------|----------|--------------|
| <a href="#">GC-927</a> | Build World                                            | Story         | BIG WORLD & GLOWCORE PHAS... | DONE   | NP       | 8            |
| <a href="#">GC-208</a> | Fix screen border                                      | Bug           |                              | DONE   | YS       | -1           |
| <a href="#">GC-352</a> | Add Scene for rendering Icons                          | Story         | GRAPHICS                     | DONE   | CS       | 5            |
| <a href="#">GC-363</a> | Generate Icons for existing Items                      | Story         | GRAPHICS                     | DONE   | CS       | 2            |
| <a href="#">GC-205</a> | Make Blocks Rotatable                                  | Story         | GAME FRAMEWORK & COMPLET...  | DONE   | NP       | 5            |
| <a href="#">GC-209</a> | Make Marketing Material for Steam Page                 | Story         | PLAYTESTING AND RELEASE      | DONE   | NP       | 5            |
| <a href="#">GC-204</a> | Prepare Presentation                                   | Documentation |                              | DONE   | CS       | 34           |
| <a href="#">GC-236</a> | Camera is not on player position when loading savefile | Bug           |                              | DONE   | YS       | -1           |
| <a href="#">GC-211</a> | Inventory item sprites invisible when loading savefile | Bug           |                              | DONE   |          | -1           |
| <a href="#">GC-212</a> | UseAction is triggered in Inventory                    | Bug           |                              | DONE   | NP       | -1           |
| <a href="#">GC-233</a> | Fix Node Outlines                                      | Bug           |                              | DONE   | YS       | -1           |
| <a href="#">GC-244</a> | Add Axe Machine                                        | Story         | RESOURCE AUTOMATION          | DONE   | NP       | 8            |
| <a href="#">GC-235</a> | Add Replant-Machine                                    | Story         | RESOURCE AUTOMATION          | DONE   | NP       | 5            |
| <a href="#">GC-238</a> | Add Rotator-Block                                      | Story         | RESOURCE AUTOMATION          | DONE   | NP       | 5            |
| <a href="#">GC-219</a> | Axe broken when loading game                           | Bug           |                              | DONE   |          | -1           |
| <a href="#">GC-238</a> | Update NFR Status                                      | Documentation |                              | DONE   | CS       | 3            |
| <a href="#">GC-237</a> | Improve Border Fog                                     | Story         | GRAPHICS                     | DONE   | NP       | 5            |
| <a href="#">GC-238</a> | Document Management Summary                            | Documentation | DOCUMENTATION                | DONE   | CS       | 3            |
| <a href="#">GC-23</a>  | Game Ending                                            | Story         | GAME FRAMEWORK & COMPLET...  | DONE   | NP       | 5            |
| <a href="#">GC-245</a> | Add Soundeffects                                       | Story         | GAME FRAMEWORK & COMPLET...  | DONE   | CS       | 25           |
| <a href="#">GC-89</a>  | Add Rock Biome                                         | Story         | BIOMES & MATTERSCALES        | DONE   | NP       | 3            |
| <a href="#">GC-236</a> | webgl                                                  | Bug           |                              | DONE   | CS       | -1           |
| <a href="#">GC-220</a> | Player stuck inside GlowCore                           | Bug           |                              | DONE   |          | -1           |
| <a href="#">GC-222</a> | GlowCore Inventory is not saved                        | Bug           |                              | DONE   | NP       | -1           |
| <a href="#">GC-224</a> | Make Inventory Stacks Splittable                       | Story         |                              | DONE   | CS       | -1           |
| <a href="#">GC-225</a> | Saved Sound Settings are not loaded                    | Bug           |                              | DONE   |          | -1           |
| <a href="#">GC-226</a> | Add Crafting Station Themes                            | Story         | GRAPHICS                     | DONE   | CS       | -1           |

Work items completed outside of sprint

| Key | Summary | Work type | Epic | Status | Assignee | Story points |
|-----|---------|-----------|------|--------|----------|--------------|
|-----|---------|-----------|------|--------|----------|--------------|

Image 44: Sprint 7 Backlog

## 8.4 Risk Management

This chapter describes potential risks to the project and how we plan to mitigate them. We identified 8 risks, which are listed in the table below. Additionally, these risks are visualized in two risk matrices, one showing the initial risk assessment and the other showing the risk assessment after mitigation strategies are applied.

| No. | Risk                           | Description                                                                                                                                                                                                                                  | Mitigation Strategy                                                                                                                                                                                                  |
|-----|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R01 | Lack of familiarity with Unity | None of our team members have any significant experience with the Unity Engine. As such, this could heavily impact our development times.                                                                                                    | We made ourselves familiar with the engine before the actual start of the project, to get an idea of the workflows.                                                                                                  |
| R02 | Unity CI/CD Complexity         | We are depending on Unity's build system being robust enough to work in the context of a CI/CD pipeline. We also potentially rely on pre-existing GitHub Actions for Unity, which may not be well-maintained or compatible with our project. | We set up the CI/CD pipeline very early in the project and did earlier tests. We tried out pre-existing GitHub Actions to make sure they satisfy our needs.                                                          |
| R03 | Scope/Feature Creep            | As we are developing a game, we run the risk of encountering scope creep, i.e. continuously planning new features, which would result in us eventually not being able to keep up with the workload.                                          | We created a detailed project plan with long-term and short-term planning, along with well-defined milestones and a clear MVP.                                                                                       |
| R04 | Bad Code Quality               | In large codebases it is easy to end up with badly designed systems that don't fit together with the rest of the code. This would hinder further development.                                                                                | We do mandatory code reviews before changes can be merged. Additionally, we will set up CI/CD pipelines to automatically run code quality and formatting checks on pull requests.                                    |
| R05 | Team Member is MIA             | A team member might be MIA for an extended period of time due to various reasons, such as an illness or damaged hardware. This would reduce our capacity and slow down development.                                                          | We can't reduce the possibility of this occurring, but we can distribute work across the remaining team members. We also avoid knowledge monopolies and always aim to share discoveries with the other team members. |
| R06 | Personal Conflicts             | There is always the potential for personal conflicts within the team, which would complicate development.                                                                                                                                    | We openly communicate any interpersonal issues and try to resolve conflicts as early as possible.                                                                                                                    |
| R07 | Data Loss                      | There is the potential of local storage media failing, resulting in significant data loss.                                                                                                                                                   | We mitigate this by frequently committing our changes to GitHub, and keeping untracked files in cloud based locations.                                                                                               |
| R08 | Workload Underestimation       | We might be underestimating the amount of work required for specific features and tasks.                                                                                                                                                     | We plan with a buffer and try to be conservative with our estimates. We also try to break down large tasks into smaller ones, which are easier to estimate.                                                          |

**Table 52:** Risk Analysis



### 8.4.1 Risk Matrix

The below table shows the risk assessment of the identified risks, before any mitigation strategies are applied. The risks are categorized based on their probability of occurrence and their potential severity.

| Probability | Severity   |          |          |              |
|-------------|------------|----------|----------|--------------|
|             | Negligible | Marginal | Critical | Catastrophic |
| Certain     |            |          |          |              |
| Likely      |            |          | R04      | R01          |
| Possible    |            | R02/R06  |          | R03/R08      |
| Unlikely    |            |          |          | R05          |
| Rare        |            |          |          | R07          |
| None        |            |          |          |              |

**Table 53:** Risk Matrix

### 8.4.2 Risk Matrix (Mitigated)

The below table shows the risk assessment of the identified risks, after mitigation strategies have been applied.

| Probability | Severity   |          |             |              |
|-------------|------------|----------|-------------|--------------|
|             | Negligible | Marginal | Critical    | Catastrophic |
| Certain     |            |          |             |              |
| Likely      |            |          |             |              |
| Possible    |            |          |             |              |
| Unlikely    |            | R06      | R01/R04/R05 | R08          |
| Rare        | R07        | R02      |             |              |
| None        |            |          |             | R03          |

**Table 54:** Risk Matrix (Mitigated)

### 8.4.3 Assessed Risk Matrix (14.04.2026)

The below table shows the risk assessment of the identified risks, after mitigation strategies have been applied.

| Probability | Severity   |          |          |              |
|-------------|------------|----------|----------|--------------|
|             | Negligible | Marginal | Critical | Catastrophic |
| Certain     |            |          |          |              |
| Likely      |            |          |          |              |
| Possible    |            |          | R01      |              |
| Unlikely    |            | R06/R02  | R04/R05  | R08          |
| Rare        | R07        |          |          |              |
| None        |            |          |          | R03          |

**Table 55:** Assessed Risk Matrix (14.04.2026)

#### 8.4.4 Adjustment of Current Risk Assessment

In the current risk matrix, we increased the probability by one level for both R01 (Lack of familiarity with Unity) and R02 (Unity CI/CD Complexity).

For R01, we observed that working with Unity involves more aspects than initially expected, especially issues that only become visible during actual implementation. The project description already pointed out that Unity could be a risk, so we were aware of this from the beginning. However, during development we realized that some parts are more complex in practice. Despite this, we are still very motivated to use Unity for our project and try to reduce the risk through clear communication and well-defined implementation strategies.

For R02, there are recurring challenges with the CI/CD setup in Unity, which is why we increased the probability here as well. In particular, there is a risk of running out of GitHub Actions minutes. At the moment, we expect the available minutes to be sufficient, but if needed, we already have a plan to set up and use a self-hosted runner on our own machine.

Overall, both risks are now considered slightly more likely, but appropriate mitigation measures are in place.

# 9 Time Tracking Report

## 9.1 Time Tracking

Every member tracks time on Jira on the corresponding task/story. Time spent in meetings is documented in the meeting protocols. The time is logged in 15-minute blocks in the format e.g. 3h 15m.

## 9.2 Time Statistics

The time is grouped by member and by sprint. Per sprint and person the time is rounded to the nearest hour.

| Sprint | Cedric Cathomas | Nathanael Fässler | Yoris Kucera | Dominik Wyss | Total |
|--------|-----------------|-------------------|--------------|--------------|-------|
| 1      | 23h             | 29h               | 21h          | 24h          | 97h   |
| 2      | 22h             | 20h               | 19h          | 23h          | 84h   |
| 3      | 18h             | 13h               | 19h          | 15h          | 65h   |
| 4      | 25h             | 13h               | 15h          | 17h          | 70h   |
| 5      | 7h              | 14h               | 18h          | 15h          | 54h   |
| 6      | 20h             | 23h               | 20h          | 19h          | 82h   |
| 7      | 8h              | 17h               | 10h          | 18h          | 53h   |
| Total  | 123h            | 129h              | 122h         | 131h         | 505h  |

**Table 56:** Time Statistics

# 10 Personal Reports

## 10.1 Nathanael Fässler

This project was definitely a personal highlight of this semester and of my time at OST in general. We created a video game from start to finish and experienced every aspect of game development. We began planning in the previous semester, brainstormed ideas for gameplay and game rules, created our own 3D models, programmed the game systems, conducted user tests and finally published it on Steam.

Game development is my favorite field within software engineering, so I am very grateful that we were even allowed to pursue such an unconventional SE Project and that it worked out so well.

Making a game in Unity introduced challenges that most other groups didn't have. Testing game code is very difficult and the architecture is not as obvious as in projects that just consist of a frontend and backend. At the same time I find writing code for a game much more exciting than implementing business logic. The programming involved a lot of algorithmic problem solving and since our game is 3D, many programming tasks involved vector mathematics. It felt rewarding to finally apply the theoretical concepts learned in DMI (Diskrete Mathematik für Informatik) and AlgDat (Algorithmen und Datenstrukturen).

The initiative to make a game and the initial idea came from me, so we decided that I would take the role of the Product Owner. I was responsible for the planning and I learned how to manage a backlog effectively. I underestimated how difficult it is to create tasks with sufficiently clear requirements and instructions and I see potential to improve my communication skills. Since I mainly focused on the product and planning, I was not involved in the CI/CD process which I consider a missed opportunity and something I would like to learn more about in the future.

In retrospect I would definitely choose to make a game again. However, for future students I would only recommend developing a game in SE Project if the following conditions are met:

1. The team members are already very familiar with the game engine or have enough time to learn it **before** the semester starts.
2. The members define one person who is responsible for decisions about the game's contents or they have very good communication skills and are able to make decisions time efficiently.
3. The game idea needs to be very small but designed to be extensible. The game should be simple enough to be developed within a few weeks, while still allowing additional features and content to be added. Ideally, the project is always in a state where it could be considered a finished product.

## 10.2 Dominik Wyss

When our team first discussed possible project ideas, I was not very enthusiastic about developing a game. In general, game projects were not strongly recommended, and none of us had any previous experience with game development. From my perspective, it also seemed unlikely that the project would be very useful for my future career, since becoming a game developer has never been one of my goals.

Despite these concerns, I spent a lot of time discussing ideas and planning the project together with the team. Once we decided to move forward with the idea, I wanted to contribute as much as possible and help make the project successful. Even though I do not see myself working on game projects again in the future, I have always been interested in games as a hobby. Over the years, I have played many different games and completed quite a few of them. Because of this, I was able to bring ideas and observations from successful games into discussions about our own project. As a computer science student, I also tend to look at games differently and think about how certain mechanics and systems could be implemented in software and adapted to our own requirements.

One aspect that surprised me positively was the collaboration within the team. Teamwork is often difficult, especially when people have different opinions about how something should be implemented. There were several situations where not everyone agreed with a decision. Sometimes a team member strongly preferred one solution, but after discussing it, the team decided to follow a different approach. This was not always easy to accept, but in most cases people respected the final decision and continued working towards the same goal.

I think this was one of the reasons why our project was successful. Even when opinions differed, the team remained constructive and focused on finding a solution instead of proving who was right. We consistently attended meetings,

completed our tasks, and supported each other throughout the semester. Looking back, I was impressed that we managed to maintain this level of commitment until the very end and successfully deliver a finished product.

For future projects, I would personally prefer to work on a more traditional software application. I am more interested in solving technical problems through code, working with algorithms, or designing software systems. While I appreciate the experience of developing a game, I realized that I am less interested in areas such as player movement, visual design, or building a complete 3D world. My interests are more focused on software that is driven by functionality and technical problem-solving.

Overall, the project challenged some of my initial expectations. Even though I would not choose a game project again, I learned a lot about teamwork, planning, and working on a type of software that was completely new to me. The experience showed me that a team can still work effectively even when members have different opinions and preferences. These lessons will be valuable in future software projects regardless of the field I work in.

## 10.3 Cedric Cathomas

I had never worked with Unity before and had never created a game, so the whole project was new territory for me. C# was already my favorite language, which gave me at least some familiar ground to stand on. I was genuinely excited, because I love video games and building one finally let me peek behind the curtain and see how everything actually connects. At the same time I was a little nervous, since I had no experience and knew it would mean extra effort to learn everything from scratch. But I committed to it, and preparing well before the semester started kept that excitement going.

As the team's UI Architect I implemented our entire UI and the dependencies that connect it to the game logic. I built most of the things that make up the UI, such as the inventory, the hotbar, the crafting stations, the GlowCore and the whole menu and settings, and I was also able to contribute to some of the logic elements, which I really enjoyed because it showed me how a game actually works under the hood. Almost everything was new to me, since it was my first time building both a game and of course a game UI. Unity itself, the scenes, and the canvas were completely unfamiliar, but the preparation beforehand and the time we invested as a team to learn the basics helped a lot. It was a very rewarding process to see how good a UI can look once you understand how it is done. The architecture also gave me the chance to apply the principles we learned in SEP2: I really learned how to make use of dependency injection together with the open-closed and single-responsibility principles to create a clean and extensible architecture where we could add new UIs as they came. What I value most is that I now know how those principles actually look and feel to use in production. I will never forget what dependency injection is or why it matters. I was also surprised how useful the tooling around the project is, from the workflows and GitHub Actions to the test runner, the sprites and the audio system. On top of that I took on the role of project manager, making sure everything was done on time and that everyone could do their work. I enjoyed seeing the project from that different perspective, even though I prefer doing the tasks over managing them. My personal highlight was the menu scene and the extendable, reusable inventory I built. It was a great moment when the controls finally worked together with the game and once the audio system was in place we suddenly had a full game, with a main menu, settings, a pause screen and all the UIs you know from the classic games of my childhood. Everything fit together and it felt like we had made it.

I have had work experience where teamwork mattered, but honestly it is no comparison to how well we worked together in our team of four. I did not know it was possible for everyone to contribute so much and to be so dedicated to helping each other out, especially in a uni project. Responses were quick most of the time, the communication was always there and there were always ideas, questions and discussions. Seeing the others so motivated motivated everyone in turn and that is what pushed me personally to give my best. The sprint plannings and meetings gave us a great overview and helped us keep track of every change and every ongoing problem. We did tend to plan too much, but that was because we wanted to do a lot without really knowing how long each task would take, which made sense given the unfamiliar tools and the fact that almost every task involved something new. A game also has so many different areas that a team of four, not working on it full-time, can quickly reach its limits. The clearest problems we had were the merge conflicts in the main scene. Coordinating our changes was very hard, especially in the beginning, because of all the dependencies and to this day I do not know how we could have avoided them completely. We did our best by communicating the changes that were likely to cause conflicts, but in the end that is simply a problem you face when building a game in a larger team where tasks sometimes overlap. Compared to my old job, this team was definitely better and for such a big and unusual project I think we really got the best out of it.

Of course there is always room for improvement. Sometimes reviews took too long, but considering we only worked on the project around eight hours a week, it is hard to expect someone to look at changes instantly, especially the bigger ones. It was also difficult to understand the actual goal or implementation of a task at the beginning, which needed a lot of additional communication and rework. Ideally tasks would be planned through well enough not to raise so many questions, but that is

hard when everyone also has to program, plan and document at the same time. I personally did not expect the game to be so complex or to have so many different systems, from the UI and the audio system to the scriptable objects and the logic, and at some point it became quite overwhelming. That is the main reason I would not build a game again in this setting. To do it cleanly it would need a dedicated product owner with an exact vision, a separate architect, developers who implement based on clear descriptions and a lot more time. The testing alone is very time-consuming and having to learn all the new concepts on top of that was one step too many. Still, I do not regret it, because I was able to learn so much and take away so many things for my career.

Overall I am very happy with our product and with everything I was able to learn. We made a fantastic game with a genuinely addictive drive and a lot of possibilities to try out. The path was not an easy one, but the result was definitely worth it and each of us carried a lot of responsibility in our own area. As honest as I can be, I would not do it again, because the project pushed me to my limits with its complexity and all the dependencies and potential issues, which made it at times too stressful. This only worked because everyone truly contributed. With a different team I could imagine this being impossible to pull off so well, so in that sense we were lucky to have a Product Owner who burned for the project and carried us along with his passion. It took a lot of energy, but it gave back even more and the result surpassed my expectations. I am very proud of what we built.

## 10.4 Yoris Kucera

When the team first started collecting project ideas, I was immediately drawn to the thought of building a game. I had made a few attempts at game development on my own in the past, none of which I ever really finished, so the idea of seeing a complete game through from start to finish, this time with a committed team behind it, was genuinely exciting to me. It felt like a chance to finally turn something I had only ever dabbled in into a real product.

That initial enthusiasm shifted fairly quickly once we were deep into development. I came to realize that game development is not really a field I am passionate about, and the main reason is how constrained you are by the architecture that Unity lays out for you. So much of the work is about fitting your code into the engine's component model, its scene structure, and its lifecycle, rather than designing systems the way you would approach them in a more traditional software project. For someone who enjoys the design and structuring side of software engineering, constantly having to bend around the engine's conventions was often more frustrating than rewarding.

Despite that, the role I took on ended up being the part of the project I valued most. As the team's Software Architect I was responsible for a large share of the architectural decisions in the codebase. I spent a lot of time thinking about how our systems should be separated, how components should communicate, and how to keep things extensible so that new features could be added without rewriting what already existed. Working through those decisions, and seeing the rest of the team build on top of the structure we agreed on, was where I learned the most and where I felt I contributed the most value.

The part I genuinely enjoyed the most, however, was designing the low-res rendering architecture and the highlight/outline system. This was where I got to write shaders and work much closer to the rendering pipeline, and it turned out to be the kind of self-contained, technical problem solving I really like. There was a lot of satisfaction in getting the low-res look exactly right and in making interactable objects clearly stand out with a clean outline. Unlike a lot of the gameplay work, these systems gave me room to actually engineer a solution end to end, and I was proud of how they came together.

Overall I am happy with the final product and with what we managed to build as a team. We delivered a complete, mostly polished game, and I learned a lot along the way, both about software architecture and about the parts of development I do and do not enjoy. That said, my biggest takeaway is that, in the future, I would most likely not choose to make a game in a project like this again. I appreciate the experience and do not regret it, but I now know that my interests lie more in traditional software engineering, where I have the freedom to design systems on my own terms rather than within the boundaries an engine sets for me.

# 11 Meeting Minutes

## 11.1 Weekly Scrum / 28.02.2026

| Field        | Value                                         |
|--------------|-----------------------------------------------|
| Date         | 28.02.2026                                    |
| Time         | 11:00–14:00                                   |
| Sprint       | Sprint 1                                      |
| Location     | Online                                        |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera |
| Excused      | Cedric Cathomas                               |
| Absent       | -                                             |
| Meeting lead | Yoris Kucera                                  |
| Minutes by   | Nathanael Fässler                             |
| Links        | -                                             |

### 11.1.1 Agenda

| Agenda item         | Owner     | Result / Decision                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GitHub Setup        | Yoris     | <ul style="list-style-type: none"> <li>We switched from GitLab to GitHub</li> </ul> <p>The following rules were discussed:</p> <ul style="list-style-type: none"> <li>Pull Request requires one approval that is not the same person as the author.</li> <li>Merging Methods are Merge and Squash.</li> <li>Every Jira Task corresponds to one branch.</li> </ul> |
| CICD Pipeline Setup | Dominik   | <ul style="list-style-type: none"> <li>Web build and .exe build succeeded.</li> <li>Pages Architecture with main and dev branches decided</li> </ul>                                                                                                                                                                                                              |
| Timetracking        | Dominik   | <ul style="list-style-type: none"> <li>We won't use GitLab timetracking because we switched to GitHub</li> <li>Timetracking is in Jira: Everyone tracks time for individual Tasks/Stories.</li> </ul>                                                                                                                                                             |
| Unity/Coding Setup  | Yoris     | Coding Guidelines decided                                                                                                                                                                                                                                                                                                                                         |
| Steam Release       | Nathanael | <ul style="list-style-type: none"> <li>Registered on Steamworks</li> <li>Up to 30 days waiting time until application is processed</li> <li>Maybe we need to make a contract with OST.</li> </ul>                                                                                                                                                                 |
| Testing             | Nathanael | Unit-Test and Integration-Test Strategy discussed.                                                                                                                                                                                                                                                                                                                |
| Long/Shorttermplan  | Nathanael | <ul style="list-style-type: none"> <li>Hierarchy: There are Epics which contain Stories.</li> <li>Everything that is not a game-feature is a task.</li> <li>Longtermplan with epics according to the Jira Timeline</li> <li>Shorttermplan per sprint on the Jira board</li> </ul>                                                                                 |

|  |  |                                                                                                                                                                                                                                                                                            |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"> <li>• Priority is defined by the order in the backlog in Jira</li> <li>• MVP is defined</li> <li>• We will estimate Tasks/Stories with Story-points</li> <li>• On Jira we can create Bug-Tickets which we will not estimate with Storypoints</li> </ul> |
|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 11.1.2 Todos

| Todo                           | Owner     | Due        |
|--------------------------------|-----------|------------|
| Finish CICD Pipeline           | Dominik   | 06.03.2026 |
| Ask Advisor about contract     | Nathanael | 04.03.2026 |
| Ask Advisor about project plan | Nathanael | 04.03.2026 |

## 11.2 Weekly Scrum and Sprint Retrospective / 06.03.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 06.03.2026                                                     |
| Time         | 14:00-16:00                                                    |
| Sprint       | Sprint 1                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Nathanael Fässler                                              |
| Minutes by   | Cedric Cathomas                                                |
| Links        | -                                                              |

### 11.2.1 Agenda

| Agenda item                                                                                                  | Owner             | Result / Decision                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Board Review                                                                                                 | All               | <ul style="list-style-type: none"> <li>• Walked through all current board tasks</li> <li>• Updated status on completed tasks</li> </ul>                                                              |
| Done Tasks and New Functionalities                                                                           | All               | <ul style="list-style-type: none"> <li>• Reviewed completed work from Sprint 1</li> <li>• Discussed new functionalities and rules</li> </ul>                                                         |
| Open Questions                                                                                               | All               | <ul style="list-style-type: none"> <li>• Discussed open questions regarding project setup and workflow</li> </ul>                                                                                    |
| Unity Graphics in Repository                                                                                 | Nathanael Fässler | <ul style="list-style-type: none"> <li>• Only .fbx files are to be committed to the repository for 3D graphics in Unity</li> </ul>                                                                   |
| Documentation Versioning                                                                                     | Cedric Cathomas   | <ul style="list-style-type: none"> <li>• Tags for documentation versions start with "d" (e.g. d1.1)</li> <li>• Version is updated automatically on pull request to main via GitHub Action</li> </ul> |
| How do we manage smaller tasks that are not worth creating a new task on Jira. (small documentation changes) | Dominik Wyss      | We create ticket for small backlog tasks that do not need an own Jira item so we can                                                                                                                 |



|                                                |                 |                                                                                                                             |
|------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------|
|                                                |                 | reference it on our pull request and use the name of that ticket for the branch                                             |
| Decide if we need a separate pdf for protocols | Cedric Cathomas | We decided that there's no use in having a separate page just for meeting minutes, so we have one thing less to worry about |

## 11.2.2 Sprint 1 Retrospective

### 11.2.2.1 What Went Well

- Good and fast communication
- Good planning on Jira
- Good preplanning visible
- Product Owner did a good job
- Good and understandable selection of planning and working tools
- Pipeline and GitHub are working
- Typst is set up
- General setup went very well

### 11.2.2.2 What Didn't Go Well

- Problems with GitLab
- Didn't have enough time for some tasks

### 11.2.2.3 Improvements

- Better time management
- Better definitions of done
- Use GitHub from the beginning

## 11.2.3 Todos

| Todo                                                                              | Owner             | Due        |
|-----------------------------------------------------------------------------------|-------------------|------------|
| Create ticket for small backlog tasks that do not need an own Jira item           | Nathanael Fässler | 07.03.2026 |
| Create all table references so documentation looks nice and page breaks are clean | Cedric Cathomas   | 07.03.2026 |
| Adapt milestone dates to single dates (e.g. 16.02.2026 instead of a time frame)   | Cedric Cathomas   | 07.03.2026 |
| clean up documentation (spaces that are too big)                                  | Cedric Cathomas   | 07.03.2026 |
| Change date format in documentation to use dots                                   | Cedric Cathomas   | 07.03.2026 |
| Create task for combined build and documentation pipeline                         | Nathanael Fässler | 07.03.2026 |
| Use figures for images so they appear in the list of figures                      | Nathanael Fässler | 07.03.2026 |
| remove separate meeting minutes pdf typst code                                    | Cedric Cathomas   | 07.03.2026 |

## 11.3 Sprint Planning / 06.03.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 06.03.2026                                                     |
| Time         | 16:00–18:00                                                    |
| Sprint       | Sprint 2                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Yoris Kucera                                                   |
| Minutes by   | Nathanael Fässler                                              |
| Links        | -                                                              |

### 11.3.1 Sprint goal

- Build first working prototype.
- Define Functional Requirements and Non-Functional Requirements.
- Create Domain Model

## 11.4 Sprint Review (Advisor) / 10.03.2026

| Field        | Value                                                                        |
|--------------|------------------------------------------------------------------------------|
| Date         | 10.03.2026                                                                   |
| Time         | 15:00–16:00                                                                  |
| Sprint       | Sprint 2                                                                     |
| Location     | Building 8                                                                   |
| Participants | Thomas Bocek, Dominik Wyss, Yoris Kucera, Nathanael Fässler, Cedric Cathomas |
| Excused      | -                                                                            |
| Absent       | -                                                                            |
| Meeting lead | Yoris Kucera                                                                 |
| Minutes by   | Nathanael Fässler                                                            |
| Links        | -                                                                            |

### 11.4.1 Agenda

| Agenda item                               | Owner               | Result / Decision                                                                                                                    |
|-------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Status of Work                            | Yoris               | Feedback was good                                                                                                                    |
| Project Plan                              | Yoris               | Good impression. Advisor liked that we asked him at 02.03.2026 on our initiative                                                     |
| Showcase Pipeline                         | Dominik             | Positive that Dominik sent the pipeline beforehand                                                                                   |
| Feedback                                  | Thomas              | Overall very good                                                                                                                    |
| Questions about Grading & Review Schedule | Dominik & Nathanael | <ul style="list-style-type: none"> <li>• To get a better grade we have to do more than what is required on the checklist.</li> </ul> |

|  |  |                                                                                                                                                                               |
|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"> <li>The group can choose when the Review Meetings take place.</li> <li>Meeting 5 has to be before 05.06.2026 (final submission)</li> </ul> |
|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 11.4.2 Todos

| Todo                                                                 | Owner           | Due        |
|----------------------------------------------------------------------|-----------------|------------|
| Update README on GitLab to indicate that project has moved to GitHub | Yoris           | 20.03.2026 |
| Configure Pipeline to build documentation                            | Not assigned    | 20.03.2026 |
| Add AGENTS.md                                                        | Cedric Cathomas | 20.03.2026 |

### 11.4.3 Next dates

| Next date               | When             |
|-------------------------|------------------|
| Review 2 - Requirements | 24.03.2026 15:00 |

## 11.5 Weekly Scrum / 13.03.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 13.03.2026                                                     |
| Time         | 14:30-15:30                                                    |
| Sprint       | Sprint 2                                                       |
| Location     | Online                                                         |
| Participants | Cedric Cathomas, Nathanael Fässler, Dominik Wyss, Yoris Kucera |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Dominik Wyss                                                   |
| Minutes by   | Yoris Kucera                                                   |
| Links        | -                                                              |

### 11.5.1 Agenda

| Agenda item                  | Owner        | Result / Decision                                                                                                                                                                                        |
|------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Time Tracking                | Dominik Wyss | <ul style="list-style-type: none"> <li>New time tracking report system integrated in Jira (Worklogs Report)</li> <li>Task does not need to be assigned for time to be tracked</li> </ul>                 |
| Documentation Build Pipeline | Yoris Kucera | <ul style="list-style-type: none"> <li>Pipeline for building documentation on GitHub Actions is set up and working</li> <li>Automatically deployed to pages on push to dev and main branches.</li> </ul> |
| Domain Model                 | Yoris Kucera | <ul style="list-style-type: none"> <li>First version of domain model created</li> <li>State machine for enemy created</li> </ul>                                                                         |

|                             |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Functional Requirements     | Nathanael Fässler                | <ul style="list-style-type: none"> <li>• Documented Functional Requirements, Actors and Personas</li> <li>• Documented Use Cases</li> <li>• Use Cases formatting could be improved</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Player and Camera           | Nathanael Fässler                | <ul style="list-style-type: none"> <li>• Player controller implemented and working, controls with WASD</li> <li>• Camera implemented and working, controls with mouse movement</li> <li>• Camera follows player and rotates around it, fixed height and distance</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Non-Functional Requirements | Cedric Cathomas                  | <ul style="list-style-type: none"> <li>• Documented Non-Functional Requirements</li> <li>• Priority Matrix for NFRs</li> <li>• NFRs grouped by category and numbered (e.g. NFR101, NFR102, NFR403, etc.)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| GitHub CI/CD Failing        | Dominik Wyss/Yoris Kucera        | <ul style="list-style-type: none"> <li>• Pipeline fails because of lack of disk space on the runner</li> <li>• Possible solutions:               <ul style="list-style-type: none"> <li>▸ Clean up even more storage</li> <li>▸ Make repository public</li> <li>▸ Use self-hosted runner with more storage</li> <li>▸ Rent server at OST</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| UI Sketches                 | Dominik Wyss/<br>Cedric Cathomas | <ul style="list-style-type: none"> <li>• Sketches for Main Menu               <ul style="list-style-type: none"> <li>▸ Scene from the game shown in the background</li> <li>▸ Buttons for New Game, Continue, Save Files, Settings, Exit</li> </ul> </li> <li>• Sketch for New Game menu               <ul style="list-style-type: none"> <li>▸ Give name to world</li> </ul> </li> <li>• Sketch for Save File Menu               <ul style="list-style-type: none"> <li>▸ List of save files with name and date</li> <li>▸ Button to delete save file</li> <li>▸ Show glowcore level?</li> <li>▸ Separate list for auto saves and manual saves?</li> </ul> </li> <li>• Sketch for Settings Menu               <ul style="list-style-type: none"> <li>▸ Sliders for music and sound effects volume</li> <li>▸ Slider for mouse sensitivity?</li> <li>▸ Graphics options?</li> </ul> </li> <li>• Sketch for Pause Menu</li> </ul> |

## 11.5.2 Todos

| Todo                                        | Owner           | Due        |
|---------------------------------------------|-----------------|------------|
| Fix WebGL Build Pipeline                    | Yoris Kucera    | 20.03.2026 |
| Unify NFRs, FRs and Domain Model            | Yoris Kucera    | 20.03.2026 |
| Adjust README on GitLab                     | Dominik Wyss    | 20.03.2026 |
| Agents.md                                   | Cedric Cathomas | 20.03.2026 |
| Explicitly address review feedback in Teams | Dominik Wyss    | 20.03.2026 |

## 11.6 Weekly Scrum and Sprint Retrospective / 20.03.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 20.03.2026                                                     |
| Time         | 13:00–15:00                                                    |
| Sprint       | Sprint 2                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Yoris Kucera                                                   |
| Minutes by   | Nathanael Fässler                                              |
| Links        | -                                                              |

### 11.6.1 Agenda

| Agenda item                           | Owner   | Result / Decision                                                                                                                                                                          |
|---------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Planned Inventory System Architecture | Yoris   | <ul style="list-style-type: none"> <li>ScriptableObject Item represents InventoryItem and WorldItem at the same time.</li> <li>ItemStack ScriptableObject hold Item and amount.</li> </ul> |
| World Grid & GlowCore expansion       | Cedric  | <ul style="list-style-type: none"> <li>Showed simple prototype for tree generation on border expansion.</li> <li>Grid represented by 2D Array.</li> </ul>                                  |
| Node Interaction System               | Dominik | <ul style="list-style-type: none"> <li>Setting for Interaction Range for every Node.</li> <li>Problem with InputSystem to track how long a button is held.</li> </ul>                      |

### 11.6.2 What went well

- Had fun beginning the work in Unity.
- Good cooperation: Questions by team-members are answered quickly.
- The planning starts to pay off.

### 11.6.3 What didn't

- Unclear dependencies of tasks.
- CI/CD work can be annoying.

### 11.6.4 Improvements

- Keep Jira Board up to date.
- Better coordinate the task assignment.

## 11.6.5 Todos

| Todo                          | Owner     | Due        |
|-------------------------------|-----------|------------|
| Send documentation to Advisor | Dominik   | 21.03.2026 |
| Make Meeting Minutes          | Nathanael | 23.03.2026 |
| Finish and merge prototype    | Yoris     | 23.03.2026 |

## 11.7 Sprint Planning / 20.03.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 20.03.2026                                                     |
| Time         | 15:00–16:00                                                    |
| Sprint       | Sprint 3                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Yoris Kucera                                                   |
| Minutes by   | Nathanael Fässler                                              |
| Links        | -                                                              |

### 11.7.1 Sprint goal

- Create working MVP.
- Implement the foundational Item and Node Architecture.

## 11.8 Sprint Review (Advisor) / 24.03.2026

| Field        | Value                                                                        |
|--------------|------------------------------------------------------------------------------|
| Date         | 24.03.2026                                                                   |
| Time         | 15:00–16:00                                                                  |
| Sprint       | Sprint 3                                                                     |
| Location     | Building 8                                                                   |
| Participants | Thomas Bocek, Dominik Wyss, Yoris Kucera, Nathanael Fässler, Cedric Cathomas |
| Excused      | -                                                                            |
| Absent       | -                                                                            |
| Meeting lead | Yoris Kucera                                                                 |
| Minutes by   | Cedric Cathomas                                                              |
| Links        | GitHub repository + Jira board (internal team links)                         |

## 11.8.1 Agenda

| Agenda item                                        | Owner        | Result / Decision                                                                       |
|----------------------------------------------------|--------------|-----------------------------------------------------------------------------------------|
| Documentation status (FR, architecture, protocols) | Yoris Kucera | Overall quality was positively received; minor consistency updates required (see Todos) |
| Current game state demo                            | Yoris Kucera | Game is playable as planned for pre-alpha release                                       |
| Mockups                                            | Yoris Kucera | Gave a clear picture of how our UI will look                                            |
| Code quality (linter/status)                       | Yoris Kucera | Everything was good                                                                     |
| Advisor feedback                                   | Thomas Bocek | Very good feedback, overall impression better than last week                            |

## 11.8.2 Todos

| Todo                                                      | Owner           | Due        |
|-----------------------------------------------------------|-----------------|------------|
| FR: align color definitions and structure as in the NFRs  | Cedric Cathomas | 27.03.2026 |
| Enemy state machine: add explicit color legend to diagram | Yoris Kucera    | 27.03.2026 |

## 11.8.3 Next dates

| Next date                     | When             |
|-------------------------------|------------------|
| Review 3 - End of Elaboration | 14.04.2026 15:00 |

## 11.9 Weekly Scrum / 27.03.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 27.03.2026                                                     |
| Time         | 14:00–14:45                                                    |
| Sprint       | Sprint 3                                                       |
| Location     | Online                                                         |
| Participants | Cedric Cathomas, Nathanael Fässler, Dominik Wyss, Yoris Kucera |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Dominik Wyss                                                   |
| Minutes by   | Nathanael Fässler                                              |
| Links        | -                                                              |

## 11.9.1 Agenda

| Agenda item                                    | Owner             | Result / Decision                                                                                                                                                     |
|------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| New Nodes and Items created with new 3D Models | Dominik Wyss      | <ul style="list-style-type: none"> <li>Scriptable Object Architecture works</li> </ul>                                                                                |
| Pixelate Shader                                | Yoris Kucera      | <ul style="list-style-type: none"> <li>Team thinks it looks good</li> <li>Showcased parameters of the shader</li> <li>Camera angle and zoom needs refining</li> </ul> |
| Inventory UI                                   | Cedric Cathomas   | <ul style="list-style-type: none"> <li>Prototype is working</li> </ul>                                                                                                |
| Node and Tile hovering Architecture            | Nathanael Fässler | <ul style="list-style-type: none"> <li>Will be implemented with events</li> </ul>                                                                                     |

## 11.9.2 Todos

| Todo                                        | Owner        | Due        |
|---------------------------------------------|--------------|------------|
| Explicitly address review feedback in Teams | Dominik Wyss | 03.04.2026 |

## 11.10 Weekly Scrum and Sprint Retrospective / 03.04.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 03.04.2026                                                     |
| Time         | 13:00–14:00                                                    |
| Sprint       | Sprint 3                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Cedric Cathomas                                                |
| Minutes by   | Dominik Wyss                                                   |
| Links        | -                                                              |

### 11.10.1 Agenda

| Agenda item                                                                  | Owner             | Result / Decision                                                                                                                   |
|------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Add new nodes, items, tools                                                  | Dominik Wyss      | <ul style="list-style-type: none"> <li>New ScriptableObjects and Prefabs</li> <li>Updated MainWorld to demo new features</li> </ul> |
| Pixelized Shader                                                             | Yoris Kucera      | <ul style="list-style-type: none"> <li>Working shader with editable fields</li> </ul>                                               |
| Building System                                                              | Nathanael Fässler | <ul style="list-style-type: none"> <li>Chest as first building block</li> <li>Valid/invalid placement indicator</li> </ul>          |
| Inventory UI with system architecture changes                                | Cedric Cathomas   | <ul style="list-style-type: none"> <li>Inventory and handslot</li> </ul>                                                            |
| Feedback Review 2: FR coloring indicator same as NFR                         | Cedric Cathomas   |                                                                                                                                     |
| Feedback Review 2: Enemy state machine: add explicit color legend to diagram | Yoris Kucera      |                                                                                                                                     |



## 11.10.2 What went well

- Good communication within the team
- Strong interest and engagement in the project
- Attention to detail
- Review comments were questioned and discussed, not just blindly accepted
- Everything came together into a working game

## 11.10.3 What didn't

- Conflicting ideas about how to implement features and resulting dependencies sometimes disrupted the workflow.
- Tickets should be described more clearly to support conceptual implementation
- Planned too much
- The 3D player model was more difficult than expected

## 11.10.4 Todos

| Todo                                                                                                               | Owner   | Due        |
|--------------------------------------------------------------------------------------------------------------------|---------|------------|
| Send message to advisor about feedback from last review                                                            | Dominik | 10.04.2026 |
| Send message to advisor about review 3, point A5. Remove A5 from checklist because we don't have external systems. | Dominik | 10.04.2026 |

## 11.11 Sprint Planning / 03.04.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 03.04.2026                                                     |
| Time         | 14:00–15:30                                                    |
| Sprint       | Sprint 4                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Cedric Cathomas                                                |
| Minutes by   | Dominik Wyss                                                   |
| Links        | -                                                              |

### 11.11.1 Sprint goal

- Finish tickets from old sprint
- Working prototype with all core mechanics
- Focus on diagram and documentation changes for review 3

## 11.12 Sprint Review (Advisor) / 14.04.2026

| Field        | Value                                                                        |
|--------------|------------------------------------------------------------------------------|
| Date         | 14.04.2026                                                                   |
| Time         | 15:00–16:00                                                                  |
| Sprint       | Sprint 4                                                                     |
| Location     | Building 8                                                                   |
| Participants | Thomas Bocek, Dominik Wyss, Yoris Kucera, Nathanael Fässler, Cedric Cathomas |
| Excused      | -                                                                            |
| Absent       | -                                                                            |
| Meeting lead | Yoris Kucera                                                                 |
| Minutes by   | Nathanael Fässler                                                            |
| Links        | GitHub repository + Jira board (internal team links)                         |

### 11.12.1 Agenda

| Agenda item              | Owner             | Result / Decision                                                                                               |
|--------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------|
| Showcase DEV-environment | Yoris Kucera      | GitHub Action minutes are scarce. We should search for a better solution.                                       |
| Test Concept             | Yoris Kucera      | Positive: The idea of Integration Tests was adapted to the domain to use Playmode tests.                        |
| C4 Architecture          | Yoris Kucera      | Well-documented and focuses on high-level topics.                                                               |
| Techstack & Patterns     | Yoris Kucera      | Use of component-based architecture makes sense. Unity inspector variables are not really dependency injection. |
| External Dependencies    | Thomas Bocek      | Missing specification of Steam as an external dependency.                                                       |
| Alpha Demo               | Nathanael Fässler | Core game loop is working and playable. Player usability has to be improved.                                    |
| Extensibility Demo       | Nathanael Fässler | Demonstration to show how a new Node can be added without writing code.                                         |
| Savefile Structure       | Yoris Kucera      | Could easily be changed to support generated worlds by including the seed in the header.                        |

### 11.12.2 Todos

| Todo                                              | Owner        | Due        |
|---------------------------------------------------|--------------|------------|
| Research solution to reduce GitHub Runner minutes | Yoris Kucera | 28.04.2026 |
| Document Steam as an external dependency          | Dominik Wyss | 28.04.2026 |

### 11.12.3 Next dates

| Next date          | When             |
|--------------------|------------------|
| Review 4 - Quality | 05.05.2026 15:00 |

### 11.13 Weekly Scrum and Sprint Retrospective / 18.04.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 18.04.2026                                                     |
| Time         | 13:00–14:00                                                    |
| Sprint       | Sprint 4                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Nathanael Fässler                                              |
| Minutes by   | Yoris Kucera                                                   |
| Links        | -                                                              |

#### 11.13.1 Agenda

| Agenda item                 | Owner                                 | Result / Decision                                                                                                                                                                                      |
|-----------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Discuss GlowCore Upgrade UI | Nathanael Fässler/<br>Cedric Cathomas | <ul style="list-style-type: none"> <li>Upgrade system with individual levels and large upgrades</li> </ul>                                                                                             |
| Review TODOs                | All                                   | <ul style="list-style-type: none"> <li>Steam Interface Documentation</li> <li>Self-Hosted GitHub Runner               <ul style="list-style-type: none"> <li>► New Task Created</li> </ul> </li> </ul> |
| Save-Data                   | Yoris Kucera                          | <ul style="list-style-type: none"> <li>Almost done</li> <li>Moved to next sprint</li> </ul>                                                                                                            |

#### 11.13.2 What went well

- Teamwork in Documentation Tasks
- Most tasks were completed this sprint
- Everyone took on tasks as appropriate
- Alpha Release

#### 11.13.3 What didn't

- Save-Data Task still not done
- A little close on time (Spring Break)
- Uneven distribution of work-types (e.g. one person doing more docs)
- PRs with only small doc changes take a long time until review

## 11.13.4 Todos

| Todo                                                                   | Owner | Due        |
|------------------------------------------------------------------------|-------|------------|
| Think about items to be added to the game and what use they could have | All   | 24.04.2026 |

## 11.14 Sprint Planning / 18.04.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 18.04.2026                                                     |
| Time         | 14:00–15:00                                                    |
| Sprint       | Sprint 5                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Nathanael Fässler                                              |
| Minutes by   | Yoris Kucera                                                   |
| Links        | -                                                              |

### 11.14.1 Sprint goal

- Document code style and testing
- Improve test coverage
- Focus on playtesting

## 11.15 Weekly Scrum / 24.04.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 24.04.2026                                                     |
| Time         | 11:00–11:30                                                    |
| Sprint       | Sprint 5                                                       |
| Location     | Online                                                         |
| Participants | Cedric Cathomas, Nathanael Fässler, Dominik Wyss, Yoris Kucera |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Dominik Wyss                                                   |
| Minutes by   | Cedric Cathomas                                                |
| Links        | -                                                              |

## 11.15.1 Agenda

| Agenda item                                         | Owner             | Result / Decision                                                                                                                                                                                                                                     |
|-----------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR and UC current Testing state                    | Cedric Cathomas   | <ul style="list-style-type: none"> <li>partially implemented for certain UC's and NFR's is fine</li> <li>Need to synchronize deadlines</li> <li>new color code N/A is needed</li> </ul>                                                               |
| Integration Tests                                   | Nathanael Fässler | <ul style="list-style-type: none"> <li>Not all wished integration tests will be possible to write.</li> </ul>                                                                                                                                         |
| New Tasks in Backlog due to discussion with Advisor | Dominik Wyss      | <ul style="list-style-type: none"> <li>Dominik explained wished changes of the advisor</li> <li>4 new tasks got created in Jira</li> <li>further descriptions of the tasks can be found on the corresponding ticket or we can ask Dominik.</li> </ul> |

## 11.15.2 Todos

| Todo                                                         | Owner           | Due        |
|--------------------------------------------------------------|-----------------|------------|
| Create new color code for the verification of UC's and NFR's | Cedric Cathomas | 01.05.2026 |

## 11.16 Weekly Scrum and Sprint Retrospective / 30.04.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 30.04.2026                                                     |
| Time         | 20:30–20:40                                                    |
| Sprint       | Sprint 5                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Nathanael Fässler                                              |
| Minutes by   | Yoris Kucera                                                   |
| Links        | -                                                              |

### 11.16.1 What went well

- Time management
- Proper prototype is cool
- Teamwork
- New outline looks good
- Savefile

## 11.16.2 What didn't

- Time management
- Tasks still not done 1 day before sprint end
- Extra tasks from advisor
- Random features in one PR that were not planned
- BIG PRs

## 11.17 Sprint Planning / 01.05.2026

| Field        | Value                                            |
|--------------|--------------------------------------------------|
| Date         | 01.05.2026                                       |
| Time         | 13:00–15:00                                      |
| Sprint       | Sprint 6                                         |
| Location     | Online                                           |
| Participants | Nathanael Fässler, Dominik Wyss, Cedric Cathomas |
| Excused      | Yoris Kucera                                     |
| Absent       | -                                                |
| Meeting lead | Nathanael Fässler                                |
| Minutes by   | Cedric Cathomas                                  |
| Links        | -                                                |

### 11.17.1 Sprint goal

- Extend game content with phased progression
- Add new resources and mechanics
- Improve game playability and tutorial pacing
- Implement main menu, refine GlowCore UI and crafting system

## 11.18 Sprint Review (Advisor) / 05.05.2026

| Field        | Value                                                                        |
|--------------|------------------------------------------------------------------------------|
| Date         | 05.05.2026                                                                   |
| Time         | 15:00–16:00                                                                  |
| Sprint       | Sprint 6                                                                     |
| Location     | Building 8                                                                   |
| Participants | Thomas Bocek, Dominik Wyss, Yoris Kucera, Nathanael Fässler, Cedric Cathomas |
| Excused      | -                                                                            |
| Absent       | -                                                                            |
| Meeting lead | Dominik Wyss                                                                 |
| Minutes by   | Nathanael Fässler                                                            |
| Links        | GitHub repository + Jira board (internal team links)                         |

## 11.18.1 Agenda

| Agenda item              | Owner             | Result / Decision                                                                                  |
|--------------------------|-------------------|----------------------------------------------------------------------------------------------------|
| GlowCore v0.2.0 demo     | Dominik Wyss      | Works, but still needs improvements.                                                               |
| GitHub selfhosted runner | Yoris Kucera      | Is working. We have to be careful if we make the repository public.                                |
| Quality measures         | Dominik Wyss      | There are files with less than 10 lines of code. In our case it makes sense to keep them separate. |
| User Tests               | Nathanael Fässler | The first two user tests provided valuable feedback.                                               |
| Code Review              | Dominik Wyss      | WorldGrid.cs was reviewed. No unclean code found.                                                  |
| Steampage                | Dominik Wyss      | Looks good.                                                                                        |

## 11.18.2 Todos

| Todo                         | Owner        | Due        |
|------------------------------|--------------|------------|
| Document deployment on steam | not assigned | 19.05.2026 |

## 11.18.3 Next dates

| Next date               | When             |
|-------------------------|------------------|
| Review 5 - Architecture | 19.05.2026 15:00 |

## 11.19 Weekly Scrum / 08.05.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 08.05.2026                                                     |
| Time         | 20:00–20:30                                                    |
| Sprint       | Sprint 6                                                       |
| Location     | Online                                                         |
| Participants | Cedric Cathomas, Nathanael Fässler, Dominik Wyss, Yoris Kucera |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Nathanael Fässler                                              |
| Minutes by   | Yoris Kucera                                                   |
| Links        | -                                                              |

## 11.19.1 Agenda

| Agenda item           | Owner             | Result / Decision                                                          |
|-----------------------|-------------------|----------------------------------------------------------------------------|
| New Crafting Stations | Nathanael Fässler | <ul style="list-style-type: none"> <li>Anvil</li> <li>Furnace</li> </ul>   |
| Chest UI              | Cedric Cathomas   | <ul style="list-style-type: none"> <li>Chest has its own UI now</li> </ul> |

## 11.19.2 Todos

| Todo                             | Owner        | Due        |
|----------------------------------|--------------|------------|
| Fix Unity Crash in Test Workflow | Yoris Kucera | 09.05.2026 |

## 11.20 Weekly Scrum and Sprint Retrospective / 15.05.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 15.05.2026                                                     |
| Time         | 13:00-14:00                                                    |
| Sprint       | Sprint 6                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Yoris Kucera                                                   |
| Minutes by   | Cedric Cathomas                                                |
| Links        | -                                                              |

### 11.20.1 Agenda

| Agenda item               | Owner                              | Result / Decision                                                                                                                                                                                     |
|---------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UI design and title scene | Cedric Cathomas                    | <ul style="list-style-type: none"> <li>• Showed how the implemented UI looks and works</li> </ul>                                                                                                     |
| Documentation             | Nathanael Fässler and Yoris Kucera | <ul style="list-style-type: none"> <li>• Presented Documentation changes</li> <li>• Deployment diagram</li> <li>• Design decisions</li> <li>• Game design</li> <li>• Frontend Architecture</li> </ul> |
| Audio system              | Dominik Wyss                       | <ul style="list-style-type: none"> <li>• Demo of how to implement new audio effects</li> <li>• Presented Audio manager singleton</li> </ul>                                                           |
| Biome and Environment     | Dominik Wyss                       | <ul style="list-style-type: none"> <li>• Demo of how to implement biomes and environment objects</li> </ul>                                                                                           |

### 11.20.2 What went well

- Game looks much better, overall impression improved a lot, looks like a real game now
- Border looks nice
- Good teamwork

### 11.20.3 What didn't

- Discord communication tone
- Non documented documentation tasks



## 11.20.4 Todos

| Todo             | Owner           | Due        |
|------------------|-----------------|------------|
| NFR Verification | Cedric Cathomas | 17.05.2026 |
| Time tracking    | All             | 17.05.2026 |

## 11.21 Sprint Planning / 15.05.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 15.05.2026                                                     |
| Time         | 14:00–15:00                                                    |
| Sprint       | Sprint 7                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Cedric Cathomas, Yoris Kucera |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Yoris Kucera                                                   |
| Minutes by   | Cedric Cathomas                                                |
| Links        | -                                                              |

### 11.21.1 Sprint goal

- Finalise Audio system
- Finalise Game biomes and environment objects
- Presentation preparation
- Final game features

## 11.22 Sprint Review (Advisor) / 19.05.2026

| Field        | Value                                                                        |
|--------------|------------------------------------------------------------------------------|
| Date         | 19.05.2026                                                                   |
| Time         | 15:00–16:00                                                                  |
| Sprint       | Sprint 7                                                                     |
| Location     | Building 8                                                                   |
| Participants | Thomas Bocek, Dominik Wyss, Yoris Kucera, Nathanael Fässler, Cedric Cathomas |
| Excused      | -                                                                            |
| Absent       | -                                                                            |
| Meeting lead | Yoris Kucera                                                                 |
| Minutes by   | Dominik Wyss                                                                 |
| Links        | GitHub repository + Jira board (internal team links)                         |

## 11.22.1 Agenda

| Agenda item                   | Owner             | Result / Decision                                              |
|-------------------------------|-------------------|----------------------------------------------------------------|
| Linux Build                   | Yoris Kucera      | Game doesn't start because of error.                           |
| GlowCore near finish demo     | Yoris Kucera      | Great working game, but it still needs some finishing touches. |
| New GlowCore levels           | Nathanael Fässler | New progress for the game, looks good.                         |
| New Biomes and their creation | Nathanael Fässler | Great map for the world.                                       |
| Sound                         | Nathanael Fässler | Sound is very nice, brings the game to life.                   |
| Automation                    | Nathanael Fässler | New automation feature is impressive.                          |

## 11.22.2 Todos

| Todo                      | Owner        | Due        |
|---------------------------|--------------|------------|
| Fix Linux build           | not assigned | 05.06.2026 |
| Shorten the documentation | not assigned | 05.06.2026 |

## 11.23 Weekly Scrum / 22.05.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 22.05.2026                                                     |
| Time         | 13:00–14:00                                                    |
| Sprint       | Sprint 7                                                       |
| Location     | Online                                                         |
| Participants | Cedric Cathomas, Nathanael Fässler, Dominik Wyss, Yoris Kucera |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Cedric Cathomas                                                |
| Minutes by   | Nathanael Fässler                                              |
| Links        | -                                                              |

## 11.23.1 Agenda

| Agenda item                  | Owner             | Result / Decision                       |
|------------------------------|-------------------|-----------------------------------------|
| Assignment of last tasks     | All               | All important tasks have been assigned. |
| Showcase Automation Features | Nathanael Fässler | Features are fully working now.         |

## 11.23.2 Todos

| Todo                   | Owner    | Due        |
|------------------------|----------|------------|
| Write Personal Reports | Everyone | 02.06.2026 |

## 11.24 Weekly Scrum and Sprint Retrospective / 29.05.2026

| Field        | Value                                                          |
|--------------|----------------------------------------------------------------|
| Date         | 29.05.2026                                                     |
| Time         | 13:00–13:40                                                    |
| Sprint       | Sprint 7                                                       |
| Location     | Online                                                         |
| Participants | Nathanael Fässler, Dominik Wyss, Yoris Kucera, Cedric Cathomas |
| Excused      | -                                                              |
| Absent       | -                                                              |
| Meeting lead | Dominik Wyss                                                   |
| Minutes by   | Yoris Kucera                                                   |
| Links        | -                                                              |

### 11.24.1 Agenda

| Agenda item    | Owner             | Result / Decision                                                                                                                                                                                                    |
|----------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Improved Fog   | Dominik Wyss      | <ul style="list-style-type: none"> <li>World border is more clear now because of improved fog</li> </ul>                                                                                                             |
| Complete World | Nathanael Fässler | <ul style="list-style-type: none"> <li>Game World is now built completely</li> <li>New map item to help navigate</li> <li>There are rivers now</li> <li>New potions</li> <li>Smoother camera and movement</li> </ul> |
| Item Icons     | Cedric Cathomas   | <ul style="list-style-type: none"> <li>All items have their own icons now</li> <li>New Unity scene to generate these icons</li> <li>Integrated directly into item prefabs</li> </ul>                                 |

### 11.24.2 What went well

- Game looks good
- “Mandatory” tasks are almost all completed
- Icons look cool
- Automation is good, game has proper progression

### 11.24.3 What didn't

- Still many open tasks
- Merge conflicts in Unity Scene files
- Pull Requests took a long time to get approved
- Still a few annoying bugs

### 11.24.4 Todos

| Todo                                  | Owner             | Due        |
|---------------------------------------|-------------------|------------|
| Turn in final documentation on Moodle | All               | 05.06.2026 |
| Write personal Reports                | All               | 02.06.2026 |
| Release Game on Steam                 | Nathanael Fässler | 05.06.2026 |

# List of Figures

|          |                                                                          |    |
|----------|--------------------------------------------------------------------------|----|
| Image 1  | Use Case Diagram                                                         | 7  |
| Image 2  | Domain Model Diagram                                                     | 23 |
| Image 3  | Enemy State Machine Diagram                                              | 24 |
| Image 4  | Main Menu                                                                | 25 |
| Image 5  | New Game                                                                 | 26 |
| Image 6  | Pause Menu                                                               | 26 |
| Image 7  | Audio Settings                                                           | 27 |
| Image 8  | Display Settings                                                         | 27 |
| Image 9  | Unapplied Changes                                                        | 28 |
| Image 10 | Inventory                                                                | 28 |
| Image 11 | Crafting Table                                                           | 28 |
| Image 12 | More Crafting Stations                                                   | 29 |
| Image 13 | GlowCore Upgrade                                                         | 29 |
| Image 14 | Chest                                                                    | 29 |
| Image 15 | C4 System Context Diagram                                                | 30 |
| Image 16 | C4 Container Diagram                                                     | 31 |
| Image 17 | C4 Component Diagram - Gameplay Logic                                    | 33 |
| Image 18 | C4 Component Diagram - Presentation Layer                                | 34 |
| Image 19 | Interface Overview Diagram                                               | 35 |
| Image 20 | Technologies Diagram                                                     | 38 |
| Image 21 | Deployment Diagram                                                       | 44 |
| Image 22 | CI/CD Pipeline Overview                                                  | 45 |
| Image 23 | Definition of Ready and Definition of Done - ticket lifecycle flowchart  | 46 |
| Image 24 | GC-186 - Example of a ticket satisfying all Definition of Ready criteria | 47 |
| Image 25 | GC-186 - Example of a ticket satisfying all Definition of Done criteria  | 47 |
| Image 26 | Lines of Code over Time                                                  | 50 |

|                                               |           |
|-----------------------------------------------|-----------|
| <b>Image 27 Test Coverage Chart .....</b>     | <b>53</b> |
| <b>Image 28 Savefile Structure .....</b>      | <b>58</b> |
| <b>Image 29 Initial Jira Timeline .....</b>   | <b>69</b> |
| <b>Image 30 Updated Jira Timeline .....</b>   | <b>69</b> |
| <b>Image 31 Initial Backlog .....</b>         | <b>70</b> |
| <b>Image 32 Sprint 1 Backlog .....</b>        | <b>71</b> |
| <b>Image 33 Sprint 2 Burndown Chart .....</b> | <b>72</b> |
| <b>Image 34 Sprint 2 Backlog .....</b>        | <b>72</b> |
| <b>Image 35 Sprint 3 Burndown Chart .....</b> | <b>73</b> |
| <b>Image 36 Sprint 3 Backlog .....</b>        | <b>73</b> |
| <b>Image 37 Sprint 4 Burndown Chart .....</b> | <b>74</b> |
| <b>Image 38 Sprint 4 Backlog .....</b>        | <b>74</b> |
| <b>Image 39 Sprint 5 Burndown Chart .....</b> | <b>75</b> |
| <b>Image 40 Sprint 5 Backlog .....</b>        | <b>75</b> |
| <b>Image 41 Sprint 6 Burndown Chart .....</b> | <b>76</b> |
| <b>Image 42 Sprint 6 Backlog .....</b>        | <b>76</b> |
| <b>Image 43 Sprint 7 Burndown Chart .....</b> | <b>77</b> |
| <b>Image 44 Sprint 7 Backlog .....</b>        | <b>77</b> |

# List of Tables

|                 |                                                                   |           |
|-----------------|-------------------------------------------------------------------|-----------|
| <b>Table 1</b>  | <b>Glossary</b>                                                   | <b>5</b>  |
| <b>Table 2</b>  | <b>UC Status Color Legend</b>                                     | <b>8</b>  |
| <b>Table 3</b>  | <b>UC Overview – All Use Cases at a Glance</b>                    | <b>8</b>  |
| <b>Table 4</b>  | <b>UC01 - Install Game</b>                                        | <b>9</b>  |
| <b>Table 5</b>  | <b>UC02 - Save Game</b>                                           | <b>9</b>  |
| <b>Table 6</b>  | <b>UC03 - Break Nodes</b>                                         | <b>9</b>  |
| <b>Table 7</b>  | <b>UC04 - Expand Map</b>                                          | <b>9</b>  |
| <b>Table 8</b>  | <b>UC05 - Store Items</b>                                         | <b>10</b> |
| <b>Table 9</b>  | <b>UC06 - Craft Items</b>                                         | <b>10</b> |
| <b>Table 10</b> | <b>UC07 - Build Nodes</b>                                         | <b>10</b> |
| <b>Table 11</b> | <b>UC08 - Experience Story</b>                                    | <b>10</b> |
| <b>Table 12</b> | <b>UC09 - Fight Enemies</b>                                       | <b>11</b> |
| <b>Table 13</b> | <b>UC10 - Automate Resource Gathering</b>                         | <b>11</b> |
| <b>Table 14</b> | <b>UC11 - Change Settings</b>                                     | <b>11</b> |
| <b>Table 15</b> | <b>NFR Status Color Legend</b>                                    | <b>12</b> |
| <b>Table 16</b> | <b>NFR Overview – All Non-Functional Requirements at a Glance</b> | <b>13</b> |
| <b>Table 17</b> | <b>NFR Priority Matrix – MVP Relevance vs. Importance</b>         | <b>14</b> |
| <b>Table 18</b> | <b>NFR101 – World Load Time</b>                                   | <b>14</b> |
| <b>Table 19</b> | <b>NFR102 – In-Game Transition Time</b>                           | <b>15</b> |
| <b>Table 20</b> | <b>NFR103 – Frame Rate</b>                                        | <b>15</b> |
| <b>Table 21</b> | <b>NFR104 – Save File Size</b>                                    | <b>15</b> |
| <b>Table 22</b> | <b>NFR105 – Minimal Hardware Requirements</b>                     | <b>16</b> |
| <b>Table 23</b> | <b>NFR201 – Player Restriction Visibility</b>                     | <b>16</b> |
| <b>Table 24</b> | <b>NFR202 – Light Upgrade Progress Indication</b>                 | <b>16</b> |
| <b>Table 25</b> | <b>NFR203 – New Player Learnability</b>                           | <b>17</b> |
| <b>Table 26</b> | <b>NFR204 – Input Support</b>                                     | <b>17</b> |

|                                                        |           |
|--------------------------------------------------------|-----------|
| <b>Table 27 NFR205 – Game Language</b>                 | <b>17</b> |
| <b>Table 28 NFR206 – Visual Style Consistency</b>      | <b>18</b> |
| <b>Table 29 NFR301 – Save File Portability</b>         | <b>18</b> |
| <b>Table 30 NFR302 – Game Progress Persistence</b>     | <b>19</b> |
| <b>Table 31 NFR303 – Save Data Integrity</b>           | <b>19</b> |
| <b>Table 32 NFR401 – Extensibility of Game Systems</b> | <b>19</b> |
| <b>Table 33 NFR402 – Automated Test Coverage</b>       | <b>20</b> |
| <b>Table 34 NFR501 – Multi-Platform Support</b>        | <b>20</b> |
| <b>Table 35 NFR502 – Engine and Rendering Pipeline</b> | <b>20</b> |
| <b>Table 36 NFR503 – Display Resolution Support</b>    | <b>21</b> |
| <b>Table 37 NFR-to-Architecture Traceability</b>       | <b>22</b> |
| <b>Table 38 Interface Contracts</b>                    | <b>36</b> |
| <b>Table 39 UT01</b>                                   | <b>54</b> |
| <b>Table 40 UT01 - Additional Feedback</b>             | <b>54</b> |
| <b>Table 41 UT02</b>                                   | <b>55</b> |
| <b>Table 42 UT02 - Additional Feedback</b>             | <b>55</b> |
| <b>Table 43 Team availability for review meetings</b>  | <b>60</b> |
| <b>Table 44 Project Management Roles</b>               | <b>62</b> |
| <b>Table 45 Developer Roles</b>                        | <b>63</b> |
| <b>Table 46 Planned Meetings Overview</b>              | <b>64</b> |
| <b>Table 47 Planning Tools</b>                         | <b>65</b> |
| <b>Table 48 Work Items</b>                             | <b>65</b> |
| <b>Table 49 Initial Milestones</b>                     | <b>66</b> |
| <b>Table 50 Updated Milestones</b>                     | <b>67</b> |
| <b>Table 51 Epics</b>                                  | <b>68</b> |
| <b>Table 52 Risk Analysis</b>                          | <b>78</b> |
| <b>Table 53 Risk Matrix</b>                            | <b>79</b> |
| <b>Table 54 Risk Matrix (Mitigated)</b>                | <b>79</b> |
| <b>Table 55 Assessed Risk Matrix (14.04.2026)</b>      | <b>79</b> |

**Table 56 Time Statistics** ..... **81**